መሠረተ ሕይወት

Meserete Hiwot Super-App

System Documentation

Mezmur · Education · Programs & Attendance · Selection · Member Services · Administration

Document version	1.0
Date	2026-10-16
Status	Initial release — verified against source code
System release covered	1.5.3+16 (production) · source tree 1.5.3+19
Organisation	Meserete Hiwot Sunday School, Debre Enqu St. Lideta Lemaryam Church
Classification	Internal — authorised stakeholders; no credentials included

Product: መሠረተ ሕይወት (Meserete Hiwot) Sunday School super-app — internal code name *Mezmur Zelideta* (package `com.mezmur.zelideta`); the API identifies itself as *Mezmur Zemseretee Hiwot API*.

Organisation: Meserete Hiwot Sunday School (መሠረተ ሕይወት ሰንበት ትምህርት ቤት), Debre Enqu St. Lideta Lemaryam Church, Ethiopian Orthodox Tewahdo Church.

Document version: 1.0 · **Date:** 2026-10-16 · **Status:** Initial release — reviewed against the source code; items marked *requires verification* could not be confirmed from the code alone.

System version covered: production release **1.5.3+16** (latest release recorded in the project logs) and the source tree at **1.5.3+19**, which contains unreleased work that is identified separately throughout.

Audience: future developers, technical administrators, the project owner and authorised stakeholders. Credentials, keys and the restricted system-console design are deliberately excluded; see *Appendix F*.

Contents

1. Document control 5
2. Executive summary 6
3. System purpose, scope, objectives and target users 7
3.1 Purpose 7
3.2 Scope 7
3.3 Objectives 7
3.4 Target users 7
4. Functional overview and feature inventory 8
4.1 Accounts & security 8
4.2 Members, HR & departments 8
4.3 Mezmur 9
4.4 Programs & attendance 9
4.5 Education 10
4.6 Holiday selection & Mezmur services 11
4.7 Contributions, property & library 11
4.8 Projects & collaboration 12
4.9 Communication 12
4.10 Administration, reports & audit 12
4.11 Platform 13
5. System architecture and component relationships 14
6. Technology stack, frameworks, libraries and external dependencies 16
6.1 Backend (`laravel/composer.json`) 16
6.2 Mobile / web client (`mobile/mobile/pubspec.yaml`) 16
6.3 External services (all optional, configured by environment variables) 16
7. Backend architecture and major modules 18
7.1 Service catalogue 18
8. Frontend / mobile architecture and navigation 20
8.1 Screen map 20
8.2 Representative screens (rendered from the app's own widgets with sample data) 21
9. Database documentation 23
9.1 Tables by module 23
9.2 Important constraints and conventions 29
9.3 Migrations 30
10. Authentication, sessions, device restrictions and account recovery 31
11. Roles, departments, permissions and administrative hierarchy 32
11.1 Roles (from the database) 32
11.2 Departments 32

11.3 Member groups 33
11.4 Permission catalogue (63 keys) and the roles that hold them 33
12. Detailed module-by-module functionality 36
12.1 Admin panel 36
12.2 Members and HR 36
12.3 Programs (see section 14 for attendance) 36
12.4 Banners, pages and guidance 37
13. Mezmur, lyrics, audio, file uploads and media management 38
14. Programs, attendance, punctuality, scoring and attendance reports 39
14.1 Released behaviour (1.5.3+16) 39
14.2 Unreleased (source tree, target builds +18/+19) 39
15. Education, classes and course attendance 40
16. Holiday selection: eligibility, review, approval and finalisation 41
17. Contributions, finance, property and borrowing workflows 43
17.1 Monthly contributions 43
17.2 Items, borrowing, fines and library 43
17.3 Department property requests 44
18. Departments, chat, projects, tasks, budgets and expenses 45
19. Notifications, announcements, reminders and delivery mechanisms 46
20. Ethiopian calendar and localisation support 47
21. Offline behaviour, synchronisation and conflict handling 48
22. Reporting, exports, audit logs and document generation 49
23. API documentation 50
23.1 Conventions 50
23.2 Request/response examples 50
23.3 Route reference 51
24. Important workflows (diagrams) 69
25. Security, privacy, authorisation, data validation and file-upload protections 72
26. Installation, environment configuration, deployment and hosting 73
26.1 Requirements 73
26.2 Environment variables (names only) 73
26.3 First installation 73
26.4 Client builds 73
26.5 Release routine used by the project 74
27. Database setup, migrations, backups and recovery 75
28. Background jobs, scheduled processes, queues and notification delivery 76
29. Logging, troubleshooting, known errors and maintenance 77
30. Testing approach and verified test coverage 78

31. Current limitations, technical debt, incomplete features and risks 79
32. Recommended future improvements (proposals, not existing functionality) 80
33. Glossary 81
Appendix A. Screen and route map of the admin panel 82
Appendix B. Settings keys 83
Appendix C. Data dictionary 84
Appendix D. Unreleased schema changes in the source tree 135
Appendix E. Source documents consulted 136
Appendix F. Restricted information 137

1. Document control

Item	Value
Document	Meserete Hiwot Super-App — System Documentation
Version / date	1.0 / 2026-10-16
Prepared from	laravel/ (backend), mobile/mobile/ (Flutter app), database schema of the integration-test database (production dump + all applied migrations), project logs (MEMORY.md, GUIDANCE_IMPLEMENTATION_AUDIT_AND_DEBUG_LOG.md, SELECTION.md, prompt.md)
Method	Static reading of code, routes (`php artisan route:list`), live schema introspection of `mh_app_test`, execution of both automated test suites
Companion files	Feature_and_Module_Inventory.xlsx · IMPLEMENTATION_STATUS.md · System_Architecture.png/.svg · diagrams/ · Amharic presentation

2. Executive summary

Meserete Hiwot is a *super-app* for a Sunday-school community: one Flutter application (Android, web; iOS configured) backed by one Laravel REST API and a MariaDB/MySQL database. It joins the community's main activities in one place:

- **Mezmur** — a hymn library with categories, lyrics (including time-synced lyrics), an audio player and offline downloads.
- **Education** — classes, courses, teachers, marks with an approval workflow, assignments, quizzes, and an online learning platform (categories → subjects → units → lessons).
- **Programs & attendance** — church programs with recurring schedules and reminders, and an HR-controlled attendance session with QR scanning, signatures and offline capture.
- **Holiday Mezmur service selection** — a transparent, configurable scoring of members (attendance, exams, Mezmur attendance, contributions, department participation) with department review and finalisation.
- **Member services** — digital member ID with QR, monthly contributions, property borrowing and the library, department membership, projects and tasks, department chat, and notifications.
- **Administration** — 63 permissions grouped into roles, a permission-filtered admin panel, audit and activity logs, and 15 report types exportable as PDF, Excel and CSV.

The feature inventory lists **98 features**: 79 live in 1.5.3+16, 6 implemented in source but not released, 7 partial, 4 planned/missing, 1 withdrawn and 1 configured but unverified. The API exposes **564 documented routes** (15 of them belong to unreleased work; a small restricted set is excluded) and the database has **128 tables** in the test copy of the schema.

Key risks (section 31): plaintext credentials in local project files; a production data dump stored in the project folder; unreleased schema changes in the source tree that make 37 integration tests fail against the un-migrated test database; shared hosting without a cron or queue worker (jobs run after HTTP requests).

3. System purpose, scope, objectives and target users

3.1 Purpose

To give the Sunday school one trustworthy digital place for its spiritual content (Mezmur, lessons), its organisation (departments, programs, projects) and its accountability (attendance, contributions, property, selection results), replacing paper lists and scattered messaging.

3.2 Scope

In scope (verified in code): every module listed in section 4. Out of scope: payments collection through a payment gateway (contributions are *recorded*, not charged), public self-service web pages beyond share links and the APK download page, and any integration not listed in section 6.

3.3 Objectives

- Make hymns, lyrics and audio available on every member's phone, including offline.
- Record attendance reliably and fairly: one accountable taker, a signature per mark, no unverifiable 'late' judgement, offline capture without data loss.
- Run education end-to-end: registration, teaching assignments, marks review and publication, online lessons and quizzes.
- Select members for holiday Mezmur service by published, configurable rules, with department review and an audit trail.
- Give leaders reports and exports; give members visibility of their own records (attendance calendar, contributions, results).
- Protect personal data: least-privilege permissions, encryption at rest for sensitive fields, device binding, audit logging.

3.4 Target users

User	What they use
Members (students, youth & adults, married, distance)	Mezmur, programs, reminders, own attendance, member card, contributions, library, education (when enabled), department chat
Teachers	Their courses: lessons, quizzes, marks entry and submission, assignments
Attendance takers	The roster of programs they are assigned to (online or offline)
Department leaders / admins	Department members, participation scores, projects, property requests, chat moderation
HR	Attendance control: takers, signature exemptions, corrections, offline conflict queue
Finance & Property	Contributions, rates, property requests, project budget and expense approval
Education administrators / registrars	Classes, courses, teachers, registration, marks review, content
Mezmur administrators	Mezmur content, wedding services, selection categories and Mezmur report
High administrators / administrators	Users, roles, settings, releases, notifications, reports, audit

4. Functional overview and feature inventory

Status legend: **LIVE** = Live (released 1.5.3+16); **UNREL** = Implemented in source – not yet released; **PARTIAL** = Partially implemented; **MISSING** = Not implemented (planned); **WITHDRAWN** = Withdrawn (removed by request); **CONFIG** = Configured – not verified. The full inventory with evidence is also in *Feature_and_Module_Inventory.xlsx*.

4.1 Accounts & security

Feature	Status	Evidence (source)	Notes
Registration with 4/6-digit PIN, member group, membership profile	LIVE	routes/api/auth.php · AuthService::register	
Sign-in by phone or e-mail + PIN; 5 failures → 15-minute lock	LIVE	AuthService::login (MAX_FAILED, LOCK_MINUTES)	
One trusted device per member; administrators on several devices	LIVE	AuthService::login · Middleware::resolve	
Access JWT + rotating refresh tokens; app sessions do not time out	LIVE	Tokens.php · config mz.jwt	Decision 2026-10-13: automatic logout removed
Session list, end one / others / all sessions	LIVE	Sessions.php · /settings/sessions	
PIN reset by 6-digit code via Telegram or e-mail	LIVE	AuthService::forgotPassword/resetPassword	
Admin device reset, force logout, disable, soft delete	LIVE	UserAdmin.php	
Account approval states (pending / active / disabled), can_login flag	LIVE	AuthService::login	
Biometric lock and screenshot blocking on personal screens	LIVE	local_auth · widgets/secure_screen.dart	
Telegram account linking	LIVE	Notifications/Telegram.php · routes/api/telegram.php	

4.2 Members, HR & departments

Feature	Status	Evidence (source)	Notes
Membership form (profile) with encrypted personal fields	LIVE	MemberProfile.php · Crypto/EncryptedColumns.php	
Member ID numbers with group prefix; digital ID card with QR	LIVE	MemberIds.php · Members.php · /settings/me	
Member groups (types) management	LIVE	MemberGroups.php	
Profile change requests with approval and permanent history	LIVE	ProfileChanges.php	
7 departments, membership, role labels, department admins	LIVE	Departments.php · DepartmentAdmins.php	
Monthly department participation score by the department leader	LIVE	Departments::saveParticipation	
Professions catalogue	LIVE	Professions.php	
Bulk import of users / Mezmurs / categories (xlsx, csv, paste; dry run)	LIVE	Import.php	

4.3 Mezmur

Feature	Status	Evidence (source)	Notes
Mezmur library: hierarchical categories, search, recent / popular, today's category	LIVE	Mezmurs.php · Categories.php	
Lyrics with adjustable font size; time-synced lyrics (LRC) and LRC maker for admins	LIVE	core/utils/lrc.dart · admin/lrc_maker.dart	
Audio player: background playback, lock-screen controls, queue, shuffle, repeat, speed, sleep timer	LIVE	services/audio_player_service.dart	
Favourites and offline downloads (lyrics always, audio optional)	LIVE	favorites_service.dart · download_service.dart	
YouTube link playback	LIVE	features/mezmurs/youtube_view.dart	
Admin: create / edit / publish Mezmurs, audio & cover upload, bulk category assignment, drafts	LIVE	content_admin.dart · Mezmurs.php	
Share links (/m/{id}) with preview page and app deep link	LIVE	routes/web/public.php	

4.4 Programs & attendance

Feature	Status	Evidence (source)	Notes
Programs: once / weekly / monthly / twice monthly / Ethiopian calendar days; occurrence overrides	LIVE	ProgramSchemas.php · Schedule.php	
Audience (all or participants), registration, capacity, member-group targeting	LIVE	Programs.php	
Program reminders (2 d / 1 d / 6 h / 1 h) and 15-day interval reminders; mandatory reminders	LIVE	ReminderJobs.php · program_alarms.dart	
Attendance sessions: one HR-assigned taker, opens 15 min before, Start / End, auto-end	LIVE	AttendanceSessions.php	
QR or manual marking; signature for every Present mark (group exemptions)	LIVE	Attendance::set · attendance.signature setting	
No 'late' status; capture time and elapsed seconds since Start recorded	LIVE	Attendance.php (LATE_REMOVED)	
Absence excuse requests with deadline and monthly / quarterly limits; admin cancel of approved excuse	LIVE	Permissions.php	
Corrections after closure with review	LIVE	Attendance::requestCorrection/reviewCorrection	

Feature	Status	Evidence (source)	Notes
Offline attendance with encrypted queue and HR conflict review queue	LIVE	Sync.php · /admin/attendance/conflicts	
Member attendance calendar and statistics	LIVE	Attendance::calendar/stats · /settings/attendance	
Class attendance through programs linked to classes (edu_attendance frozen)	UNREL	migration 2026_10_20_000002 · Settings::eduAttendanceCutover	Release target +19 (F4)
Elapsed-time interval percentages per attendance category for selection	UNREL	Selections::validateIntervalRules/scoreOccurrenceInterval	Release target +19 (F6)

4.5 Education

Feature	Status	Evidence (source)	Notes
Classes, courses (subjects), schedules, staff, registration windows, regular / distance streams	LIVE	Edu.php · EduRegistration.php	
Teacher assignments with history; only active teachers enter marks	LIVE	EduTeachers.php · edu_course_teachers	
Marks workflow draft → submitted → approved / rejected → published, amendments with reason	LIVE	EduMarks.php · edu_score_history	
Assignments (individual / group), submissions and grading	PARTIAL	Edu.php	No publish time, deadline enforcement or late rule (F9)
Student requests (class / stream / mode change, add / drop) and complaints (EC-YYYY-NNNNNN)	LIVE	EduRequests.php · EduComplaints.php	
Learning platform: categories → subjects → units → lessons with materials and progress	LIVE	EduLearning.php · features/edu/learn	
Quizzes: 4 question types, timer, attempts, manual grading, send to score sheet	LIVE	EduQuizzes.php	
Question banks / random draws	PARTIAL	EduQuizzes.php	Banks not implemented (F12)
Module gate: Education can be switched off per member (403 MODULE_DISABLED)	LIVE	Http/ModuleGate.php	
Academic years, terms, Grades 1–12 levels, exclusive enrolment, course offerings	UNREL	EduAcademicYears.php · migration 2026_10_20_000001	Release target +18 (F1–F3)
Course outcomes and pass marks	MISSING	pass_mark column added in unreleased migration only	F10
Failed courses and online retakes	MISSING	—	F11
Certificates and verification	MISSING	—	F13 – needs wording/signatory decision (D7)
Student dashboard	PARTIAL	EduDashboard::forUser	F14
Academic and engagement scores for admins	MISSING	—	F15

4.6 Holiday selection & Mezmur services

Feature	Status	Evidence (source)	Notes
Holiday selection: 5 weighted components, minimums, pass mark, bonus, per-program weights	LIVE	Selections.php (DEFAULT_CONFIG)	
Lifecycle draft → open → review → finalized (+ cancel); required department reviews; overrides with reason	LIVE	Selections::setStatus/submitReview/override	
Service categories (ቁረብ / መዝሙር), rankings, publication, member feedback	LIVE	Selections::assignCategories/publish/submitFeedback	
Selection and Mezmur-selection reports (PDF / XLSX / CSV)	LIVE	ReportExport.php · SelectionReports.php	
Contribution month range for selection	UNREL	routes/api/selections.php start_month/end_month	Was a defect (F5); fix in source
Selection configuration versions	UNREL	Selections::recordConfigVersion	F6
Department-based criteria (50/30/10/10)	WITHDRAWN	backups/department-criteria-2026-10-15	Removed 2026-10-15; tables kept unused
Wedding (other) Mezmur services with manual participant selection	LIVE	Weddings.php	
Daily choir eligibility computation	LIVE	Choir::refresh · job choir:eligibility 02:15	

4.7 Contributions, property & library

Feature	Status	Evidence (source)	Notes
Monthly contributions by Ethiopian month, rates per member type, receipts, verification	LIVE	Payments.php	
'Paid after due month' label (grace days) — financial rule kept	LIVE	payments.policy grace_days	
Items / property inventory with categories, codes, departments, history	LIVE	Items.php	
Borrowing: direct, bulk, approval, renewal, return conditions, signatures	LIVE	Borrowings.php	
Fines: per day / week, grace, cap, override with reason, daily recalculation	LIVE	Fines.php · job fines:recalculate	
Library: digital PDF books (encrypted on device), physical book requests	PARTIAL	Library.php · book_vault.dart	No sub-categories, favourites, reading progress (F16)
Physical library with per-copy IDs	PARTIAL	items quantity-based	F17 – decision D6 needed

Feature	Status	Evidence (source)	Notes
Department property requests with review, issue, partial issue, return	LIVE	PropertyRequests.php	

4.8 Projects & collaboration

Feature	Status	Evidence (source)	Notes
Department projects, teams, tasks with several assignees, statuses with reasons	LIVE	Projects/ProjectService.php · TaskService.php	
Budgets by category, expenses with receipts, over-budget control, adjustments (Finance approval)	LIVE	Projects/BudgetService.php	
Issues, suggestions (convert to task), comments, files, activity feed, reports	LIVE	Projects/FeedbackService.php · ReportService.php	
Department chat: group + private threads, media for leaders, meeting links, Task Bot	LIVE	Chat.php · Projects/Bot.php	
Task deadline reminders	LIVE	ProjectReminders.php · scheduled_notifications	

4.9 Communication

Feature	Status	Evidence (source)	Notes
In-app notification centre with topics and per-user preferences	LIVE	Notifications.php (8 topics)	
Push notifications (Firebase Cloud Messaging)	LIVE	Fcm.php · push_service.dart	Requires Firebase credentials in the environment
E-mail delivery (SMTP)	LIVE	Mailer.php	Requires SMTP settings
Admin broadcasts to one / selected / filtered / all, resend, delivery history	LIVE	Broadcasts.php · Targeting.php	
Banners (advertisements) and internal pages	LIVE	Promotions.php	
Notification delivery health screen	LIVE	Notifications::health · /admin/notification-health	
In-app guidance: 56 bilingual articles, tours, contextual help	LIVE	GuidanceService.php · guidance_catalog.dart	

4.10 Administration, reports & audit

Feature	Status	Evidence (source)	Notes
Role-based access: 63 permissions, roles, direct grants and revocations	LIVE	Auth/Catalog.php · Admins.php	
Admin panel with 10 permission-filtered groups	LIVE	features/admin/admin_home.dart	
Dashboard and reports	LIVE	Reports.php	
15 export kinds in PDF / XLSX / CSV, each audited	LIVE	ReportExport.php	
Audit log and user activity log	LIVE	Auth/Audit.php · Activity.php	

Feature	Status	Evidence (source)	Notes
System settings with validated schemas	LIVE	Settings.php	
App releases: APK hosting, download page, mandatory update / minimum build	LIVE	Releases.php · update_service.dart	

4.11 Platform

Feature	Status	Evidence (source)	Notes
Ethiopian calendar (Amete Mihret) for periods, dates, reports and program days	LIVE	Support/Ethiopian.php · ethiopian_date_picker.dart	
Amharic (template) and English localisation (2,754 keys)	LIVE	lib/l10n/app_am.arb · app_en.arb	
Light / dark theme, church branding and module watermarks	LIVE	core/theme · widgets/brand.dart	
Offline: cache of GET responses, offline pack, sync queue for listed operations	PARTIAL	local_db.dart · sync_service.dart	Platform-wide offline incl. admins planned (F7)
Offline queue preserved on sign-out	UNREL	features/auth/session.dart logout(discard)	Was a defect (F0a)
Background jobs without cron (after-response tick)	LIVE	Jobs/Jobs.php	
At-rest encryption of sensitive columns	LIVE	Crypto/Seal.php · DataEncryptor.php	
Web app (PWA) build	LIVE	mobile/mobile/web · deployed per records	
iOS app	CONFIG	ios/ project, Info.plist	No build or store release verified
Shared Settings filtered by module; Education edit pen; role guidance for students/teachers; responsive checks of admin tables	PARTIAL	prompt.md F18–F21	Polish items

5. System architecture and component relationships

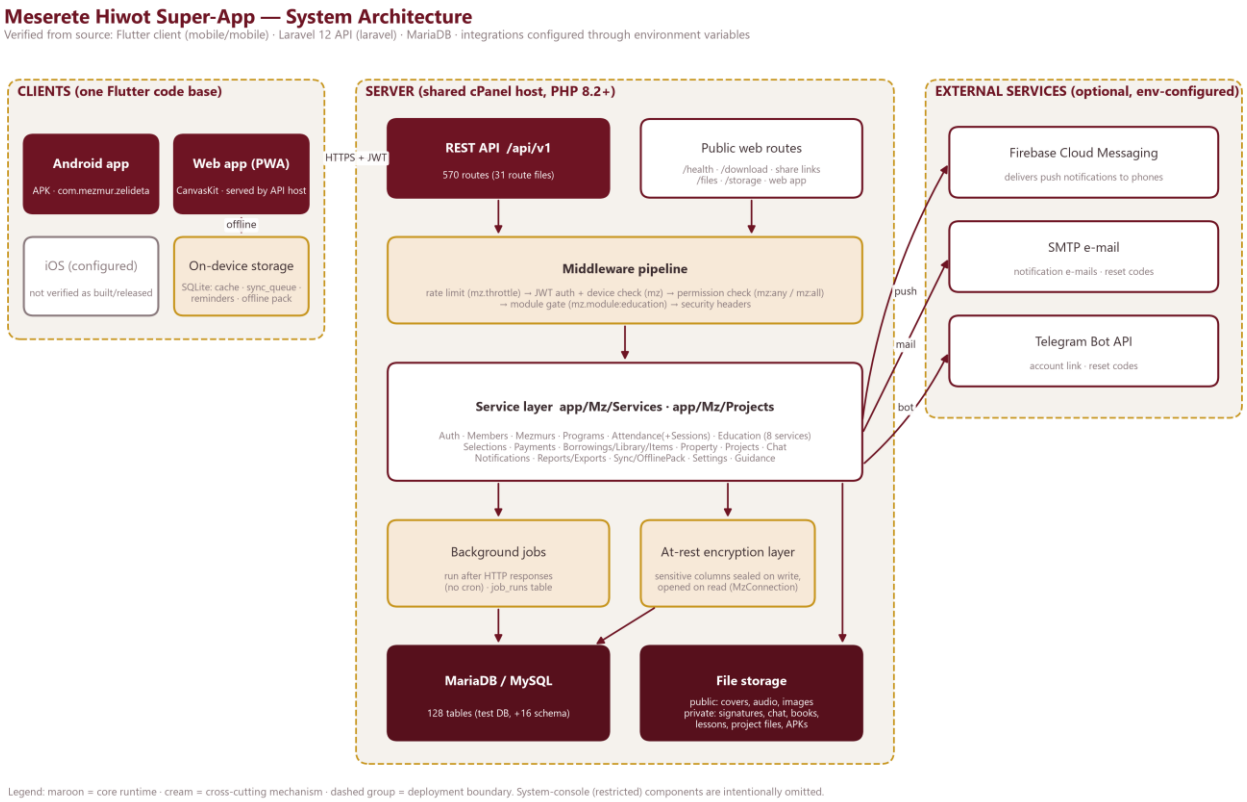


Figure 5.1 — System architecture (editable source: System_Architecture.svg).

Request path. The Flutter client calls `https://<host>/api/v1/...` over HTTPS with a bearer JWT. Every API route passes the global rate limiter, then (for protected routes) the `mz` middleware, which verifies the token, the account status, the token version, the trusted device (members) and revocation; permission middleware (`mz:any,...` / `mz:all,...`) and, for Education, the module gate follow. Route closures in `routes/api/*.php` validate input with the project's schema library (`App\Mz\Support\Z`) and call static service classes; services re-check authorisation for row-level rules, write audit records and return arrays that `Http::ok` wraps in the standard envelope.

Data path. Services use Laravel's query builder through a custom connection (`MzConnection`/`MzBuilder`) that seals configured sensitive columns on write and opens them on read. Files are stored under a configurable storage root, split into *public* (covers, audio, images) and *private* (signatures, chat, books, lesson materials, project files, APKs) areas; private files are streamed only by authorised routes.

Background work. There is no cron requirement: after an HTTP response is sent, at most once a minute one request takes a database named lock and runs due jobs (section 28).

Client side. The app keeps an on-device SQLite database (cache of GET responses, downloaded Mezmurs, favourites, reminders, the sync queue and the offline pack). Offline-capable operations are queued and replayed through `POST /api/v1/sync` (section 21).

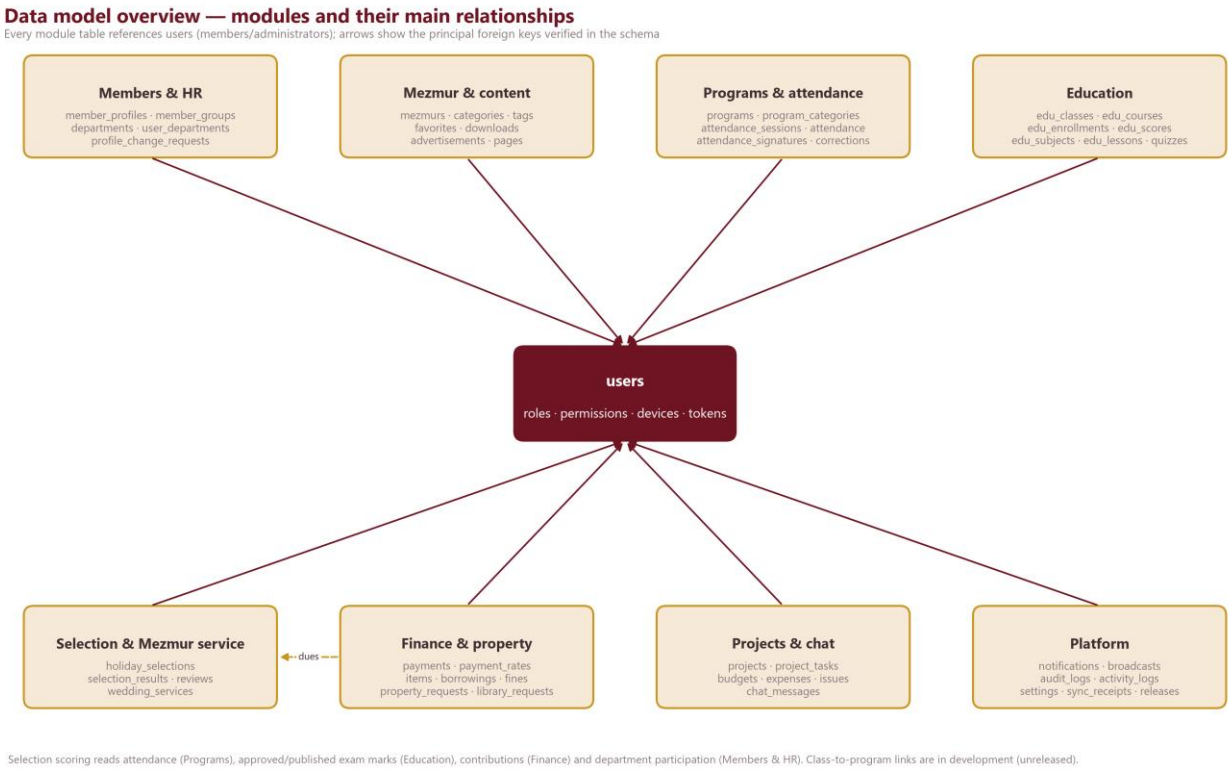


Figure 5.2 — Module-level data relationships.

6. Technology stack, frameworks, libraries and external dependencies

6.1 Backend (`laravel/composer.json`)

Component	Version / detail
Language	PHP ^8.2 (development machine: PHP 8.3.29)
Framework	Laravel framework ^12.0 (installed 12.69.3)
PDF generation	dompdf/dompdf ^3.1 (report exports, Amharic via Noto Sans Ethiopic)
Other runtime	laravel/tinker
Dev / test	phpunit ^11.5, mockery, fakerphp, laravel/pint, pail, sail, collision
In-house libraries	JWT (Support/Jwt.php), validation DSL (Support/Z.php, V.php), XLSX reader/writer (Support/Xlsx.php), Ethiopian calendar (Support/Ethiopian.php), at-rest encryption (Crypto/*), TOTP (Auth/Totp.php)
Database	MariaDB / MySQL (services use MySQL-specific SQL; tests run on MariaDB 10.4)

6.2 Mobile / web client (`mobile/mobile/pubspec.yaml`)

Area	Packages
SDK	Flutter 3.44.1 (stable), Dart SDK ^3.12.1; app version 1.5.3+19 in source
State & navigation	flutter_riverpod ^3.4.3, go_router ^18.0.1
Networking & storage	dio, flutter_secure_storage, sqflite (+ sqflite_common_ffi_web for web), shared_preferences, path_provider, connectivity_plus, device_info_plus, uuid, intl
Audio & media	just_audio, just_audio_background, audio_session, video_player, youtube_player_iframe, pdfrx (PDF reader)
Notifications	firebase_core, firebase_messaging, flutter_local_notifications, timezone, flutter_timezone
Security & device	local_auth (biometrics), pointycastle (AES-GCM), mobile_scanner (QR), qr_flutter, signature (signature pad)
Files & sharing	image_picker, file_picker, photo_manager, share_plus, url_launcher, cached_network_image, flutter_cache_manager, package_info_plus
Fonts	Noto Sans Ethiopic (400–700), Noto Serif Ethiopic (500–700) bundled as assets
Dev / test	flutter_test, flutter_lints, mocktail, sqflite_common_ffi, flutter_launcher_icons

6.3 External services (all optional, configured by environment variables)

Service	Use	Configuration keys (names only)
Firebase Cloud Messaging (HTTP v1)	Push notifications	FIREBASE_PROJECT_ID, FIREBASE_CLIENT_EMAIL, FIREBASE_PRIVATE_KEY
SMTP server	Notification e-mails, PIN-reset codes	SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASSWORD, SMTP_FROM
Telegram Bot API	Account linking, reset codes, temporary passwords	TELEGRAM_BOT_TOKEN, TELEGRAM_BOT_USERNAME, TELEGRAM_WEBHOOK_SECRET

Service	Use	Configuration keys (names only)
Android App Links	Share links open the app	ANDROID_PACKAGE, ANDROID_CERT_SHA256

Note: `.env.production.example` still lists legacy names (FCM_SERVER_KEY, FIREBASE_CREDENTIALS, MAIL_) while `config/mz.php` reads the names above — requires verification when configuring a new server.*

7. Backend architecture and major modules

The backend is a port of an earlier Node.js backend (comments reference the original files) organised as follows:

Folder	Content
routes/api/*.php (31 files)	Route closures under /api/v1, one file per area; input validation and permission middleware
routes/web.php, routes/web/public.php	/health, the API root, public files, APK download page, share links, Android asset links
routes/console.php	Artisan commands: mz:jobs, mz:encrypt-data, mz:repair-text and one restricted account-bootstrap command (Appendix F)
app/Mz/Services (58 classes)	Business logic as static service classes
app/Mz/Projects (11 classes)	Project management: access, projects, tasks, budgets, feedback, files, reports, reminders, chat Task Bot
app/Mz/Auth	AuthContext, permission catalogue, audit, holders, passwords (Argon2id), tokens, TOTP
app/Mz/Http	Middleware (auth/permissions), ModuleGate, Throttle, SecurityHeaders, Upload, Ctx (request helpers)
app/Mz/Db	MzConnection / MzBuilder (encryption hooks), named locks, UTC handling
app/Mz/Crypto	Seal (envelope encryption), EncryptedColumns (map), DataEncryptor (bulk encrypt/rotate)
app/Mz/Notifications	Notifications, FCM, Mailer, Telegram, deep links
app/Mz/Jobs	Jobs runner, reminder jobs, scheduled notifications
app/Mz/Support	Validation (Z, V, *Schemas), Ethiopian calendar, dates, XLSX, JWT, crypto helpers, pagination, error type, text repair
database/migrations (26 files)	Baseline schema + additive, reversible migrations (2 are unreleased)
tests/Unit, tests/Integration	PHPUnit suites (section 30)

7.1 Service catalogue

Services	Responsibility
AuthService, Sessions, Account, Users, UserAdmin	Registration, sign-in, refresh, logout, PIN change/reset; sessions; profile & preferences; user administration
Admins, DepartmentAdmins, Permissions (excuses)	Administrators, roles and permission catalogue; department administrators; absence-excuse requests
Members, MemberIds, MemberProfile, MemberGroups, ProfileChanges, Professions	Member identity/QR card, ID numbering, membership form, member types, change requests, professions
Departments	Departments, membership, leaders, monthly participation
Mezmurs, Categories, Promotions	Hymns, category tree, banners and internal pages
Programs, Schedule, Reminders	Programs and recurrence, occurrences/overrides, personal reminders
Attendance, AttendanceSessions	Rosters, marking, signatures, corrections, statistics; sessions, takers, conflict queue
Edu, EduRegistration, EduTeachers, EduMarks, EduRequests, EduComplaints, EduJourney, EduDashboard, EduLearning, EduQuizzes, EduAcademicYears*	Education (*unreleased)
Selections, SelectionReports, Choir, Weddings	Holiday selection, Mezmur report, choir eligibility, wedding services
Payments	Contributions and rates
Items, Borrowings, Fines, Library, PropertyRequests	Inventory, loans, fines, library, department property
Chat	Department chat
Broadcasts, Targeting	Admin messaging and recipient selection

Services	Responsibility
Reports, ReportExport	Dashboard/report queries and PDF/XLSX/CSV exports
Sync, OfflinePack	Offline operation replay and offline data pack
Settings, Storage, Releases, GuidanceService, Import, Activity	Settings, file storage, app releases, guidance, bulk import, activity tracking

8. Frontend / mobile architecture and navigation

Structure (`mobile/mobile/lib`). `app/` (router, shell, top bar, navigation layout), `core/` (config, network client, storage, theme, localisation helpers, utilities), `features/<module>/` (screens), `repositories/` (API access per area), `services/` (audio, downloads, sync, push, reminders, offline pack, updates), `models/`, `widgets/` (brand, dialogs, Ethiopian date picker, QR, secure screen, pickers), `l10n/` (ARB files).

State and routing. Riverpod providers; GoRouter with one shell (`StatefulShellRoute.indexedStack`) and section-aware bottom navigation. Mezmur section: Home · Categories · My library · Programs · Settings (notifications in the top bar). Education section (approved navigation option A): Home · Learn · Classes · Progress · More, with Settings as a top-bar gear. Admin routes live under `/admin`; routes without the needed permission are hidden and the server rejects them regardless.

Network client. Dio wrapper with bearer auth, single-flight token refresh, envelope unwrapping, error-code normalisation (`ApiException`), retries of GETs on connection drops and a per-user last-good-response cache for offline display.

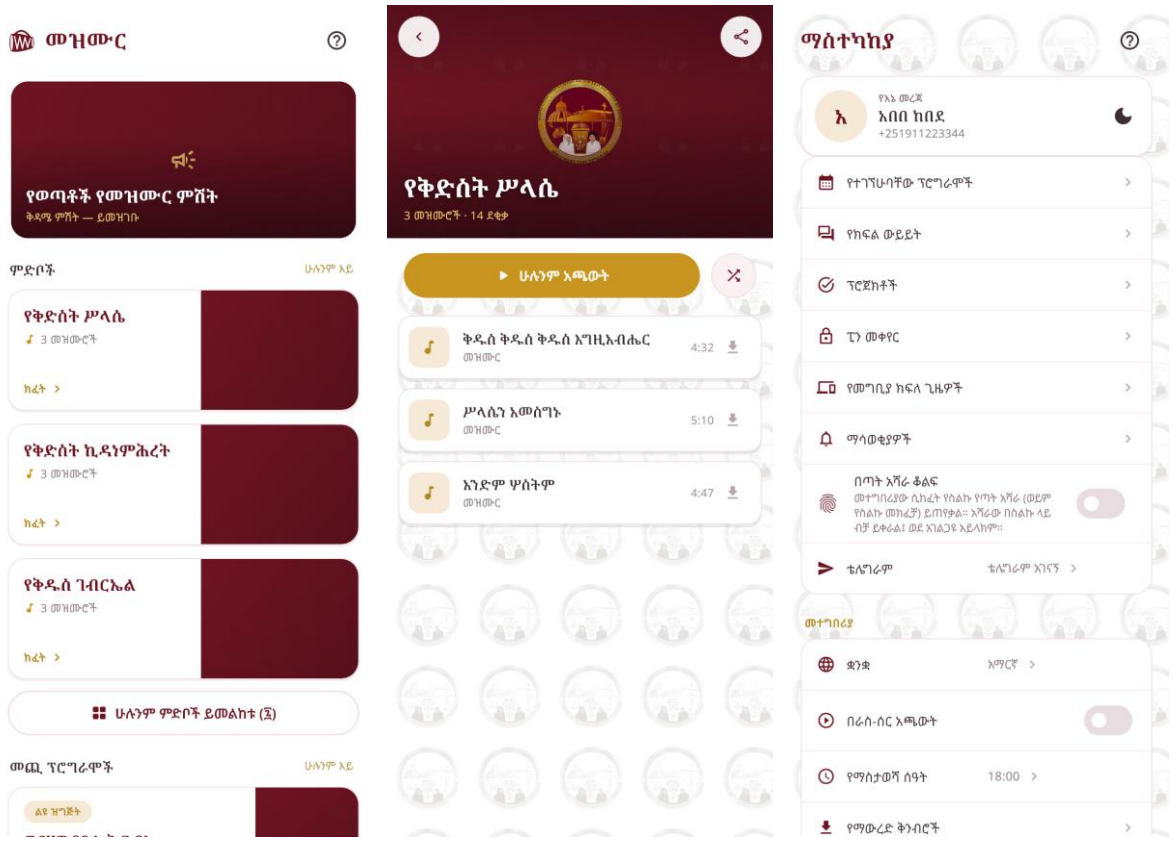
Branding. Maroon `#6E1423` and gold `#C8961E` on a warm canvas `#F5F2ED`, dark theme `#1A1113`; Noto Ethiopic fonts; module watermarks (kebero pattern for Mezmur, lectern/Ge'ez pattern for Education, church-seal pattern elsewhere).

8.1 Screen map

Area	Routes	Purpose
Shell / auth	/splash, /login, /register, /forgot, /change-password, /update	Start-up, sign-in with PIN, registration, PIN reset, forced PIN change, forced update
Hub	/hub	Super-app landing: Education, Mezmur, About me cards; Take-attendance card for assigned takers
Mezmur	/home, /home/search, /home/categories, /home/category/:id, /home/mezmurs, /home/mezmur/:id, /home/page/:id	Mezmur home, search, category tree, lists, lyrics + player, internal pages
Player	/player (mini player in shell)	Now playing: queue, shuffle, repeat, speed, sleep timer, synced lyrics
Library	/library, /edu/library, /library-request/:id	My collection (favorites, downloads), books, book requests
Programs	/programs, /programs/:id	Program list/calendar, detail, registration, reminders, excuse requests
Notifications	/notifications	In-app notification centre
Education	/edu, /edu/learn/*, /edu/classes, /edu/progress, /edu/more, /edu/search	Home · Learn · Classes · Progress · More
Education (study)	/edu/class/:id, /edu/course/:id, /edu/quiz/:id, /edu/attempt/:id, /edu/services, /edu/complaint/:id	Class & course pages, quizzes, self-service requests, complaints
Education (staff)	/edu/manage/, */edu/courses, /edu/registrations, /edu/register, /edu/review/*, /edu/content/*, /edu/quiz-edit/:id, /edu/attendance/:id, /edu/journey/:id, /edu/calendar	Class/course management, registration, marks review, content authoring, quiz editor, journey
Settings / About me	/settings, /settings/me, /settings/profile, /settings/payments, /settings/attendance, /settings/sessions, /settings/password, /settings/notifications, /settings/downloads, /settings/categories, /settings/borrowings, /settings/about, /settings/guidance	Member card + QR, profile, contributions, attendance calendar, sessions, PIN, notification & download preferences

Area	Routes	Purpose
Department chat	/settings/chat, /settings/chat/:id, /settings/chat/:id/private, /settings/chat/:id/user/:uid	Department group chat and private threads
Projects	/projects, /projects/my-tasks, /projects/approvals (+ project/task detail)	Projects, tasks, budgets, issues, suggestions, reports
Guidance	/guidance, /settings/guidance	Help centre, tours, contextual help
Admin panel	/admin and 50 sub-routes	10 permission-filtered groups (see section 12)

8.2 Representative screens (rendered from the app's own widgets with sample data)



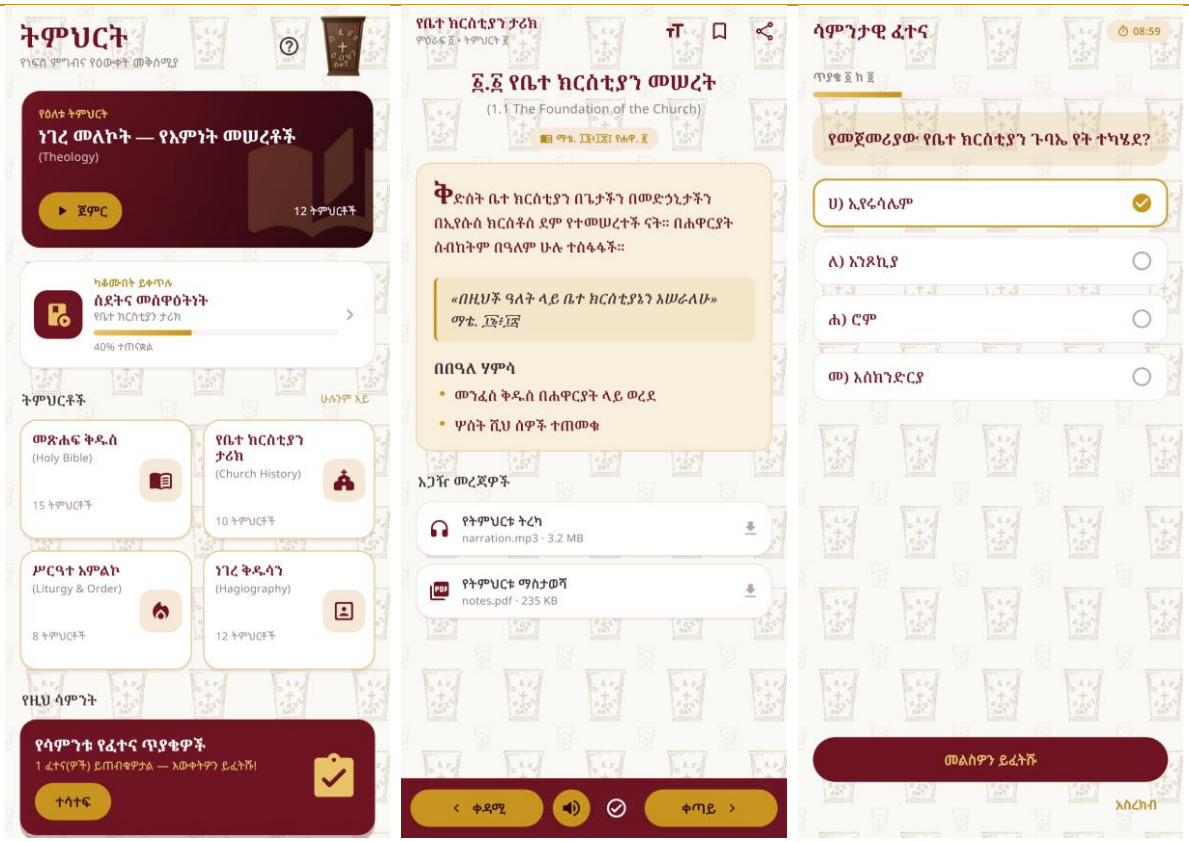


Figure 8.1 — Screens produced by `test/screenshots_test.dart` *and* `test/edu_screenshots_test.dart` *(sample data, not production records).*

9. Database documentation

The schema was read directly from the integration-test database `mh_app_test` (production dump `hersolot_mh_prod.sql` + every migration up to `2026_10_19_000001`, i.e. the 1.5.3+16 schema): **128 tables**. Engine InnoDB, charset utf8mb4 (two legacy tables were converted by `2026_10_17_000001`). The full data dictionary (every column, type, nullability, key and default) is *Appendix C*; foreign keys are listed there per table.

9.1 Tables by module

Identity, access & sessions

Table	Cols	References	Purpose
`users`	35	roles, users	Every account (members and administrators): identity, sign-in credentials (hashed), status, trusted device, member number and group.
`roles`	11	departments	Administrative roles with a level; department roles carry department_id. One hidden system role exists (restricted).
`permissions`	4	—	The permission catalogue (63 keys) mirrored from app/Mz/Auth/Catalog.php.
`role_permissions`	2	permissions, roles	Permissions granted to each role.
`user_permissions`	4	permissions, users	Direct permission grants to a single account.
`user_permission_revocations`	4	permissions, users	Per-account revocations that override role grants.
`refresh_tokens`	12	users	Rotating refresh tokens (stored as hashes) grouped by session family; device and client metadata.
`revoked_access_tokens`	3	—	Access-token ids (jti) revoked before expiry (sign-out of one session).
`password_resets`	7	users	Hashed 6-digit reset codes with expiry and attempt counter.
`device_history`	8	users	Device bind / login / rejected-login / reset events per account.
`user_preferences`	11	users	Theme, language, audio, reminder time, lyrics font scale and notification topics.
`user_preferred_categories`	2	categories, users	
`telegram_link_tokens`	6	users	
`telegram_state`	2	—	
`fcm_tokens`	8	users	Firebase push tokens per device.

Members, HR & departments

Table	Cols	References	Purpose

Table	Cols	References	Purpose
member_profiles	39	users	The membership form (birth, baptism, address, emergency contact, education, profession, confession father...). Many columns are encrypted at rest.
member_groups	10	users	Member types (Children, Youth & Adult, Married, distance...) used for targeting, attendance and payments.
id_sequences	2	—	Counters for permanent member numbers.
profile_change_requests	13	users	Member requests to change name, phone or profile fields, awaiting approval.
profile_change_history	9	profile_change_requests, users	Append-only history of applied profile changes (old → new).
professions	7	—	
departments	10	users	The 7 church departments (Amharic + English names, code prefix).
user_departments	7	departments, users	Department membership and role label of each member.
department_admins	7	departments, roles, users	Department administrators and their department role.
department_participation	9	departments, users	Monthly participation score per member per department (recorded by the department leader).
department_participation_history	9	departments, users	

Mezmur & content

Table	Cols	References	Purpose
mezmurs	25	categories, users	Hymns: title, lyrics, optional LRC timing, audio/cover, category, status, counters, content version.
categories	14	categories, users	Hierarchical Mezmur categories (parent_id, depth), with images and 'today' highlighting (study_on).
tags	2	—	
mezmur_tags	2	mezmurs, tags	
favorites	3	mezmurs, users	Member favourite Mezmurs.
downloads	5	mezmurs, users	Mezmurs downloaded for offline use (with or without audio).
advertisements	15	pages, programs, users	Home banners linked to pages or programs, scheduled by date.
ad_attachments	10	advertisements	
pages	17	programs, users	Internal content pages with call-to-action and optional program link.

Table	Cols	References	Purpose
page_images	5	pages	

Programs, attendance & excuses

Table	Cols	References	Purpose
programs	37	program_categories, users	Church programs/events: schedule, recurrence, audience, registration, attendance and reminder settings.
program_categories	11	users	Program types with colour, default reminder offsets and repeat interval.
program_participants	7	programs, users	Invited / registered participants of participant-only programs.
program_attendance_takers	4	programs, users	The one default attendance taker of each program.
program_occurrence_overrides	10	programs, users	Cancelled or rescheduled occurrences of recurring programs.
program_reminder_choices	4	programs, users	
reminders	13	borrowings, programs, users	Personal reminders (local-first, synced) for programs, borrowings or free text.
attendance	21	programs, users	One mark per member per program occurrence: status, capture time, elapsed seconds, method, signature link, sync metadata.
attendance_sessions	17	programs, users	One session per program occurrence: taker, status, start/end times and sources.
attendance_signatures	9	—	Signature images (private files, SHA-256) captured for Present marks.
attendance_sync_conflicts	20	—	HR review queue for offline attendance the server could not apply.
attendance_closures	5	programs, users	Early closure of an occurrence's attendance.
attendance_corrections	16	programs, users	Correction requests after closure, with review outcome.
permission_requests	16	programs, users	Members' requests to be excused from a program occurrence (approve, reject, cancel).

Education

Table	Cols	References	Purpose
edu_assessments	8	edu_courses	Assessments of a course (max score, weight, date).
edu_assignments	11	edu_courses, users	Assignments (individual or group) with attachment and due date.
edu_attendance	9	edu_classes, users	Legacy class attendance (to be frozen read-only when class attendance moves to programs).

Table	Cols	References	Purpose
edu_categories	12	—	Learning categories (Holy Bible, Church History, Liturgy & Order, Hagiography).
edu_class_staff	3	edu_classes, users	
edu_classes	17	users	Classes (term, academic year, semester, department, streams, registration window, capacity).
edu_complaint_messages	7	—	
edu_complaints	17	—	Academic complaints with reference EC-YYYY-NNNNNN, private attachment and status.
edu_course_enrollments	7	—	
edu_course_teachers	9	—	Teacher assignments with status history (only active teachers enter marks).
edu_courses	12	edu_classes, users	Subjects taught in a class (optionally linked to a curriculum subject).
edu_enrollments	8	edu_classes, users	Class enrolment with program (regular / distance) and stream.
edu_featured	13	—	Featured banner entries on the Education home.
edu_group_members	2	edu_groups, users	
edu_groups	5	edu_courses	
edu_journey	13	—	Append-only student journey events.
edu_lesson_materials	14	edu_lessons	Lesson files (private) or external links (video).
edu_lesson_progress	6	edu_lessons, users	Per-member lesson start/completion and bookmarks.
edu_lessons	17	edu_subjects, edu_units	Lessons: body, scripture references, objectives, teacher notes, publication status.
edu_quiz_answers	7	edu_quiz_attempts, edu_quiz_questions	
edu_quiz_attempts	13	edu_quizzes, users	Attempts with auto/manual score and transfer to the score sheet.
edu_quiz_questions	9	edu_quizzes	Quiz questions with type, options, answer, points.
edu_quizzes	17	edu_assessments, edu_courses, edu_lessons	Quizzes (practice or linked to an assessment), timing, attempts, review policy.
edu_requests	17	—	Student requests (class/stream/study-mode change, add/drop) applied on approval.
edu_schedule	11	edu_classes, edu_courses	
edu_score_history	11	—	Every mark change (old/new score and status, actor, reason).
edu_scores	15	edu_assessments, users	Marks with draft → submitted → approved/rejected → published workflow.
edu_subjects	16	edu_categories	Curriculum subjects with visibility (enrolled-only / all) and sequential option.
edu_submissions	13	edu_assignments, edu_groups, users	Student submissions with grading and feedback.
edu_units	8	edu_subjects	Units inside a subject.

Selection, choir & Mezmur services

Table	Cols	References	Purpose

Table	Cols	References	Purpose
holiday_selections	17	users	Holiday Mezmur service selections: period, member groups, JSON configuration, status, finalisation and publication.
selection_results	23	holiday_selections, users	Per-applicant computed result: components, total, qualified, overrides, service category and rank.
selection_reviews	10	—	Each required department's review decision.
selection_feedback	9	—	Members' feedback on a published result.
selection_config_versions	15	—	Created by the withdrawn department-criteria feature (unused); reused by in-development config versioning.
selection_scores	12	—	Withdrawn department-criteria feature (unused, kept for history).
selection_evaluators	6	—	Withdrawn department-criteria feature (unused, kept for history).
choir_eligibility	6	users	Daily-computed choir eligibility score per member (choir.policy).
choir_selections	10	users	
wedding_services	15	users	Wedding Mezmur services; legacy eligibility columns kept but ignored.
wedding_service_members	9	users, wedding_services	Members selected for a wedding service.

Contributions (finance)

Table	Cols	References	Purpose
payments	19	users	Monthly contributions by Ethiopian period; one payment may cover 1–12 months; status and verification.
payment_rates	7	users	Monthly rate per member type (student / worker) with effective date.

Property, borrowing & library

Table	Cols	References	Purpose
items	29	departments, users	Property and library items with stock, condition, domain (education/mezmur), department and borrowing rules.
item_categories	9	—	
borrowings	28	borrowing_batches, items, users	Individual loans: quantities, due date, return condition, fine and fine overrides.
borrowing_batches	5	users	

Table	Cols	References	Purpose
borrowing_signatures	7	borrowings, users	Signatures captured when an item is handed over.
return_conditions	6	—	
library_requests	13	borrowings, items, users	Members' requests for physical books, linked to the resulting borrowing.
property_requests	16	departments, items, users	Department property requests through review, issue and return.
property_request_events	9	property_requests, users	

Projects & tasks

Table	Cols	References	Purpose
project_budget_adjustments	11	projects, users	
project_budget_categories	6	—	
project_budget_lines	3	project_budget_categories, projects	
project_comments	7	projects, users	
project_expenses	20	project_budget_categories, projects, users	
project_files	10	project_issues, project_tasks, projects, users	
project_issues	15	departments, project_tasks, projects, users	
project_members	5	projects, users	
project_suggestions	13	departments, project_tasks, projects, users	
project_task_assignees	4	project_tasks, users	
project_tasks	25	departments, programs, project_issues, projects, users	Tasks with assignees, deadlines, status reasons, progress and versioning (offline conflicts).
projects	24	departments, users	Department projects: leader, dates, status, priority, progress, budgets.

Department chat

Table	Cols	References	Purpose
chat_messages	18	departments, project_tasks, users	Department group and private messages, attachments, meeting links and Task-Bot posts.
chat_reads	6	departments, users	

Notifications

Table	Cols	References	Purpose
notifications	19	notification_broadcasts, users	In-app notifications per user with push / e-mail delivery status and dedupe key.

Table	Cols	References	Purpose
notification_broadcasts	13	users	Admin messages to one, many, a filtered group or everyone.
scheduled_notifications	15	users	Time-based notifications stored ahead and released by the dispatcher job.

Guidance

Table	Cols	References	Purpose
guidance_articles	20	—	Help-centre articles (bilingual), contextual help per screen and tours.
user_guidance_progress	8	—	

Platform, audit & operations

Table	Cols	References	Purpose
settings	4	users	Validated JSON settings (fines, payments, attendance, borrowing, reminders, member ID format...).
audit_logs	10	users	Administrative audit trail (actor, action, entity, metadata, IP, user agent).
activity_logs	9	users	Member activity events (logins, plays, downloads...).
job_runs	5	—	Last run of each background job.
sync_receipts	5	users	Idempotency receipts of offline operations (op_id per user).
app_releases	14	users	Published APKs: version, size, SHA-256, notes, mandatory flag, minimum supported build.
migrations	3	—	

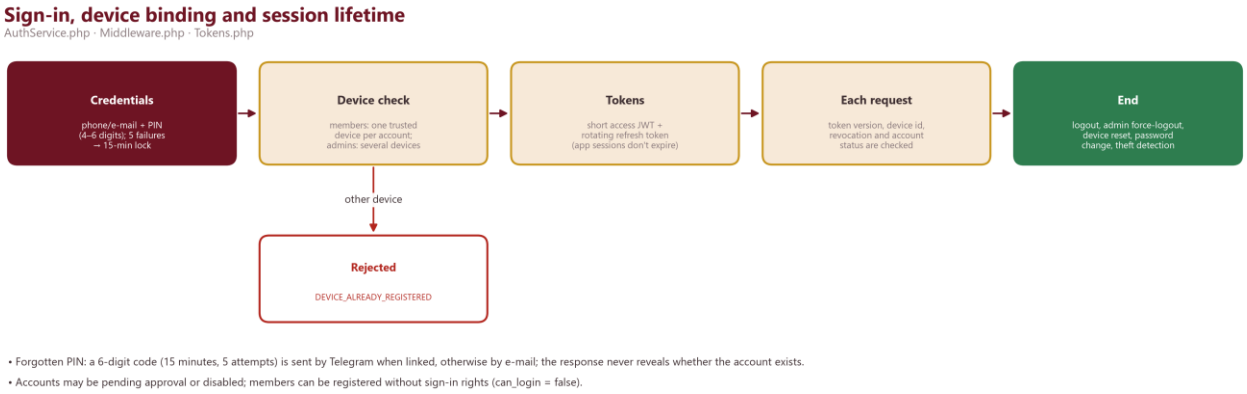
9.2 Important constraints and conventions

- **Uniqueness that enforces business rules:** one attendance mark per member per occurrence; one attendance session per program + occurrence date; `attendance.client_event_id` UNIQUE (idempotent offline replay); `sync_receipts (user_id, op_id)`; notification `dedupe_key` (reminders are idempotent); one default attendance taker per program.
- **Soft deletion** is used for Mezmurs, items and reminders (`deleted_at`); users can be soft-deleted by administrators.
- **History tables are append-only:** `edu_score_history`, `profile_change_history`, `department_participation_history`, `property_request_events`, `edu_journey`, `audit_logs`.
- **Encrypted columns** (section 25) cannot be used in WHERE / ORDER BY / JOIN; names, phone and e-mail stay searchable by design.
- **Dates:** stored in UTC (`Db/UtcDateTime`), displayed in Africa/Addis_Ababa; payable periods are Ethiopian `YYYY-MM` strings.
- **Unused tables kept on purpose:** `selection_config_versions`, `selection_scores`, `selection_evaluators` (withdrawn department criteria) — no data was deleted.

9.3 Migrations

Migration	Purpose
`2026_10_05_000000_baseline_schema`	Baseline (matches the imported production dump)
`2026_10_06_000001_notifications_auth_programs`	Notifications, auth and program additions
`2026_10_06_000002_project_management`	Projects, tasks, budgets, issues, suggestions, files
`2026_10_07_000001_library_and_registration`	Library requests, education registration
`2026_10_08_000001_encrypted_columns`	Column widths for sealed values
`2026_10_09_000001_profile_requests_admin_tools`	Profile change requests and admin tools
`2026_10_10_000001_member_groups_page_links`	Member groups, page links
`2026_10_11_000001_course_schedules_completion`	Course schedules and completion
`2026_10_12_000001_notification_center_property`	Notification centre, property requests
`2026_10_13_000001_create_professions_table`	Professions
`2026_10_13_000001_holiday_selection_review_and_categories`	Selection reviews and service categories
`2026_10_13_000002_add_mandatory_reminder_to_programs`	Mandatory reminders
`2026_10_14_000001_create_guidance_system_tables`	Guidance articles and progress
`2026_10_15_000001_programs_twice_monthly_recurrence`	Twice-monthly recurrence
`2026_10_16_000001_app_sessions_without_timeout`	App sessions without timeout
`2026_10_16_000002_permission_request_cancellation`	Cancelling approved excuses
`2026_10_16_000003_program_type_and_interval_reminders`	Program types, 15-day interval reminders
`2026_10_16_000004_education_course_management`	Teachers, marks workflow, requests, complaints, journey
`2026_10_16_000005_selection_department_criteria`	Department criteria tables (feature later withdrawn)
`2026_10_17_000001_items_borrowings_utf8mb4`	Charset fix for Amharic in items/borrowings
`2026_10_17_000002_sessions_never_expire`	Session expiry clean-up
`2026_10_17_000003_reseed_untouched_selection_criteria`	Now a no-op (kept for history)
`2026_10_18_000001_education_learning_platform`	Learning platform tables, education.content permission
`2026_10_19_000001_attendance_sessions`	Attendance sessions, signatures, conflicts, HR permissions
`2026_10_20_000001_academic_structure`	**Unreleased** — academic years, terms, levels, exclusive enrolment, course offerings
`2026_10_20_000002_program_edu_attendance`	**Unreleased** — program↔class links, class attendance cut-over, config column

10. Authentication, sessions, device restrictions and account recovery



- **Credentials.** Identifier = phone (normalised) or e-mail. All accounts now use a 4–6 digit PIN; a legacy password signs in once and forces a PIN change (`must_change_password`). Hashing: Argon2id (19 MiB memory, 2 iterations) with transparent re-hash; unknown identifiers still run a dummy verify to keep timing constant.
- **Lockout.** 5 consecutive failures lock the account for 15 minutes. Rate limits per IP: global 300 requests/60 s; auth 30/15 min; reset 10/hour.
- **Account states.** `pending` (ACCOUNT_PENDING), `disabled` (ACCOUNT_DISABLED); members may exist without sign-in rights (`can_login = false`).
- **Device binding (members).** The first successful sign-in binds the device id; a different device is refused (DEVICE_ALREADY_REGISTERED) and a device already bound to another member is refused (DEVICE_IN_USE). Every request re-checks that the token's device equals the bound device. Administrators can reset the device (`users.reset_device`), which also forces logout. Administrators may be signed in on several devices.
- **Tokens.** Short-lived access JWT (default 15 min, claims include token version, device id and session id) and an opaque rotating refresh token stored hashed per session family. App refresh tokens do not expire (decision of 2026-10-13, `JWT_REFRESH_TTL_DAYS=0`). A refresh replayed within 120 s or while its replacement is unused continues the session; reuse after the replacement was used is treated as theft.
- **Revocation.** Logout revokes the session family and the current access token id; password change, device reset and admin force-logout bump `token_version`, which invalidates every JWT immediately. Settings → Sessions lists sessions and ends one, the others, or all.
- **Recovery.** `POST /auth/forgot-password` always answers the same way. A 6-digit code (HMAC-stored, 15-minute validity, 5 attempts) is sent through Telegram if linked, otherwise by e-mail; `POST /auth/reset-password` sets a new PIN (administrators need a stronger password, checked in the service).
- **App-side protection.** Tokens in secure storage; optional biometric lock; FLAG_SECURE on member data, profile and QR screens.

11. Roles, departments, permissions and administrative hierarchy

Authorisation is permission-based. A user's effective permissions = permissions of their role + their department-admin roles + direct grants − per-user revocations (`user_permission_revocations` always win). Routes declare required permissions; services re-check them and apply row-level rules (e.g. a teacher may only grade their own active courses; a department admin sees only their departments' projects).

11.1 Roles (from the database)

Role key	Name (as stored)	English description	Level	Department-bound	# permissions
`high_admin`	ከፍተኛ አስተዳዳሪ	High administrator	50	no	59
`mezmur_lead`	የመዝሙር ክፍል መሪ	Mezmur department lead	20	no	19
`moderator`	ሞደሬተር (ትምህርት)	Moderator (education)	20	no	9
`hr_manager`	የሰው ሀብት አስተዳዳሪ	Human-resources manager	20	no	11
`payments_manager`	የክፍያ አስተዳዳሪ	Payments manager	20	no	4
`dept_education`	Education and Training Admin	Education & Training department admin	15	yes	19
`dept_hymns`	Hymns and Arts Admin	Hymns & Arts department admin	15	yes	17
`dept_hr`	Human Resources Admin	Human Resources department admin	15	yes	20
`dept_programs`	Program and Conference Admin	Program & Conference department admin	15	yes	11
`dept_finance`	Finance and Property Admin	Finance & Property department admin	15	yes	14
`dept_planning`	Planning and Development Admin	Planning & Development department admin	15	yes	9
`dept_info`	Information and Communications Admin	Information & Communications department admin	15	yes	11
`edu_registrar`	የትምህርት ተመዝጋቢ አስተዳዳሪ	Education registrar	15	no	2
`admin`	አስተዳዳሪ	Administrator	10	no	56

A hidden top-level authority role exists for platform recovery; it is excluded from this document (see Appendix F).

11.2 Departments

Code	Amharic	English
EDU	ትምህርት እና ሥልጠና ክፍል	Education and Training Department
HYM	መዝሙር እና ሥነ-ጥበብ ክፍል	Hymns and Arts Department
HR	የሰው ኃብት አስተዳደር ክፍል	Human Resources Management Department
PRG	መ/ግ እና ጉባኤ ዝግጅት ክፍል	Program and Conference Preparation Department
FIN	ሒሳብ እና ንብረት ክፍል	Finance and Property Department
PLN	ዕቅድ እና ልማት ክፍል	Planning and Development Department
INF	መረጃ እና የውስጥ ግንኙነት ክፍል	Information and Internal Communications Department

11.3 Member groups

Key	Amharic	English
children	ሕጻናት	Children
youth_adult	ወጣቶች እና አዋቂዎች	Youth & Adult
married	ባለትዳሮች	Married
g_g	የርቀት	—

11.4 Permission catalogue (63 keys) and the roles that hold them

Permission	Group	Meaning	Roles holding it
admins.manage	admins	Create and configure lower-level administrators	high_admin, admin
admins.view	admins	View administrator accounts	high_admin, admin
choir.manage	choir	Manage wedding ceremony choir services (ሙዝሙር አገልግሎት)	high_admin, mezmur_lead, dept_hymns, admin
selection.configure	choir	Create and configure holiday selections (period, weights, rules)	high_admin, admin
selection.exams	choir	Set whether exam results are required for a holiday selection	high_admin, dept_education, dept_hymns, admin
selection.finalize	choir	Finalize holiday selections	high_admin, admin
selection.review	choir	Review holiday selection results and record exceptions	high_admin, mezmur_lead, dept_education, admin
selection.view	choir	View holiday selections and their results	high_admin, mezmur_lead, dept_education, dept_hymns, dept_finance, admin
ads.manage	content	Manage advertisement banners	high_admin, mezmur_lead, dept_education, admin
categories.manage	content	Manage categories	high_admin, mezmur_lead, dept_hymns, admin
pages.manage	content	Manage internal pages	high_admin, mezmur_lead, dept_education, admin
chat.supervise	departments	Read and moderate every department chat	high_admin, dept_hr, admin
education.content	education	Manage subject categories, curriculum (subjects, units, lessons, materials) and the featured banner	high_admin, moderator, dept_education, admin
education.manage	education	Manage classes, courses, teachers and attendance takers	high_admin, moderator, dept_education, admin
education.register	education	Register members (new or existing) into classes and their courses	high_admin, moderator, dept_education, edu_registrar
hr.manage	hr	Assign members to departments and give them role labels	high_admin, hr_manager, dept_hr, admin
borrowing.approve	library	Approve direct borrowings that need approval	high_admin, dept_education, dept_hymns, dept_hr, dept_programs, dept_finance, dept_planning, dept_info, admin
borrowing.configure	library	Configure which items and members can borrow directly	high_admin, dept_finance, admin
borrowing.manage	library	Issue and return borrowed items	high_admin, mezmur_lead, moderator, dept_finance, admin
fines.manage	library	Configure fine policy and override fines	high_admin, dept_finance, admin
items.manage	library	Manage library items	high_admin, mezmur_lead, moderator, dept_finance, admin
mezmurs.create	mezmurs	Create Mezmurs	high_admin, mezmur_lead, dept_hymns, admin
mezmurs.delete	mezmurs	Delete Mezmurs	high_admin, mezmur_lead, admin

Permission	Group	Meaning	Roles holding it
mezmurs.update	mezmurs	Edit Mezmurs, audio and covers	high_admin, mezmur_lead, dept_hymns, admin
mezmurs.view	mezmurs	View Mezmurs in the admin panel (including drafts)	high_admin, mezmur_lead, dept_hymns, admin
notifications.send	notifications	Send notifications to individual / selected / filtered users	high_admin, mezmur_lead, moderator, hr_manager, payments_manager, dept_education, dept_hymns, dept_programs, dept_info, admin
notifications.send_all	notifications	Send notifications to all users	high_admin, dept_education, dept_hymns, dept_programs, dept_info, admin
payments.configure	payments	Set student / worker monthly payment rates	high_admin, admin
payments.manage	payments	Record monthly payments	high_admin, payments_manager, dept_planning, admin
payments.view	payments	View monthly contributions and reports (read-only)	high_admin, payments_manager, dept_finance, dept_planning, admin
attendance.assign	programs	Assign the one attendance taker of each program occurrence (HR)	hr_manager, dept_hr
attendance.close	programs	Close an attendance session before the program ends	high_admin, hr_manager, dept_hr, admin
attendance.configure	programs	Configure attendance (signature rules, sessions) — HR	high_admin, hr_manager, dept_hr, admin
attendance.correction.approve	programs	Review attendance correction requests	high_admin, hr_manager, dept_hr, admin
attendance.correction.request	programs	Submit attendance comments and correction requests	dept_education, dept_hymns, dept_programs, dept_info, admin
attendance.manage	programs	Take attendance: scan, mark and record permissions while the session is open	high_admin, mezmur_lead, hr_manager, dept_hr, admin
attendance.override	programs	Change attendance after the session has closed	hr_manager, dept_hr, admin
attendance.reports	programs	View attendance statistics and calendars of members	high_admin, mezmur_lead, moderator, hr_manager, dept_education, dept_hr, dept_info, admin
attendance.view	programs	View attendance rosters, calendars and summaries	high_admin, mezmur_lead, hr_manager, dept_education, dept_hymns, dept_hr, dept_programs, dept_info, admin
programs.manage	programs	Manage programs and participants	high_admin, mezmur_lead, dept_programs, admin
programs.reminders_enforce	programs	Make a program's reminders mandatory for its members	— (direct grant only)
program_categories.manage	programs	Create and edit program categories	high_admin, dept_programs, admin
projects.all_departments	projects	Access the projects of every department	high_admin
projects.expenses.approve	projects	Approve project budgets, expenses and budget adjustments (Finance)	high_admin, dept_finance
projects.manage	projects	Create and manage projects, tasks and teams in the departments one administers	high_admin, dept_education, dept_hymns, dept_hr, dept_programs, dept_finance, dept_planning, dept_info
projects.reports	projects	View project and task reports	high_admin, dept_education, dept_hymns, dept_hr, dept_programs, dept_finance, dept_planning, dept_info
projects.view	projects	View department projects (members always see their own department)	high_admin, dept_education, dept_hymns, dept_hr, dept_programs, dept_finance, dept_planning, dept_info, admin

Permission	Group	Meaning	Roles holding it
property.request	property	Request property for a department	high_admin, dept_education, dept_hymns, dept_hr, dept_programs, dept_finance, dept_planning, dept_info, admin
property.review	property	Review, approve, reject and issue department property requests	high_admin, dept_finance, dept_planning, admin
activity_logs.view	reports	View user activity logs	high_admin, admin
audit_logs.view	reports	View the admin audit log	high_admin, admin
reports.view	reports	View dashboard and reports	high_admin, mezmur_lead, moderator, admin
departments.manage	system	Create departments and choose their leads	high_admin, admin
guidance.manage	system	Manage in-app guidance articles, FAQs, and interactive tours	high_admin, admin
releases.manage	system	Publish app versions (download page / force update)	high_admin, admin
settings.manage	system	Manage system settings	high_admin, admin
member_groups.manage	users	Create and edit member types (groups) used by programs and attendance	high_admin, dept_hr, admin
member_id.configure	users	Configure member ID prefixes and display style	high_admin, dept_hr, admin
users.create	users	Create user accounts	high_admin, dept_hr, admin
users.disable	users	Disable / enable user accounts	high_admin, admin
users.reset_device	users	Reset a user's registered device and force logout	high_admin, admin
users.update	users	Edit user accounts	high_admin, dept_hr, admin
users.view	users	View user accounts	high_admin, mezmur_lead, moderator, hr_manager, payments_manager, dept_education, dept_hymns, dept_hr, dept_finance, dept_planning, dept_info, edu_registrar, admin

Notes: `guidance.manage` *exists in the database but not in* `Catalog.php` *(added by the guidance migration). Some export rules reference* `education.view` *and* `programs.view`*, which are not in the catalogue — those alternatives can never match (requires verification / clean-up).*

12. Detailed module-by-module functionality

12.1 Admin panel

The admin panel (`/admin`) shows only modules the user may use, in ten groups:

Group (Amharic label)	Modules
User management (የተጠቃሚዎች አስተዳደር)	Users · Member IDs · Member types · Profile change requests
Member & activity management (የአባላትና እንቅስቃሴ አስተዳደር)	Monthly contributions · Attendance · Fines · Send notification · Notification delivery
Department management (የክፍሎች አስተዳደር)	Departments · Department chat
Properties (ንብረቶች)	Items · Borrowing · Property requests
Education management (የትምህርት አስተዳደር)	Courses · Library (requests, books) · Classes · Learning content · Student registration
Mezmur management (የመዝሙር አስተዳደር)	Service management (holiday selection, other Mezmur services) · Categories · Mezmurs · Bulk import
Promotion management (የማስተዋወቅ አስተዳደር)	Program categories · Programs · Banners · Internal pages
Project & task management	Projects · My tasks · Approvals
Administration management	Administrators · Roles · System settings · App versions · In-app guidance
Reports & audit (ሪፖርትና ኦዲት)	Reports · Audit log · User activity

The admin home also shows a dashboard, 30-day sign-ups, recent activity and pending excuse requests. Several screens offer an *edit mode* pen on member-facing pages (Mezmur home, Programs).

12.2 Members and HR

- Registration collects name, phone, optional e-mail, PIN, member group and type (student / worker) and an optional membership profile; administrators can create members, including members without sign-in rights.
- Member numbers are permanent (never reused) and formatted with a group prefix (setting `member_id.format`, e.g. ሙ/ሕ/ወ/17). The member card shows the QR code (UUID payload) used by attendance and borrowing staff; it is also cached offline.
- Profile changes requested by members wait for approval; every applied change is recorded with old and new value.
- HR assigns members to departments with role labels; department leaders record a monthly participation score (max set by `participation.policy`), with history.

12.3 Programs (see section 14 for attendance)

- Fields: title, description, image, location, start/end (no end = end of day), category (program type), audience (all / participants), member groups, registration and capacity, status (draft / published / cancelled / completed), event type, 'additional' and 'counts for selection' flags.
- Recurrence: once, weekly, monthly (day of month or n-th weekday), twice monthly (N and N±14, or 1st & 3rd / 2nd & 4th weekday) and Ethiopian calendar days (1–30); occurrences are computed, never stored; individual occurrences can be cancelled or rescheduled.
- Reminders: per-program offsets (2 days, 1 day, 6 hours, 1 hour), members may choose their own offsets unless reminders are enforced; optional repeat every N days (15 by policy, 08:00–21:00 local, no backlog replay); device alarms with snooze work offline.
- Participants can be invited or registered in bulk using the targeting filter; finished one-off programs are marked completed by the nightly clean-up.

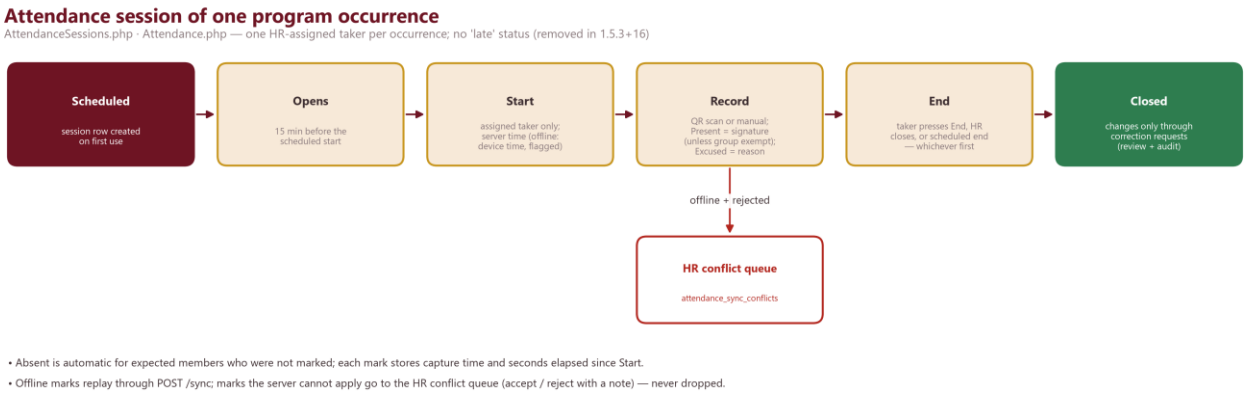
12.4 Banners, pages and guidance

Banners (advertisements) are scheduled cards on the Mezmur home that link to an internal page, a program or attachments; internal pages carry a cover, gallery, event date, location and a call to action. The guidance centre contains 56 bilingual articles (37 for administrators), interactive tours (welcome, Mezmur, programs, attendance, education, projects, library, choir, admin) and contextual help buttons on major screens; progress is stored per user.

13. Mezmur, lyrics, audio, file uploads and media management

- **Catalogue.** Hierarchical categories (parent/child with depth, images, ordering, active flag, 'today' highlighting); Mezmurs with title, lyrics, optional LRC timing, description, tags, cover, audio, status (draft/published), featured flag, view/favourite/download counters and a content version used to refresh offline copies. Search uses a full-text query with title normalisation; duplicate titles are detected.
- **Member experience.** Home (banners, categories, recent, popular, upcoming programs), category pages with play-all/shuffle, lyric view with adjustable font scale, time-synced lyric highlighting when LRC exists, YouTube links, share links, favourites and downloads (lyrics always; audio and cover optional or per preference).
- **Player.** just_audio with background playback and lock-screen/notification controls (church emblem artwork when no cover), queue, shuffle, repeat off/all/one, speed, sleep timer (at a time or after the current track), mini player and full now-playing screen; playback is tracked in activity logs.
- **Administration.** Create/edit/publish Mezmurs with audio and cover upload, LRC maker, bulk category assignment, drafts, and bulk import from Excel/CSV/pasted rows (dry run first).
- **Upload protection.** A first check on declared type and size (Http/Upload.php), then the real type is detected from the file bytes before saving (Services/Storage.php). Limits (env-configurable): images 5 MB, audio 40 MB, documents 20 MB, books 60 MB, APK 250 MB, signatures 512 KB PNG. Public files are served with `X-Content-Type-Options: nosniff` and a sandboxing CSP; private files only through authorised routes with `Cache-Control: private, no-store`.

14. Programs, attendance, punctuality, scoring and attendance reports



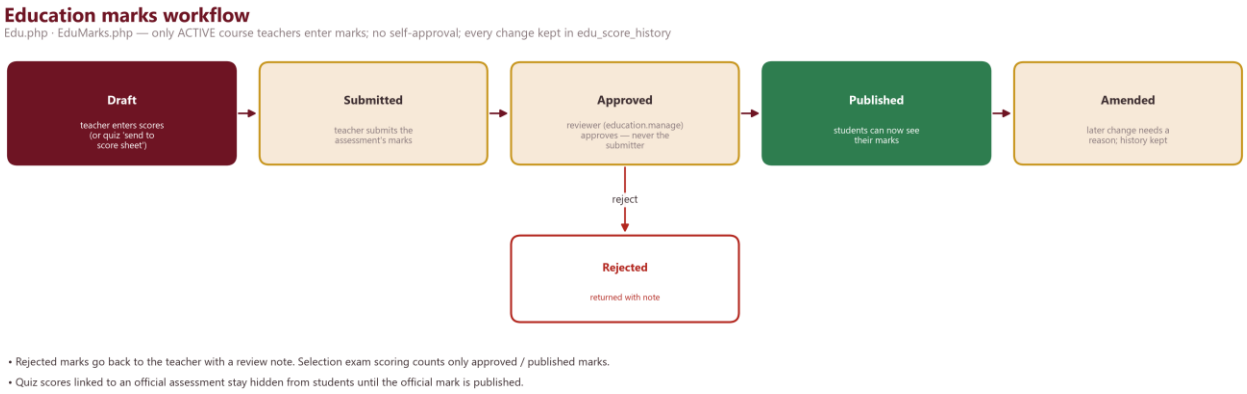
14.1 Released behaviour (1.5.3+16)

- **Control by HR.** All attendance-control permissions are held by HR roles (`dept_hr`, `hr_manager`); `attendance.assign` sets each program's one default taker and, per occurrence, a substitute (a reason is required while the session runs).
- **Session.** Created lazily per occurrence; available 15 minutes before the scheduled start; attendance time starts only when the assigned taker presses Start (server time; an offline start keeps the device time adjusted for clock skew and is flagged). The session ends when the taker presses End, HR closes it (`attendance.close`) or at the scheduled end — whichever is first, no grace period.
- **Marking.** Two steps: identify (QR UUID or manual pick, checks eligibility and existing marks without writing), then record. Only *present* and *excused* are recorded; *absent* is implied for expected members who were not marked. Present requires the member's signature (PNG) unless their member group is exempt (setting `attendance.signature`); excused requires a reason; removing a mark requires a reason. A second scan never creates a second record.
- **Punctuality.** The 'late' status was removed from attendance workflows, calculations, reports and selection evaluation (legacy 'late' rows read as present). Instead every mark stores the capture time and `elapsed_seconds` since Start, plus time source (server / device / correction) and method (QR / manual).
- **Excuses.** Members request to be excused from an occurrence before a deadline (setting `attendance.permissions`: deadline hours, monthly/quarterly limits, weight used in scoring). Approval writes an excused mark; an approved excuse can be cancelled by an administrator with a reason, removing only the mark it created.
- **Corrections.** After closure, changes go through correction requests (`attendance.correction.request`) reviewed by `attendance.correction.approve`, optionally with an arrival time; `attendance.override` allows authorised changes after closure.
- **Reports.** Member calendar and statistics, program summaries, per-occurrence roster export (PDF/XLSX/CSV) whose title and file name include the program, Ethiopian date and taker, with captured time, minutes after start, method, signature and sync columns.

14.2 Unreleased (source tree, target builds +18/+19)

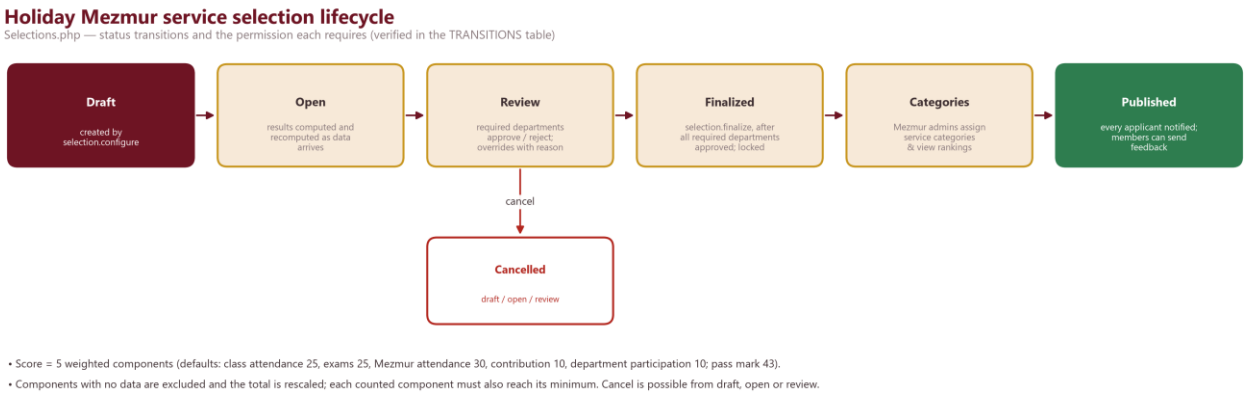
- Class attendance moves to programs linked to classes (`program_edu_classes`, `attendance_edu_classes`, cut-over setting); legacy `edu_attendance` becomes read-only.
- Selection attendance scoring by elapsed-time intervals per attendance category with an explicit 'after the last interval' percentage, and configuration versions.

15. Education, classes and course attendance



- **Structure.** Classes (name, term, academic year, semester, department, streams, registration window, regular / distance allowance, capacity, status) contain courses (subjects) with schedules (weekday, frequency, week of month, mode, location), staff and groups.
- **Registration.** Registrars (`education.register`) register new or existing members into classes and their courses, transfer and promote; members can enrol during registration windows.
- **Teachers.** `edu_course_teachers` keeps assignment history (assign / suspend / remove); only active assignments can enter marks; re-selecting a suspended teacher reactivates the assignment.
- **Marks.** Assessments (max score, weight, date) → scores in draft → submitted → approved / rejected → published; no self-approval; amendments need a reason; every change is stored in `edu_score_history`; students only see published marks. Deleting a class/course/assessment with academic records is refused (409 HAS_RECORDS — archive instead).
- **Assignments.** Individual or group assignments with attachments, submissions (text/file) and grading with feedback (publish time, deadline enforcement and late rules are not implemented).
- **Self-service.** Requests (class / stream / study-mode change, add or drop a class or subject) applied transactionally on approval; complaints with reference `EC-YYYY-NNNNNN`, messages, internal notes and private attachments; an append-only student journey.
- **Learning platform (1.5.3+15).** Categories (Holy Bible, Church History, Liturgy & Order, Hagiography) → curriculum subjects (visibility enrolled-only by default, optional sequential order) → units → lessons (body, scripture references, objectives, teacher notes, estimated minutes) with private materials (PDF, images, audio, Office files) and external video links; progress, bookmarks, continue-learning, featured banner and search. Active teachers write and publish lessons for subjects they teach; `education.content` holders edit all.
- **Quizzes.** Practice or assessment-linked quizzes with open/close times, time limit, attempt limit, scoring and shuffle options, 4 question types, autosave, manual grading; 'send to score sheet' writes draft marks into the normal workflow. Assessment-linked quiz scores are hidden until the official mark is published.
- **Class attendance (released).** Teachers / attendance takers mark class attendance (present / absent / excused, manual or QR scan; no late status); summaries feed the selection's class-attendance component. Moving this into programs is unreleased (section 14.2).
- **Module switch.** The Education module can be disabled per member/group (setting `modules.policy`); all `/edu` routes return 403 MODULE_DISABLED for them, except holders of education staff permissions.

16. Holiday selection: eligibility, review, approval and finalisation



The holiday Mezmur service selection is configured in one place, on the selection itself (the department-based criteria built on 2026-10-13/14 were withdrawn on 2026-10-15 at the church's request).

Component	Default weight	Default minimum	Measured from
Class attendance	25	10	Attendance in the member's education classes (chosen classes/courses)
Exams	25	10	Approved / published marks of the subjects the member actually takes
Mezmur attendance	30	15	Attendance at Mezmur programs (chosen categories / included programs)
Monthly contribution	10	3	Months paid ÷ months due (paid after due month counts as paid; exempt = 100 %)
Department participation	10	5	Average monthly participation score from the member's departments

- Weights must total 100; default passing score 43; default member groups: youth & adult, married.
- attendance % = Σ(program weight × status points) ÷ Σ program weights; component points = % × weight ÷ 100; total = Σ points ÷ Σ counted weights × 100. Components with no data are excluded and the total is rescaled. *Qualified* = total ≥ passing score and every counted component ≥ its minimum. Members are not ranked for qualification.
- Service bonus up to 30 points (percentage or fixed) from chosen service programs; the existing rule gives it only at 100 % Mezmur attendance and caps the component at its weight, so it cannot add points in practice — flagged as needing a business decision.
- Lifecycle and permissions: draft → open (`selection.configure`) → review (`selection.review` or configure) → finalized (`selection.finalize`, only after every required department approved); cancel from draft/open/review. Configuration locks once in review. Results are recomputed while open and at status changes. Overrides (qualified / not qualified) need a reason and are audited; finalized results are locked.
- Required review departments default to Hymns & Arts, Education & Training, and Human Resources Management.
- After finalisation: Mezmur administrators assign service categories (ለወረብ አገልግሎት የተመረጡ / ለመዝሙር አገልግሎት የተመረጡ) and view rankings; publishing notifies every applicant; members may send feedback.
- Reports: selection report (all applicants and components) and Mezmur report (`choir.manage`) in PDF/XLSX/CSV; marked provisional until finalized.

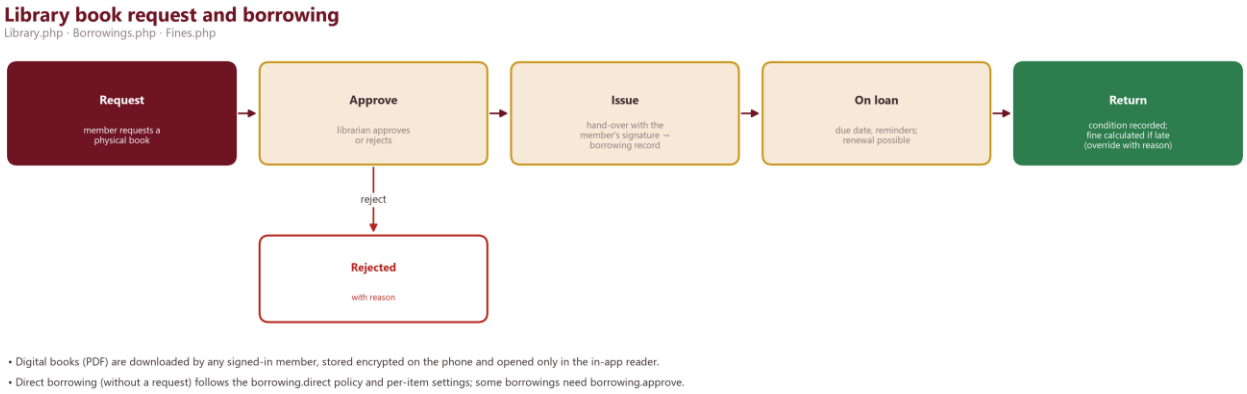
- Separate mechanisms: the *choir eligibility* score (setting `choir.policy`, recomputed nightly) and *wedding services*, whose participants are selected manually by `choir.manage` holders.

17. Contributions, finance, property and borrowing workflows

17.1 Monthly contributions

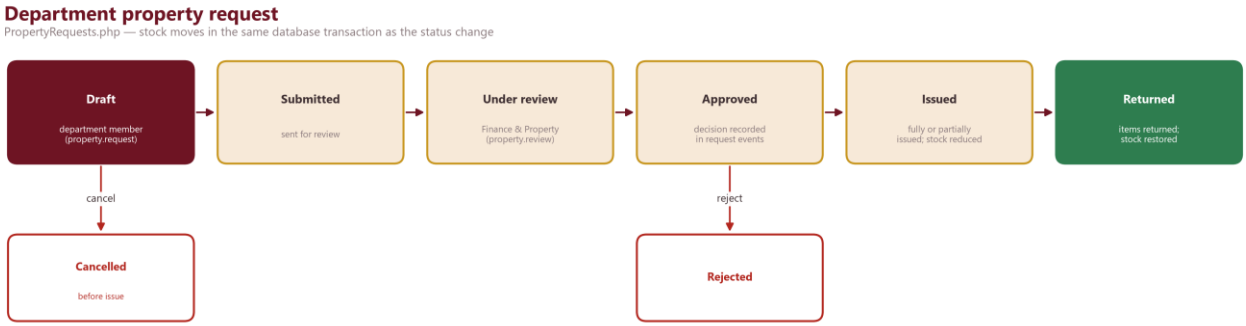
- Recorded by Ethiopian month (`YYYY-MM`, Pagume belongs to Nehase). One payment covers 1–12 consecutive months; amount = member's rate (student / worker, effective-dated in `payment_rates`) × months; children do not pay.
- A month counts as paid when covered; payments after the month + `grace_days` are labelled *Paid after due month* (financial rule only). Payments have method, reference, receipt upload, note and verification by a second administrator.
- Members see their own payment history and percentage; `payments.view` gives read-only reports; `payments.configure` sets rates; export kind `contributions`.

17.2 Items, borrowing, fines and library



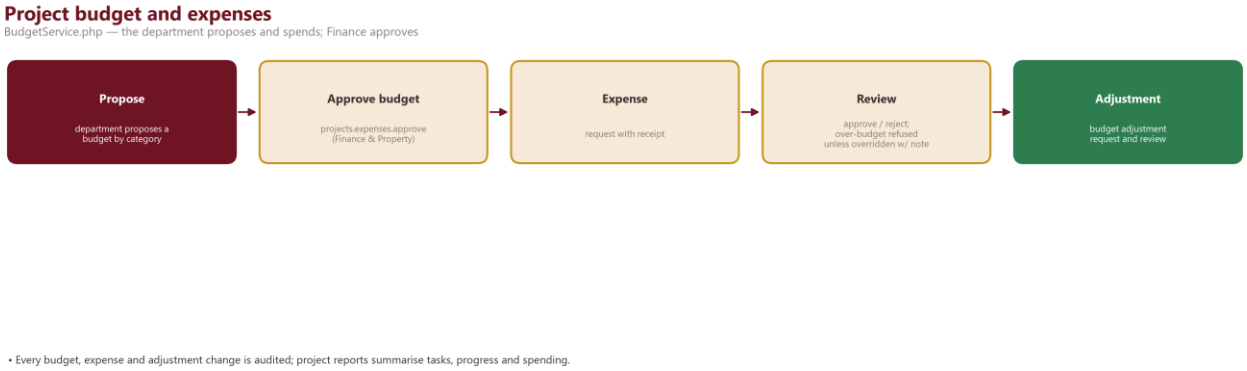
- Items carry name, author, code (auto-generated per category), category (books, department resources, electronics, equipment, furniture, instruments, office, other), quantity/available, condition, location, domain (education / mezmur), department and borrowing rules (direct borrowable, max quantity/days).
- Borrowing: individual or bulk (batch) issues, direct borrowing under the `borrowing.direct` policy (member groups, approval required, limits), approval (`borrowing.approve`), renewal, return with configurable return conditions (may mark lost / restock), signatures at hand-over.
- Fines (`fines.policy`): enabled flag, amount per day or week, grace days, maximum, currency (ETB); recalculated nightly; overrides require a reason; due and overdue reminders are sent.
- Library: digital PDF books downloaded by members are encrypted with AES-256-GCM using a keystore key and opened only in the in-app reader; physical books are requested, approved and issued with a signature, creating a normal borrowing.

17.3 Department property requests



18. Departments, chat, projects, tasks, budgets and expenses

- **Projects** belong to one department; leader, team (roles), dates, status, priority, progress (manual or computed from task weights), notes, 'allow late tasks', currency and budgets. Visibility: members see their own department; `projects.all_departments` sees all.
- **Tasks** have several assignees, deadlines within the project deadline, priority, status with reasons (pending reason, status reason), progress, weight, completion note and a version number used to detect offline conflicts; they can originate from issues, suggestions or programs. Deadline reminders are scheduled as notifications.
- **Feedback.** Issues (priority, assignee, due date, resolution), suggestions (review, convert to task), comments, files (private), activity feed and reports (summary, tasks).
- **Chat.** One private group per department plus private leader↔member threads; members send text; leaders and the supervisor (`chat.supervise`) also send media and meeting links; read markers; attachments streamed only to allowed readers; the Task Bot posts task events with action buttons that use the same task rules.



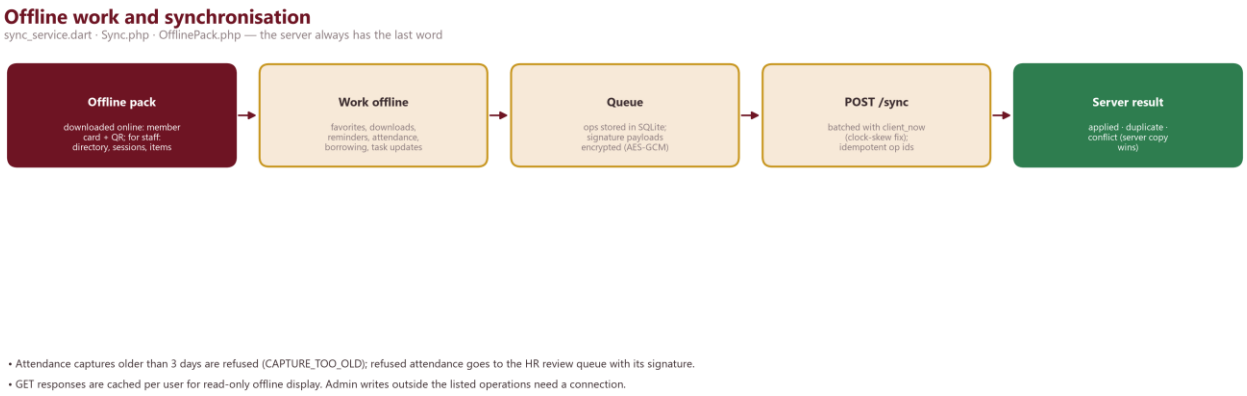
19. Notifications, announcements, reminders and delivery mechanisms

- **In-app notifications** are created in bulk with a dedupe key (idempotent), stored per user and shown in the notification centre with unread badge. Topics: program, attendance, project, department, education, property, mezmur, system; members can switch topics and channels (push / e-mail) off, except mandatory types (system, admin message, announcement, borrowing due/overdue).
- **Push** through Firebase Cloud Messaging HTTP v1 (service-account JWT exchanged for an OAuth token on the server; the app holds only client configuration). Android channels separate reminders, programs, borrowing etc. If Firebase is not configured the app works with in-app notifications only.
- **E-mail** through SMTP for notification e-mails and reset codes; **Telegram** for linked accounts (reset codes, temporary passwords).
- **Broadcasts.** Admins send to one user, a selection, a filtered group or everyone (`notifications.send_all`); a resend is a new linked broadcast; delivery history and a delivery-health screen are available.
- **Reminders.** Program reminders (job every 5 minutes), interval reminders, borrowing reminders (every 30 minutes), scheduled notifications (every minute), personal reminders scheduled locally on the phone.
- **Retention.** Read notifications older than `notifications.retention_days` (default 365) are deleted by the nightly clean-up.

20. Ethiopian calendar and localisation support

- `Support/Ethiopian.php` converts Gregorian ↔ Ethiopian (Amete Mihret), handles leap years and month lengths (13th month Pagume), and computes payable periods, ranges and month counts; the app has a matching Ethiopian date picker and formatter (covered by `ethiopian_calendar_test.dart`).
- Programs can recur on Ethiopian calendar days; contributions use Ethiopian months; report PDFs print Ethiopian dates and month names; per-occurrence attendance exports include the Ethiopian date.
- Time zone: Africa/Addis_Ababa (server setting `APP_TIMEZONE`, app uses the full tz database).
- Localisation: Amharic is the template language (`app_am.arb`) with English (`app_en.arb`), 2,754 keys; server error codes are mapped to localised messages in `core/l10n/error_text.dart`; server-side texts (notifications, PDFs) use translation files; Ethiopic fonts are bundled for the app and used by the PDF renderer.

21. Offline behaviour, synchronisation and conflict handling



Capability	Offline behaviour
Reading	Last good response of GET requests is cached per user and shown when the network fails (read-only).
Mezmur	Downloaded Mezmurs (lyrics, cover, optional audio) and favourites work fully offline.
Member card	Own member card and QR are kept in the offline pack.
Reminders	Created/edited locally, scheduled as device notifications, synced later.
Attendance (takers)	Offline pack contains member directory (no phone numbers), programs/sessions from 3 days ago to 2 weeks ahead, expected members, existing marks and timing rules; Start/End/marks are queued.
Borrowing staff	Member directory and lendable items; borrowing and returns are queued.
Projects	Task status/progress updates are queued (version check on sync).
Preferences	Changed optimistically, queued with base timestamp so a newer server copy is never silently overwritten.
Everything else	Needs a connection.

- Sync protocol: `POST /api/v1/sync` with up to 200 operations `{op_id (UUID), type, payload, client_ts}` and `client_now`; each op is idempotent through `sync_receipts`; results are applied / duplicate / conflict, and on conflict the server copy wins and is returned.
- Attendance specifics: captures older than 3 days are refused (CAPTURE_TOO_OLD); clock skew is corrected using `client_now`; business errors become entries in the HR conflict queue (with signature) instead of being dropped; HR accepts or rejects with a note and cannot resolve their own submissions.
- Queue protection: signature-bearing payloads are encrypted with AES-256-GCM using a keystore key. In 1.5.3+16 signing out deleted the queue (defect F0a); the source tree keeps queued work by default and discards it only on explicit request (unreleased).

22. Reporting, exports, audit logs and document generation

`GET /api/v1/reports/export/{kind}?format=pdf|xlsx|csv` renders one template: PDF (A4, logo, title, Ethiopian date, period, generated by/at, filters, summary, tables, header/footer, page numbers, reference id), XLSX (summary sheet + one sheet per table, frozen headers) or CSV (main table, UTF-8 with BOM). Maximum 5,000 rows per export; every export is audited.

Export kind	Permission (any of)
`users`	reports.view, hr.manage
`mezmurs`	reports.view, mezmurs.update
`attendance`	reports.view, attendance.reports
`borrowings`	reports.view, borrowing.manage
`notifications`	reports.view
`admin-activity`	audit_logs.view
`audit-logs`	audit_logs.view
`activity-logs`	activity_logs.view
`contributions`	payments.view, payments.manage
`selection`	selection.view, selection.review, selection.finalize
`projects`	projects.reports, projects.all_departments
`education`	reports.view, education.manage
`programs`	reports.view, programs.manage
`education-marks`	education.manage (wide / long layouts, missing marks shown as —)
`mezmur-selection`	choir.manage

Audit. `audit_logs` records administrative actions (actor, action, entity, metadata, IP, user agent); `activity_logs` records member events (sign-ins, failures, device changes, plays, downloads). Both have viewers and exports. Known issue: the audit-log PDF export can exhaust 512 MB memory on very large ranges (recorded 2026-10-14).

23. API documentation

23.1 Conventions

- Base URL: `https://<host>/api/v1`. JSON over HTTPS; uploads are `multipart/form-data` with a `file` field.
- Authentication: `Authorization: Bearer <access JWT>` for routes marked *Bearer token*; obtain tokens from `POST /auth/login`, renew with `POST /auth/refresh`.
- Success envelope: `{ "success": true, "code": "OK", "message": "...", "data": ..., "errors": [] }`; paginated lists add `"meta": {"page", "limit", "total", "pages"}` (query `page`, `limit`).
- Error envelope: `{ "success": false, "code": "UPPER_SNAKE_CODE", "message": "...", "data": null, "errors": [{"field", "code", "message"}] }` with HTTP 400 (bad request), 401 (UNAUTHORIZED, INVALID_TOKEN, SESSION_REVOKED), 403 (FORBIDDEN, ACCOUNT_DISABLED, MODULE_DISABLED), 404, 409 (conflicts such as DEVICE_ALREADY_REGISTERED, SELECTION_LOCKED, HAS_RECORDS), 410, 413 (FILE_TOO_LARGE), 422 (VALIDATION_ERROR), 429 (rate limit, ACCOUNT_LOCKED).
- Validation: every body/query is parsed by a strict schema (`Ctx::body(Z::object([...]))`); unknown fields are rejected on strict schemas; error codes are field-level.
- Permissions: *any of* = at least one listed permission; *all of* = every listed permission. Services apply additional row-level checks.
- An OpenAPI file is served at `/api/openapi.json` (generated from the former Node backend; *requires verification* that it matches the current routes).

23.2 Request/response examples

Sign in

```
POST /api/v1/auth/login
Content-Type: application/json

{"identifier": "0911223344", "password": "123456", "device_id": "<stable device id>", "device_name": "TECNO ..."}
```

Response `data`: access token, refresh token, expiry and the public user object (permissions, member group, flags such as `must_change_password`). Errors: INVALID_CREDENTIALS (401), ACCOUNT_LOCKED (429), ACCOUNT_PENDING, DEVICE_ALREADY_REGISTERED, DEVICE_IN_USE.

Record attendance (two steps)

```
GET /api/v1/programs/42/attendance/check?uuid=<member QR uuid>&date=2026-10-19
PUT /api/v1/programs/42/attendance/1057

{"status": "present", "via": "qr", "signature": "...", "date": "2026-10-19"}
```

Errors include LATE_REMOVED, SIGNATURE_REQUIRED, NOT_SESSION_TAKER, BEFORE_SESSION_START, SESSION_NOT_STARTED, ATTENDANCE_CLOSED, MEMBER_NOT_EXPECTED, SELF_ATTENDANCE_NOT_ALLOWED and PROGRAM_CANCELLED; a duplicate scan returns success with `duplicate: true`.

Offline sync

```
POST /api/v1/sync

{"client_now": "2026-10-19T08:05:00Z", "operations": [
  {"op_id": "<uuid>", "type": "attendance.record", "payload": {"program_id": 42, "user_id": 1057, "date": "2026-10-19", ...}, "client_ts": "2026-10-19T07:58:12Z"}
]}
```

Operation types: favorite.add/remove, download.add/remove, reminder.create/update/delete, activity, attendance.session.start/end, attendance.record, attendance.set, attendance.clear, attendance.bulk, edu_attendance.set/scan, borrowing.create, borrowing.return, project.task.update, preferences.update.

Export a report

```
GET /api/v1/reports/export/attendance?format=xlsx&program_id=42&date=2026-10-19
```

23.3 Route reference

All 564 documented routes, grouped by route file (restricted system-console routes and test-only routes are excluded; routes marked *(unreleased)* exist only in the source tree). Method, path, authentication, permission middleware and extra gates come from `php artisan route:list`.

Authentication (`api/auth.php`, 8 routes)

Method	Path	Auth	Permissions	Gates
POST	`/api/v1/auth/change-password`	Bearer token	—	rate limit: auth
POST	`/api/v1/auth/forgot-password`	Public	—	rate limit: reset
POST	`/api/v1/auth/login`	Public	—	rate limit: auth
POST	`/api/v1/auth/logout`	Optional	—	
GET	`/api/v1/auth/me`	Bearer token	—	
POST	`/api/v1/auth/refresh`	Public	—	rate limit: auth
POST	`/api/v1/auth/register`	Public	—	rate limit: auth
POST	`/api/v1/auth/reset-password`	Public	—	rate limit: reset

My account (current user) (`api/me.php`, 25 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/me`	Bearer token	—	
PATCH	`/api/v1/me`	Bearer token	—	
GET	`/api/v1/me/activity`	Bearer token	—	
POST	`/api/v1/me/activity`	Bearer token	—	
DELETE	`/api/v1/me/avatar`	Bearer token	—	
POST	`/api/v1/me/avatar`	Bearer token	—	
GET	`/api/v1/me/card`	Bearer token	—	
DELETE	`/api/v1/me/change-request`	Bearer token	—	
POST	`/api/v1/me/change-request`	Bearer token	—	
GET	`/api/v1/me/change-requests`	Bearer token	—	
DELETE	`/api/v1/me/fcm-tokens`	Bearer token	—	
POST	`/api/v1/me/fcm-tokens`	Bearer token	—	
GET	`/api/v1/me/notification-preferences`	Bearer token	—	
PUT	`/api/v1/me/notification-preferences`	Bearer token	—	
GET	`/api/v1/me/payments`	Bearer token	—	
GET	`/api/v1/me/preferences`	Bearer token	—	
PUT	`/api/v1/me/preferences`	Bearer token	—	
GET	`/api/v1/me/profile`	Bearer token	—	
POST	`/api/v1/me/push-test`	Bearer token	—	
GET	`/api/v1/me/sessions`	Bearer token	—	
POST	`/api/v1/me/sessions/revoke-all`	Bearer token	—	
POST	`/api/v1/me/sessions/revoke-others`	Bearer token	—	
DELETE	`/api/v1/me/sessions/{sid}`	Bearer token	—	

Method	Path	Auth	Permissions	Gates
DELETE	/api/v1/me/telegram	Bearer token	—	
POST	/api/v1/me/telegram/link	Bearer token	—	

Personal: notifications, reminders, favorites, downloads (api/personal.php, 27 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/attendance/corrections	Bearer token	any of: attendance.correction.approve, attendance.correction.request	
POST	/api/v1/attendance/corrections/{id}/review	Bearer token	all of: attendance.correction.approve	
GET	/api/v1/attendance/me	Bearer token	—	
GET	/api/v1/attendance/me/calendar	Bearer token	—	
GET	/api/v1/attendance/me/stats	Bearer token	—	
GET	/api/v1/attendance/signature-policy	Bearer token	any of: attendance.manage, attendance.view, attendance.configure, attendance.assign	
PUT	/api/v1/attendance/signature-policy	Bearer token	all of: attendance.configure	
GET	/api/v1/attendance/sync-conflicts	Bearer token	any of: attendance.override, attendance.correction.approve	
POST	/api/v1/attendance/sync-conflicts/{id}/resolve	Bearer token	all of: attendance.override	
GET	/api/v1/attendance/sync-conflicts/{id}/signature	Bearer token	any of: attendance.override, attendance.correction.approve	
GET	/api/v1/attendance/users/{id}	Bearer token	any of: attendance.manage, attendance.reports, attendance.view	
GET	/api/v1/attendance/users/{id}/calendar	Bearer token	any of: attendance.manage, attendance.reports, attendance.view	
GET	/api/v1/attendance/users/{id}/stats	Bearer token	any of: attendance.manage, attendance.reports, attendance.view	
GET	/api/v1/notifications	Bearer token	—	
POST	/api/v1/notifications/read-all	Bearer token	—	
GET	/api/v1/notifications/unread-count	Bearer token	—	
POST	/api/v1/notifications/{id}/read	Bearer token	—	
GET	/api/v1/permissions	Bearer token	all of: attendance.manage	
GET	/api/v1/permissions/policy	Bearer token	all of: attendance.manage	
PUT	/api/v1/permissions/policy	Bearer token	all of: attendance.manage	
GET	/api/v1/permissions/summary	Bearer token	all of: attendance.manage	
PATCH	/api/v1/permissions/{id}	Bearer token	all of: attendance.manage	
POST	/api/v1/permissions/{id}/cancel	Bearer token	all of: attendance.manage	
GET	/api/v1/reminders	Bearer token	—	
POST	/api/v1/reminders	Bearer token	—	
DELETE	/api/v1/reminders/{id}	Bearer token	—	
PATCH	/api/v1/reminders/{id}	Bearer token	—	

App bootstrap, sync and offline pack (`api/app.php`, 3 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/home`	Optional	—	
GET	`/api/v1/offline/pack`	Bearer token	—	
POST	`/api/v1/sync`	Bearer token	—	

Mezmurs, categories, home, pages, banners (member side) (`api/content.php`, 31 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/categories`	Optional	—	
POST	`/api/v1/categories`	Bearer token	all of: categories.manage	
POST	`/api/v1/categories/reorder`	Bearer token	all of: categories.manage	
DELETE	`/api/v1/categories/{id}`	Bearer token	all of: categories.manage	
GET	`/api/v1/categories/{id}`	Optional	—	
PATCH	`/api/v1/categories/{id}`	Bearer token	all of: categories.manage	
DELETE	`/api/v1/categories/{id}/image`	Bearer token	all of: categories.manage	
POST	`/api/v1/categories/{id}/image`	Bearer token	all of: categories.manage	
GET	`/api/v1/categories/{id}/mezmurs`	Optional	—	
PUT	`/api/v1/categories/{id}/today`	Bearer token	all of: categories.manage	
GET	`/api/v1/downloads`	Bearer token	—	
DELETE	`/api/v1/downloads/{mezmurId}`	Bearer token	—	
PUT	`/api/v1/downloads/{mezmurId}`	Bearer token	—	
GET	`/api/v1/favorites`	Bearer token	—	
DELETE	`/api/v1/favorites/{mezmurId}`	Bearer token	—	
PUT	`/api/v1/favorites/{mezmurId}`	Bearer token	—	
GET	`/api/v1/mezmurs`	Optional	—	
POST	`/api/v1/mezmurs`	Bearer token	all of: mezmurs.create	
GET	`/api/v1/mezmurs/admin/list`	Bearer token	all of: mezmurs.view	
GET	`/api/v1/mezmurs/admin/{id}`	Bearer token	all of: mezmurs.view	
POST	`/api/v1/mezmurs/bulk/category`	Bearer token	all of: mezmurs.update	
POST	`/api/v1/mezmurs/changes`	Bearer token	—	
GET	`/api/v1/mezmurs/lyrics-pack`	Bearer token	—	
DELETE	`/api/v1/mezmurs/{id}`	Bearer token	all of: mezmurs.delete	
GET	`/api/v1/mezmurs/{id}`	Optional	—	
PATCH	`/api/v1/mezmurs/{id}`	Bearer token	all of: mezmurs.update	
DELETE	`/api/v1/mezmurs/{id}/audio`	Bearer token	all of: mezmurs.update	
POST	`/api/v1/mezmurs/{id}/audio`	Bearer token	all of: mezmurs.update	
DELETE	`/api/v1/mezmurs/{id}/cover`	Bearer token	all of: mezmurs.update	
POST	`/api/v1/mezmurs/{id}/cover`	Bearer token	all of: mezmurs.update	
PUT	`/api/v1/mezmurs/{id}/today`	Bearer token	all of: mezmurs.update	

Programs, sessions and attendance (`api/programs.php`, 38 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/programs`	Optional	—	
POST	`/api/v1/programs`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/admin/list`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/admin/{id}`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/attendance/signatures/{attendanceId}`	Bearer token	any of: attendance.reports, attendance.override, attendance.correction.approve	
GET	`/api/v1/programs/attendance/takeable`	Bearer token	—	
GET	`/api/v1/programs/calendar`	Optional	—	
GET	`/api/v1/programs/categories`	Public	—	
POST	`/api/v1/programs/categories`	Bearer token	all of: program_categories.manage	
GET	`/api/v1/programs/categories/all`	Bearer token	all of: program_categories.manage	
DELETE	`/api/v1/programs/categories/{id}`	Bearer token	all of: program_categories.manage	
PATCH	`/api/v1/programs/categories/{id}`	Bearer token	all of: program_categories.manage	
DELETE	`/api/v1/programs/{id}`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/{id}`	Optional	—	
PATCH	`/api/v1/programs/{id}`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/{id}/attendance`	Bearer token	—	
GET	`/api/v1/programs/{id}/attendance-takers`	Bearer token	any of: attendance.view, attendance.manage, attendance.assign, programs.manage	
PUT	`/api/v1/programs/{id}/attendance-takers`	Bearer token	all of: attendance.assign	
POST	`/api/v1/programs/{id}/attendance/bulk`	Bearer token	—	
GET	`/api/v1/programs/{id}/attendance/check`	Bearer token	—	
POST	`/api/v1/programs/{id}/attendance/close`	Bearer token	all of: attendance.close	
GET	`/api/v1/programs/{id}/attendance/session`	Bearer token	—	
POST	`/api/v1/programs/{id}/attendance/session/end`	Bearer token	—	
POST	`/api/v1/programs/{id}/attendance/session/start`	Bearer token	—	
PUT	`/api/v1/programs/{id}/attendance/session/taker`	Bearer token	all of: attendance.assign	
DELETE	`/api/v1/programs/{id}/attendance/{userId}`	Bearer token	—	
PUT	`/api/v1/programs/{id}/attendance/{userId}`	Bearer token	—	
POST	`/api/v1/programs/{id}/attendance/{userId}/corrections`	Bearer token	all of: attendance.correction.request	
POST	`/api/v1/programs/{id}/image`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/{id}/my-attendance`	Bearer token	—	
PUT	`/api/v1/programs/{id}/my-reminders`	Bearer token	—	
GET	`/api/v1/programs/{id}/occurrences`	Bearer token	any of: programs.manage, attendance.manage, attendance.view	
PUT	`/api/v1/programs/{id}/occurrences/{date}`	Bearer token	all of: programs.manage	
GET	`/api/v1/programs/{id}/participants`	Bearer token	all of: programs.manage	
POST	`/api/v1/programs/{id}/participants/bulk`	Bearer token	all of: programs.manage	

Method	Path	Auth	Permissions	Gates
DELETE	/api/v1/programs/{id}/participants/{userId}	Bearer token	all of: programs.manage	
DELETE	/api/v1/programs/{id}/permission	Bearer token	—	
POST	/api/v1/programs/{id}/permission	Bearer token	—	

Education: classes, courses, marks, assignments (api/edu.php, 67 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/edu/admin/academic-years *(unreleased)*	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/academic-years *(unreleased)*	Bearer token	all of: education.manage	module: education
DELETE	/api/v1/edu/admin/academic-years/{id} *(unreleased)*	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/academic-years/{id} *(unreleased)*	Bearer token	all of: education.manage	module: education
PATCH	/api/v1/edu/admin/academic-years/{id} *(unreleased)*	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/academic-years/{id}/close *(unreleased)*	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/academic-years/{id}/current *(unreleased)*	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/academic-years/{id}/terms *(unreleased)*	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/academic-years/{id}/terms *(unreleased)*	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/classes	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/classes	Bearer token	all of: education.manage	module: education
DELETE	/api/v1/edu/admin/classes/{id}	Bearer token	all of: education.manage	module: education
PATCH	/api/v1/edu/admin/classes/{id}	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/classes/{id}/assign	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/classes/{id}/assignable	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/classes/{id}/courses	Bearer token	all of: education.manage	module: education
PUT	/api/v1/edu/admin/classes/{id}/schedule	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/classes/{id}/staff	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/classes/{id}/staff	Bearer token	all of: education.manage	module: education
DELETE	/api/v1/edu/admin/classes/{id}/staff/{userId}	Bearer token	all of: education.manage	module: education
DELETE	/api/v1/edu/admin/classes/{id}/students/{userId}	Bearer token	all of: education.manage	module: education
DELETE	/api/v1/edu/admin/courses/{id}	Bearer token	all of: education.manage	module: education
PATCH	/api/v1/edu/admin/courses/{id}	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/enrollments/{id}/promote	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/enrollments/{id}/transfer	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/levels *(unreleased)*	Bearer token	all of: education.manage	module: education

Method	Path	Auth	Permissions	Gates
POST	`/api/v1/edu/admin/levels` *(unreleased)*	Bearer token	all of: education.manage	module: education
DELETE	`/api/v1/edu/admin/levels/{id}` *(unreleased)*	Bearer token	all of: education.manage	module: education
PATCH	`/api/v1/edu/admin/levels/{id}` *(unreleased)*	Bearer token	all of: education.manage	module: education
DELETE	`/api/v1/edu/admin/terms/{id}` *(unreleased)*	Bearer token	all of: education.manage	module: education
PATCH	`/api/v1/edu/admin/terms/{id}` *(unreleased)*	Bearer token	all of: education.manage	module: education
DELETE	`/api/v1/edu/assessments/{id}`	Bearer token	—	module: education
PATCH	`/api/v1/edu/assessments/{id}`	Bearer token	—	module: education
GET	`/api/v1/edu/assessments/{id}/scores`	Bearer token	—	module: education
PUT	`/api/v1/edu/assessments/{id}/scores`	Bearer token	—	module: education
DELETE	`/api/v1/edu/assignments/{id}`	Bearer token	—	module: education
PATCH	`/api/v1/edu/assignments/{id}`	Bearer token	—	module: education
POST	`/api/v1/edu/assignments/{id}/file`	Bearer token	—	module: education
GET	`/api/v1/edu/assignments/{id}/submissions`	Bearer token	—	module: education
POST	`/api/v1/edu/assignments/{id}/submit`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/open`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}/attendance`	Bearer token	—	module: education
PUT	`/api/v1/edu/classes/{id}/attendance`	Bearer token	—	module: education
POST	`/api/v1/edu/classes/{id}/attendance/scan`	Bearer token	—	module: education
DELETE	`/api/v1/edu/classes/{id}/enroll`	Bearer token	—	module: education
POST	`/api/v1/edu/classes/{id}/enroll`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}/my-assignments`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}/my-attendance`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}/my-results`	Bearer token	—	module: education
GET	`/api/v1/edu/classes/{id}/students`	Bearer token	—	module: education
GET	`/api/v1/edu/courses/{id}`	Bearer token	—	module: education
GET	`/api/v1/edu/courses/{id}/assessments`	Bearer token	—	module: education
POST	`/api/v1/edu/courses/{id}/assessments`	Bearer token	—	module: education
GET	`/api/v1/edu/courses/{id}/assignments`	Bearer token	—	module: education
POST	`/api/v1/edu/courses/{id}/assignments`	Bearer token	—	module: education
POST	`/api/v1/edu/courses/{id}/groups`	Bearer token	—	module: education
DELETE	`/api/v1/edu/groups/{id}`	Bearer token	—	module: education
PUT	`/api/v1/edu/groups/{id}/members`	Bearer token	—	module: education
GET	`/api/v1/edu/overview`	Bearer token	—	module: education
POST	`/api/v1/edu/registrations`	Bearer token	any of: education.register, education.manage	module: education
GET	`/api/v1/edu/registrations/classes`	Bearer token	any of: education.register, education.manage	module: education
POST	`/api/v1/edu/registrations/classes/{id}/assign`	Bearer token	any of: education.register, education.manage	module: education
GET	`/api/v1/edu/registrations/classes/{id}/assignable`	Bearer token	any of: education.register, education.manage	module: education
GET	`/api/v1/edu/registrations/classes/{id}/students`	Bearer token	any of: education.register, education.manage	module: education
DELETE	`/api/v1/edu/registrations/classes/{id}/students/{userId}`	Bearer token	any of: education.register, education.manage	module: education

Method	Path	Auth	Permissions	Gates
PUT	/api/v1/edu/submissions/{id}/grade	Bearer token	—	module: education

Education: learning platform (categories, subjects, lessons, quizzes) (api/edu_learning.php, 56 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/edu/attempts/{id}	Bearer token	—	module: education
PUT	/api/v1/edu/attempts/{id}/answers	Bearer token	—	module: education
POST	/api/v1/edu/attempts/{id}/check	Bearer token	—	module: education
POST	/api/v1/edu/attempts/{id}/submit	Bearer token	—	module: education
GET	/api/v1/edu/categories	Bearer token	—	module: education
GET	/api/v1/edu/categories/{id}/subjects	Bearer token	—	module: education
POST	/api/v1/edu/content/categories	Bearer token	any of: education.content, education.manage	module: education
PUT	/api/v1/edu/content/categories/order	Bearer token	any of: education.content, education.manage	module: education
DELETE	/api/v1/edu/content/categories/{id}	Bearer token	any of: education.content, education.manage	module: education
PATCH	/api/v1/edu/content/categories/{id}	Bearer token	any of: education.content, education.manage	module: education
DELETE	/api/v1/edu/content/categories/{id}/image	Bearer token	any of: education.content, education.manage	module: education
POST	/api/v1/edu/content/categories/{id}/image	Bearer token	any of: education.content, education.manage	module: education
POST	/api/v1/edu/content/featured	Bearer token	any of: education.content, education.manage	module: education
DELETE	/api/v1/edu/content/featured/{id}	Bearer token	any of: education.content, education.manage	module: education
PATCH	/api/v1/edu/content/featured/{id}	Bearer token	any of: education.content, education.manage	module: education
POST	/api/v1/edu/content/featured/{id}/image	Bearer token	any of: education.content, education.manage	module: education
POST	/api/v1/edu/content/subjects	Bearer token	any of: education.content, education.manage	module: education
DELETE	/api/v1/edu/content/subjects/{id}	Bearer token	any of: education.content, education.manage	module: education
PATCH	/api/v1/edu/content/subjects/{id}	Bearer token	any of: education.content, education.manage	module: education
DELETE	/api/v1/edu/content/subjects/{id}/cover	Bearer token	any of: education.content, education.manage	module: education
POST	/api/v1/edu/content/subjects/{id}/cover	Bearer token	any of: education.content, education.manage	module: education
GET	/api/v1/edu/courses/{id}/quizzes	Bearer token	—	module: education
POST	/api/v1/edu/courses/{id}/quizzes	Bearer token	—	module: education
GET	/api/v1/edu/featured	Bearer token	—	module: education
DELETE	/api/v1/edu/lessons/{id}	Bearer token	—	module: education
GET	/api/v1/edu/lessons/{id}	Bearer token	—	module: education
PATCH	/api/v1/edu/lessons/{id}	Bearer token	—	module: education
POST	/api/v1/edu/lessons/{id}/links	Bearer token	—	module: education
POST	/api/v1/edu/lessons/{id}/materials	Bearer token	—	module: education
POST	/api/v1/edu/lessons/{id}/progress	Bearer token	—	module: education
DELETE	/api/v1/edu/materials/{id}	Bearer token	—	module: education
PATCH	/api/v1/edu/materials/{id}	Bearer token	—	module: education
GET	/api/v1/edu/materials/{id}/file	Bearer token	—	module: education
GET	/api/v1/edu/me/bookmarks	Bearer token	—	module: education
GET	/api/v1/edu/me/dashboard	Bearer token	—	module: education
GET	/api/v1/edu/me/progress	Bearer token	—	module: education

Method	Path	Auth	Permissions	Gates
GET	/api/v1/edu/me/quizzes	Bearer token	—	module: education
GET	/api/v1/edu/quiz-attempts/{id}	Bearer token	—	module: education
PUT	/api/v1/edu/quiz-attempts/{id}/grade	Bearer token	—	module: education
DELETE	/api/v1/edu/quizzes/{id}	Bearer token	—	module: education
GET	/api/v1/edu/quizzes/{id}	Bearer token	—	module: education
PATCH	/api/v1/edu/quizzes/{id}	Bearer token	—	module: education
POST	/api/v1/edu/quizzes/{id}/attempts	Bearer token	—	module: education
PUT	/api/v1/edu/quizzes/{id}/questions	Bearer token	—	module: education
GET	/api/v1/edu/quizzes/{id}/results	Bearer token	—	module: education
POST	/api/v1/edu/quizzes/{id}/transfer	Bearer token	—	module: education
GET	/api/v1/edu/search	Bearer token	—	module: education
GET	/api/v1/edu/subjects	Bearer token	—	module: education
GET	/api/v1/edu/subjects/authoring	Bearer token	—	module: education
GET	/api/v1/edu/subjects/{id}	Bearer token	—	module: education
POST	/api/v1/edu/subjects/{id}/lessons	Bearer token	—	module: education
PUT	/api/v1/edu/subjects/{id}/lessons/order	Bearer token	—	module: education
POST	/api/v1/edu/subjects/{id}/units	Bearer token	—	module: education
PUT	/api/v1/edu/subjects/{id}/units/order	Bearer token	—	module: education
DELETE	/api/v1/edu/units/{id}	Bearer token	—	module: education
PATCH	/api/v1/edu/units/{id}	Bearer token	—	module: education

Education: course management (teachers, marks review, requests, complaints, journey) (api/edu_academics.php, 27 routes)

Method	Path	Auth	Permissions	Gates
POST	/api/v1/edu/admin/assessments/{id}/amend	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/assessments/{id}/history	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/assessments/{id}/publish	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/assessments/{id}/review	Bearer token	all of: education.manage	module: education
PATCH	/api/v1/edu/admin/classes/{id}/students/{userId}	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/complaints	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/complaints/{id}	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/complaints/{id}/actions	Bearer token	all of: education.manage	module: education
PATCH	/api/v1/edu/admin/course-teachers/{id}	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/courses/{id}/teachers	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/courses/{id}/teachers	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/marks	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/requests	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/requests/{id}/review	Bearer token	all of: education.manage	module: education
GET	/api/v1/edu/admin/students/{userId}/journey	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/admin/students/{userId}/subjects	Bearer token	all of: education.manage	module: education
POST	/api/v1/edu/assessments/{id}/submit	Bearer token	—	module: education
GET	/api/v1/edu/complaints/{id}/attachment	Bearer token	—	module: education
GET	/api/v1/edu/me/complaints	Bearer token	—	module: education
POST	/api/v1/edu/me/complaints	Bearer token	—	module: education
GET	/api/v1/edu/me/complaints/{id}	Bearer token	—	module: education
POST	/api/v1/edu/me/complaints/{id}/messages	Bearer token	—	module: education
GET	/api/v1/edu/me/journey	Bearer token	—	module: education
GET	/api/v1/edu/me/requests	Bearer token	—	module: education
POST	/api/v1/edu/me/requests	Bearer token	—	module: education
DELETE	/api/v1/edu/me/requests/{id}	Bearer token	—	module: education

Method	Path	Auth	Permissions	Gates
GET	/api/v1/edu/teaching	Bearer token	—	module: education

Holiday service selection (api/selections.php, 20 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/choir/me	Bearer token	—	
GET	/api/v1/choir/me/education-calendar	Bearer token	—	
GET	/api/v1/selections	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
POST	/api/v1/selections	Bearer token	all of: selection.configure	
GET	/api/v1/selections/options	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
GET	/api/v1/selections/{id}	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
PATCH	/api/v1/selections/{id}	Bearer token	any of: selection.configure, selection.exams	
POST	/api/v1/selections/{id}/categories/assign	Bearer token	any of: selection.finalize, choir.manage, choir.assign	
POST	/api/v1/selections/{id}/compute	Bearer token	any of: selection.configure, selection.review	
GET	/api/v1/selections/{id}/feedback	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
POST	/api/v1/selections/{id}/feedback	Bearer token	—	
POST	/api/v1/selections/{id}/publish	Bearer token	all of: selection.finalize	
GET	/api/v1/selections/{id}/ranked-categories	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
PUT	/api/v1/selections/{id}/required-reviewers	Bearer token	all of: selection.configure	
GET	/api/v1/selections/{id}/results	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
GET	/api/v1/selections/{id}/results/{userId}	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
PUT	/api/v1/selections/{id}/results/{userId}/override	Bearer token	any of: selection.review, selection.finalize	
GET	/api/v1/selections/{id}/reviews	Bearer token	any of: selection.view, selection.configure, selection.review, selection.finalize	
POST	/api/v1/selections/{id}/reviews	Bearer token	any of: selection.review, selection.configure	
POST	/api/v1/selections/{id}/status	Bearer token	any of: selection.configure, selection.review, selection.finalize	

Wedding (other) Mezmur services (`api/weddings.php`, 9 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/weddings`	Bearer token	all of: choir.manage	
POST	`/api/v1/weddings`	Bearer token	all of: choir.manage	
GET	`/api/v1/weddings/eligible`	Bearer token	all of: choir.manage	
GET	`/api/v1/weddings/{id}`	Bearer token	all of: choir.manage	
PATCH	`/api/v1/weddings/{id}`	Bearer token	all of: choir.manage	
POST	`/api/v1/weddings/{id}/announce`	Bearer token	all of: choir.manage	
GET	`/api/v1/weddings/{id}/candidates`	Bearer token	all of: choir.manage	
POST	`/api/v1/weddings/{id}/members`	Bearer token	all of: choir.manage	
PATCH	`/api/v1/weddings/{id}/members/{userId}`	Bearer token	all of: choir.manage	

User administration (`api/users.php`, 16 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/users`	Bearer token	all of: users.view	
POST	`/api/v1/users`	Bearer token	all of: users.create	
POST	`/api/v1/users/bulk/preview`	Bearer token	all of: users.view	
GET	`/api/v1/users/change-requests`	Bearer token	all of: users.view	
POST	`/api/v1/users/change-requests/{id}/approve`	Bearer token	all of: users.update	
POST	`/api/v1/users/change-requests/{id}/reject`	Bearer token	all of: users.update	
DELETE	`/api/v1/users/{id}`	Bearer token	all of: users.disable	
GET	`/api/v1/users/{id}`	Bearer token	all of: users.view	
PATCH	`/api/v1/users/{id}`	Bearer token	all of: users.update	
GET	`/api/v1/users/{id}/devices`	Bearer token	all of: users.view	
POST	`/api/v1/users/{id}/disable`	Bearer token	all of: users.disable	
POST	`/api/v1/users/{id}/enable`	Bearer token	all of: users.disable	
POST	`/api/v1/users/{id}/force-logout`	Bearer token	all of: users.reset_device	
GET	`/api/v1/users/{id}/profile-history`	Bearer token	all of: users.view	
POST	`/api/v1/users/{id}/reset-device`	Bearer token	all of: users.reset_device	
POST	`/api/v1/users/{id}/reset-password`	Bearer token	all of: users.update	

Administrators and roles (`api/admins.php`, 12 routes)

Method	Path	Auth	Permissions	Gates
GET	`/api/v1/admins`	Bearer token	all of: admins.view	
POST	`/api/v1/admins`	Bearer token	all of: admins.manage	
GET	`/api/v1/admins/permissions`	Bearer token	all of: admins.view	
POST	`/api/v1/admins/promote`	Bearer token	all of: admins.manage	
GET	`/api/v1/admins/roles`	Bearer token	all of: admins.view	
POST	`/api/v1/admins/roles`	Bearer token	all of: admins.manage	
DELETE	`/api/v1/admins/roles/{id}`	Bearer token	all of: admins.manage	

Method	Path	Auth	Permissions	Gates
PATCH	/api/v1/admins/roles/{id}	Bearer token	all of: admins.manage	
GET	/api/v1/admins/{id}	Bearer token	all of: admins.view	
PATCH	/api/v1/admins/{id}	Bearer token	all of: admins.manage	
POST	/api/v1/admins/{id}/demote	Bearer token	all of: admins.manage	
POST	/api/v1/admins/{id}/reset-password	Bearer token	all of: admins.manage	

Member groups (types) (api/member_groups.php, 4 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/member-groups	Bearer token	—	
POST	/api/v1/member-groups	Bearer token	all of: member_groups.manage	
DELETE	/api/v1/member-groups/{key}	Bearer token	all of: member_groups.manage	
PATCH	/api/v1/member-groups/{key}	Bearer token	all of: member_groups.manage	

Member ID numbering (api/member_ids.php, 2 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/member-ids/format	Bearer token	any of: member_id.configure, users.view	
PUT	/api/v1/member-ids/format	Bearer token	all of: member_id.configure	

Professions catalogue (api/professions.php, 6 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/admin/professions	Bearer token	any of: users.manage, settings.manage	
POST	/api/v1/admin/professions	Bearer token	any of: users.manage, settings.manage	
DELETE	/api/v1/admin/professions/{id}	Bearer token	any of: users.manage, settings.manage	
PUT	/api/v1/admin/professions/{id}	Bearer token	any of: users.manage, settings.manage	
GET	/api/v1/professions	Public	—	
GET	/api/v1/professions/categories	Public	—	

Departments, HR, payments, choir (api/org.php, 29 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/departments	Bearer token	—	
GET	/api/v1/departments/admins	Bearer token	all of: admins.view	
GET	/api/v1/departments/admins/of-user/{id}	Bearer token	all of: admins.view	
PUT	/api/v1/departments/admins/of-user/{id}/permissions	Bearer token	all of: admins.manage	
GET	/api/v1/departments/led	Bearer token	—	
GET	/api/v1/departments/of-user/{id}	Bearer token	all of: hr.manage	

Method	Path	Auth	Permissions	Gates
PATCH	/api/v1/departments/{id}	Bearer token	all of: departments.manage	
GET	/api/v1/departments/{id}/admins	Bearer token	all of: admins.view	
PUT	/api/v1/departments/{id}/admins/{userId}	Bearer token	all of: admins.manage	
GET	/api/v1/departments/{id}/members	Bearer token	all of: hr.manage	
POST	/api/v1/departments/{id}/members	Bearer token	all of: hr.manage	
DELETE	/api/v1/departments/{id}/members/{userId}	Bearer token	all of: hr.manage	
PATCH	/api/v1/departments/{id}/members/{userId}	Bearer token	all of: hr.manage	
GET	/api/v1/departments/{id}/participation	Bearer token	—	
PUT	/api/v1/departments/{id}/participation	Bearer token	—	
GET	/api/v1/members/{uuid}	Bearer token	—	
GET	/api/v1/payments	Bearer token	any of: payments.view, payments.manage	
POST	/api/v1/payments	Bearer token	all of: payments.manage	
GET	/api/v1/payments/member/{id}	Bearer token	any of: payments.view, payments.manage	
GET	/api/v1/payments/policy	Bearer token	any of: payments.view, payments.manage	
PUT	/api/v1/payments/policy	Bearer token	all of: payments.configure	
GET	/api/v1/payments/rates	Bearer token	any of: payments.view, payments.manage	
POST	/api/v1/payments/rates	Bearer token	all of: payments.configure	
DELETE	/api/v1/payments/rates/{id}	Bearer token	all of: payments.configure	
PATCH	/api/v1/payments/rates/{id}	Bearer token	all of: payments.configure	
DELETE	/api/v1/payments/{id}	Bearer token	all of: payments.manage	
PATCH	/api/v1/payments/{id}	Bearer token	all of: payments.manage	
POST	/api/v1/payments/{id}/receipt	Bearer token	all of: payments.manage	
POST	/api/v1/payments/{id}/verify	Bearer token	all of: payments.manage	

Items, borrowing, fines (api/borrowing.php, 36 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/borrowings	Bearer token	all of: borrowing.manage	
POST	/api/v1/borrowings	Bearer token	all of: borrowing.manage	
POST	/api/v1/borrowings/bulk	Bearer token	all of: borrowing.manage	
GET	/api/v1/borrowings/direct-policy	Bearer token	all of: borrowing.manage	
PUT	/api/v1/borrowings/direct-policy	Bearer token	all of: borrowing.configure	
GET	/api/v1/borrowings/me	Bearer token	—	
GET	/api/v1/borrowings/me/{id}	Bearer token	—	
GET	/api/v1/borrowings/return-conditions	Bearer token	—	
POST	/api/v1/borrowings/return-conditions	Bearer token	all of: borrowing.manage	
PATCH	/api/v1/borrowings/return-conditions/{id}	Bearer token	all of: borrowing.manage	
GET	/api/v1/borrowings/{id}	Bearer token	all of: borrowing.manage	
POST	/api/v1/borrowings/{id}/approve	Bearer token	all of: borrowing.approve	
DELETE	/api/v1/borrowings/{id}/fine	Bearer token	all of: fines.manage	

Method	Path	Auth	Permissions	Gates
PUT	/api/v1/borrowings/{id}/fine	Bearer token	all of: fines.manage	
POST	/api/v1/borrowings/{id}/fine/status	Bearer token	all of: fines.manage	
POST	/api/v1/borrowings/{id}/reject	Bearer token	all of: borrowing.approve	
POST	/api/v1/borrowings/{id}/renew	Bearer token	all of: borrowing.manage	
POST	/api/v1/borrowings/{id}/return	Bearer token	all of: borrowing.manage	
GET	/api/v1/borrowings/{id}/signature	Bearer token	—	
POST	/api/v1/borrowings/{id}/signature	Bearer token	all of: borrowing.manage	
GET	/api/v1/fines/policy	Bearer token	—	
PUT	/api/v1/fines/policy	Bearer token	all of: fines.manage	
GET	/api/v1/fines/summary	Bearer token	all of: fines.manage	
GET	/api/v1/items	Bearer token	—	
POST	/api/v1/items	Bearer token	all of: items.manage	
GET	/api/v1/items/categories	Bearer token	—	
POST	/api/v1/items/categories	Bearer token	all of: items.manage	
DELETE	/api/v1/items/categories/{key}	Bearer token	all of: items.manage	
PATCH	/api/v1/items/categories/{key}	Bearer token	all of: items.manage	
GET	/api/v1/items/next-code	Bearer token	all of: items.manage	
DELETE	/api/v1/items/{id}	Bearer token	all of: items.manage	
GET	/api/v1/items/{id}	Bearer token	—	
PATCH	/api/v1/items/{id}	Bearer token	all of: items.manage	
POST	/api/v1/items/{id}/file	Bearer token	all of: items.manage	
GET	/api/v1/items/{id}/history	Bearer token	any of: items.manage, borrowing.manage, property.review	
POST	/api/v1/items/{id}/image	Bearer token	all of: items.manage	

Library (books, requests) (api/library.php, 10 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/library/books	Bearer token	—	
GET	/api/v1/library/books/{id}/file	Bearer token	—	
POST	/api/v1/library/books/{id}/request	Bearer token	—	
GET	/api/v1/library/categories	Bearer token	—	
GET	/api/v1/library/requests	Bearer token	all of: borrowing.manage	
GET	/api/v1/library/requests/me	Bearer token	—	
DELETE	/api/v1/library/requests/{id}	Bearer token	—	
POST	/api/v1/library/requests/{id}/approve	Bearer token	all of: borrowing.manage	
POST	/api/v1/library/requests/{id}/issue	Bearer token	all of: borrowing.manage	
POST	/api/v1/library/requests/{id}/reject	Bearer token	all of: borrowing.manage	

Department property requests (api/property.php, 6 routes)

Method	Path	Auth	Permissions	Gates

Method	Path	Auth	Permissions	Gates
GET	/api/v1/property-requests	Bearer token	any of: property.request, property.review	
POST	/api/v1/property-requests	Bearer token	all of: property.request	
GET	/api/v1/property-requests/{id}	Bearer token	any of: property.request, property.review	
POST	/api/v1/property-requests/{id}/issue	Bearer token	all of: property.review	
POST	/api/v1/property-requests/{id}/return	Bearer token	all of: property.review	
POST	/api/v1/property-requests/{id}/{action}	Bearer token	any of: property.request, property.review	

Projects, tasks, budgets (api/projects.php, 53 routes)

Method	Path	Auth	Permissions	Gates
POST	/api/v1/chat/{id}/messages/{messageId}/task-status	Bearer token	—	
GET	/api/v1/projects	Bearer token	—	
POST	/api/v1/projects	Bearer token	—	
GET	/api/v1/projects/approvals	Bearer token	—	
GET	/api/v1/projects/budget-categories	Bearer token	—	
POST	/api/v1/projects/budget-categories	Bearer token	—	
PATCH	/api/v1/projects/budget-categories/{id}	Bearer token	—	
PATCH	/api/v1/projects/budget/adjustments/{id}	Bearer token	—	
GET	/api/v1/projects/departments/{id}/people	Bearer token	—	
GET	/api/v1/projects/events	Bearer token	—	
GET	/api/v1/projects/expenses	Bearer token	—	
PATCH	/api/v1/projects/expenses/{id}	Bearer token	—	
GET	/api/v1/projects/expenses/{id}/receipt	Bearer token	—	
DELETE	/api/v1/projects/files/{id}	Bearer token	—	
GET	/api/v1/projects/files/{id}	Bearer token	—	
GET	/api/v1/projects/issues	Bearer token	—	
GET	/api/v1/projects/issues/{id}	Bearer token	—	
PATCH	/api/v1/projects/issues/{id}	Bearer token	—	
POST	/api/v1/projects/issues/{id}/comments	Bearer token	—	
GET	/api/v1/projects/my-tasks	Bearer token	—	
GET	/api/v1/projects/overview	Bearer token	—	
GET	/api/v1/projects/reports/summary	Bearer token	—	
GET	/api/v1/projects/reports/tasks	Bearer token	—	
GET	/api/v1/projects/suggestions	Bearer token	—	
POST	/api/v1/projects/suggestions	Bearer token	—	
GET	/api/v1/projects/suggestions/{id}	Bearer token	—	
PATCH	/api/v1/projects/suggestions/{id}	Bearer token	—	
POST	/api/v1/projects/suggestions/{id}/comments	Bearer token	—	
POST	/api/v1/projects/suggestions/{id}/convert	Bearer token	—	
DELETE	/api/v1/projects/tasks/{id}	Bearer token	—	

Method	Path	Auth	Permissions	Gates
GET	/api/v1/projects/tasks/{id}	Bearer token	—	
PATCH	/api/v1/projects/tasks/{id}	Bearer token	—	
POST	/api/v1/projects/tasks/{id}/ask-status	Bearer token	—	
PUT	/api/v1/projects/tasks/{id}/assignees	Bearer token	—	
GET	/api/v1/projects/tasks/{id}/comments	Bearer token	—	
POST	/api/v1/projects/tasks/{id}/comments	Bearer token	—	
POST	/api/v1/projects/tasks/{id}/status	Bearer token	—	
DELETE	/api/v1/projects/{id}	Bearer token	—	
GET	/api/v1/projects/{id}	Bearer token	—	
PATCH	/api/v1/projects/{id}	Bearer token	—	
GET	/api/v1/projects/{id}/activity	Bearer token	—	
GET	/api/v1/projects/{id}/activity/{logId}	Bearer token	—	
GET	/api/v1/projects/{id}/budget	Bearer token	—	
PUT	/api/v1/projects/{id}/budget	Bearer token	—	
POST	/api/v1/projects/{id}/budget/adjustments	Bearer token	—	
POST	/api/v1/projects/{id}/budget/approve	Bearer token	—	
POST	/api/v1/projects/{id}/expenses	Bearer token	—	
GET	/api/v1/projects/{id}/files	Bearer token	—	
POST	/api/v1/projects/{id}/files	Bearer token	—	
POST	/api/v1/projects/{id}/issues	Bearer token	—	
GET	/api/v1/projects/{id}/tasks	Bearer token	—	
POST	/api/v1/projects/{id}/tasks	Bearer token	—	
PUT	/api/v1/projects/{id}/team	Bearer token	—	

Department chat (api/chat.php, 8 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/chat	Bearer token	—	
GET	/api/v1/chat/messages/{id}/file	Bearer token	—	
GET	/api/v1/chat/{id}/contacts	Bearer token	—	
GET	/api/v1/chat/{id}/messages	Bearer token	—	
POST	/api/v1/chat/{id}/messages	Bearer token	—	
DELETE	/api/v1/chat/{id}/messages/{messageId}	Bearer token	—	
POST	/api/v1/chat/{id}/read	Bearer token	—	
GET	/api/v1/chat/{id}/threads	Bearer token	—	

Admin notifications / broadcasts (api/admin_notifications.php, 5 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/admin/notifications/broadcasts	Bearer token	all of: notifications.send	
GET	/api/v1/admin/notifications/broadcasts/{id}	Bearer token	all of: notifications.send	
POST	/api/v1/admin/notifications/broadcasts/{id}/resend	Bearer token	all of: notifications.send	

Method	Path	Auth	Permissions	Gates
GET	/api/v1/admin/notifications/health	Bearer token	any of: notifications.send, settings.manage	
POST	/api/v1/admin/notifications/send	Bearer token	all of: notifications.send	

Banners (ads) and internal pages (admin) (api/promotions.php, 20 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/advertisements	Public	—	
POST	/api/v1/advertisements	Bearer token	all of: ads.manage	
GET	/api/v1/advertisements/admin/list	Bearer token	all of: ads.manage	
POST	/api/v1/advertisements/reorder	Bearer token	all of: ads.manage	
DELETE	/api/v1/advertisements/{id}	Bearer token	all of: ads.manage	
PATCH	/api/v1/advertisements/{id}	Bearer token	all of: ads.manage	
POST	/api/v1/advertisements/{id}/attachments	Bearer token	all of: ads.manage	
DELETE	/api/v1/advertisements/{id}/attachments/{attachmentId}	Bearer token	all of: ads.manage	
POST	/api/v1/advertisements/{id}/image	Bearer token	all of: ads.manage	
POST	/api/v1/advertisements/{id}/links	Bearer token	all of: ads.manage	
POST	/api/v1/pages	Bearer token	all of: pages.manage	
GET	/api/v1/pages/admin/list	Bearer token	all of: pages.manage	
GET	/api/v1/pages/admin/{id}	Bearer token	all of: pages.manage	
GET	/api/v1/pages/link-targets	Bearer token	all of: pages.manage	
DELETE	/api/v1/pages/{id}	Bearer token	all of: pages.manage	
GET	/api/v1/pages/{id}	Optional	—	
PATCH	/api/v1/pages/{id}	Bearer token	all of: pages.manage	
POST	/api/v1/pages/{id}/cover	Bearer token	all of: pages.manage	
POST	/api/v1/pages/{id}/images	Bearer token	all of: pages.manage	
DELETE	/api/v1/pages/{id}/images/{imageId}	Bearer token	all of: pages.manage	

Reports, exports, audit and activity logs (api/supervision.php, 14 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/activity-logs	Bearer token	all of: activity_logs.view	
GET	/api/v1/audit-logs	Bearer token	all of: audit_logs.view	
GET	/api/v1/reports/admin-activity	Bearer token	all of: reports.view; all of: audit_logs.view	
GET	/api/v1/reports/attendance	Bearer token	all of: reports.view	
GET	/api/v1/reports/borrowings	Bearer token	all of: reports.view	
GET	/api/v1/reports/dashboard	Bearer token	all of: reports.view	
GET	/api/v1/reports/export/{kind}	Bearer token	—	
GET	/api/v1/reports/mezmurs	Bearer token	all of: reports.view	
GET	/api/v1/reports/notifications	Bearer token	all of: reports.view	
GET	/api/v1/reports/users	Bearer token	all of: reports.view	
GET	/api/v1/settings	Bearer token	all of: settings.manage	

Method	Path	Auth	Permissions	Gates
GET	/api/v1/settings/modules	Bearer token	all of: settings.manage	
PUT	/api/v1/settings/modules	Bearer token	all of: settings.manage	
PUT	/api/v1/settings/{key}	Bearer token	all of: settings.manage	

Bulk import (api/import.php, 2 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/import/templates/{kind}	Bearer token	—	
POST	/api/v1/import/{kind}	Bearer token	—	

App releases (api/releases.php, 5 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/app/version	Public	—	
GET	/api/v1/releases	Bearer token	all of: releases.manage	
POST	/api/v1/releases	Bearer token	all of: releases.manage	
DELETE	/api/v1/releases/{id}	Bearer token	all of: releases.manage	
PATCH	/api/v1/releases/{id}	Bearer token	all of: releases.manage	

In-app guidance (api/guidance.php, 12 routes)

Method	Path	Auth	Permissions	Gates
GET	/api/v1/guidance	Optional	—	
POST	/api/v1/guidance/admin	Bearer token	—	
GET	/api/v1/guidance/admin/all	Bearer token	—	
DELETE	/api/v1/guidance/admin/{id}	Bearer token	—	
PATCH	/api/v1/guidance/admin/{id}	Bearer token	—	
POST	/api/v1/guidance/admin/{id}/publish	Bearer token	—	
POST	/api/v1/guidance/admin/{id}/retire	Bearer token	—	
GET	/api/v1/guidance/contextual	Optional	—	
GET	/api/v1/guidance/progress	Bearer token	—	
POST	/api/v1/guidance/progress	Bearer token	—	
POST	/api/v1/guidance/progress/reset	Bearer token	—	
GET	/api/v1/guidance/{idOrSlug}	Optional	—	

Telegram webhook / linking (api/telegram.php, 1 routes)

Method	Path	Auth	Permissions	Gates
POST	/api/v1/telegram/webhook	Public	—	

Public web routes (no /api prefix) (web/public.php, 8 routes)

Method	Path	Auth	Permissions	Gates
GET	/.well-known/assetlinks.json	Public	—	

Method	Path	Auth	Permissions	Gates
GET	/api/openapi.json	Public	—	
GET	/download	Public	—	
GET	/download/android/{code}.apk	Public	—	
GET	/download/assets/logo.jpg	Public	—	
GET	/files/{path}	Public	—	
GET	/open/{type}/{id}	Public	—	
GET	/{kind}/{id}	Public	—	

Public web routes (no /api prefix) (web.php, 2 routes)

Method	Path	Auth	Permissions	Gates
GET	/	Public	—	
GET	/health	Public	—	

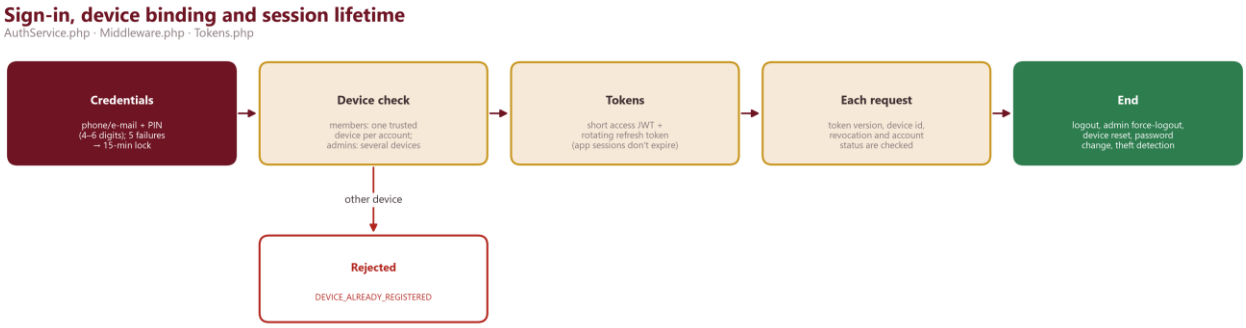
Framework: local-disk file serving (Laravel 'serve' option of the local disk) (vendor/laravel/framework/src/Illuminate/Filesystem/FilesystemServiceProvider.php, 2 routes)

Method	Path	Auth	Permissions	Gates
GET	/storage/{path}	Public	—	
PUT	/storage/{path}	Public	—	

24. Important workflows (diagrams)

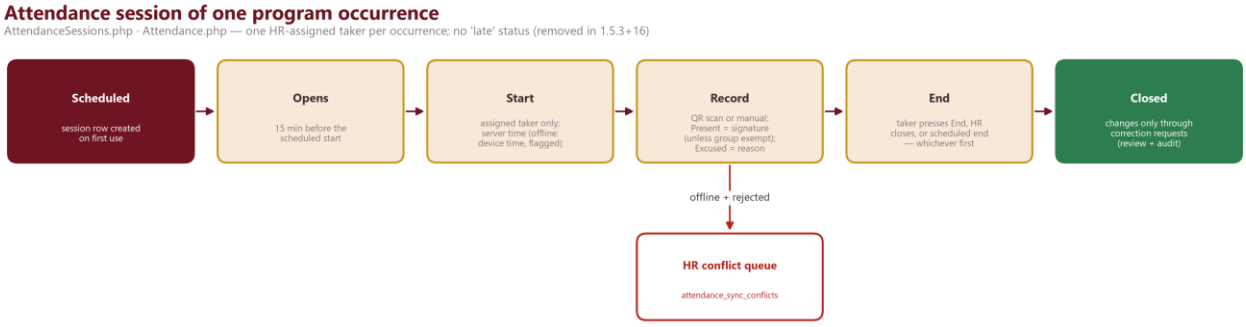
The following diagrams are drawn from the verified code paths named in each subtitle; editable SVG versions are in `diagrams/`.

Sign-in and device binding



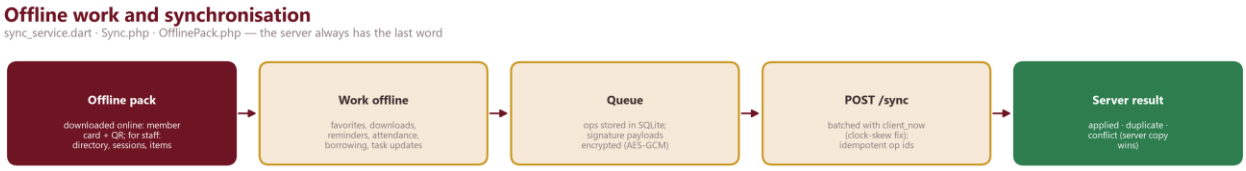
- Forgotten PIN: a 6-digit code (15 minutes, 5 attempts) is sent by Telegram when linked, otherwise by e-mail; the response never reveals whether the account exists.
- Accounts may be pending approval or disabled; members can be registered without sign-in rights (can_login = false).

Attendance session



- Absent is automatic for expected members who were not marked; each mark stores capture time and seconds elapsed since Start.
- Offline marks replay through POST /sync; marks the server cannot apply go to the HR conflict queue (accept / reject with a note) — never dropped.

Offline sync



- Attendance captures older than 3 days are refused (CAPTURE_TOO_OLD); refused attendance goes to the HR review queue with its signature.
- GET responses are cached per user for read-only offline display. Admin writes outside the listed operations need a connection.

Holiday selection

Holiday Mezmur service selection lifecycle

Selections.php — status transitions and the permission each requires (verified in the TRANSITIONS table)

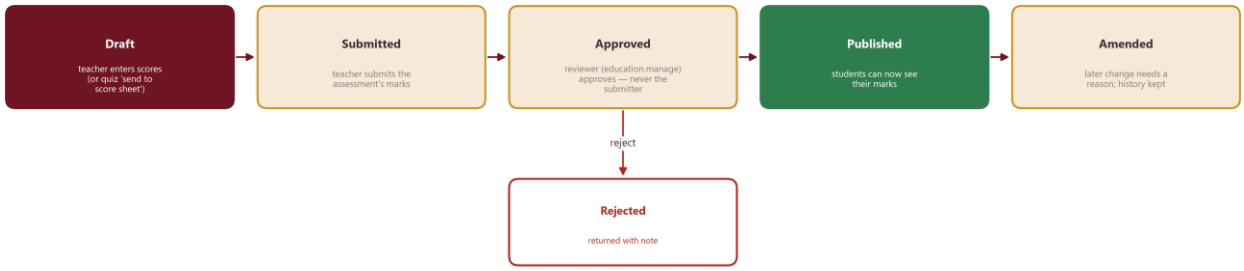


- Score = 5 weighted components (defaults: class attendance 25, exams 25, Mezmur attendance 30, contribution 10, department participation 10; pass mark 43).
- Components with no data are excluded and the total is rescaled; each counted component must also reach its minimum. Cancel is possible from draft, open or review.

Marks workflow

Education marks workflow

Edu.php · EduMarks.php — only ACTIVE course teachers enter marks; no self-approval; every change kept in edu_score_history

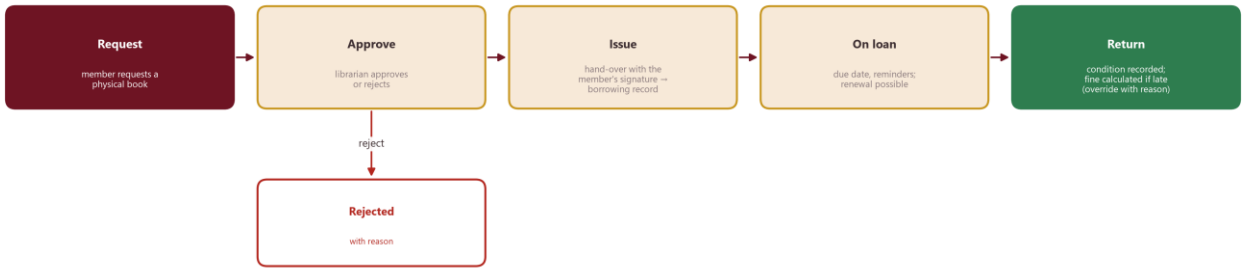


- Rejected marks go back to the teacher with a review note. Selection exam scoring counts only approved / published marks.
- Quiz scores linked to an official assessment stay hidden from students until the official mark is published.

Library and borrowing

Library book request and borrowing

Library.php · Borrowings.php · Fines.php

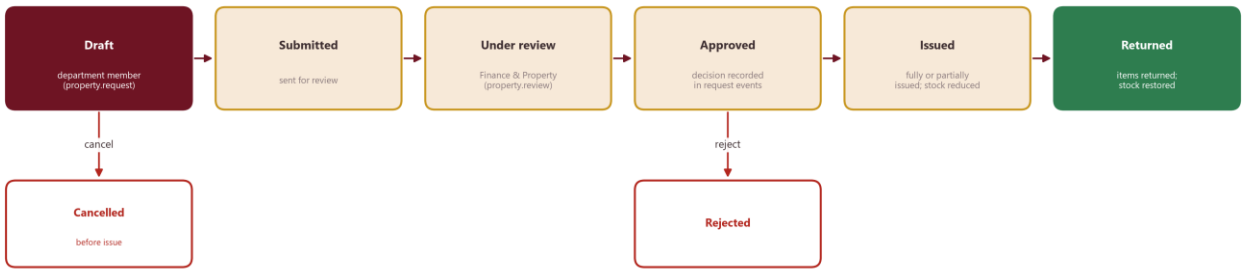


- Digital books (PDF) are downloaded by any signed-in member, stored encrypted on the phone and opened only in the in-app reader.
- Direct borrowing (without a request) follows the borrowing.direct policy and per-item settings; some borrowings need borrowing.approve.

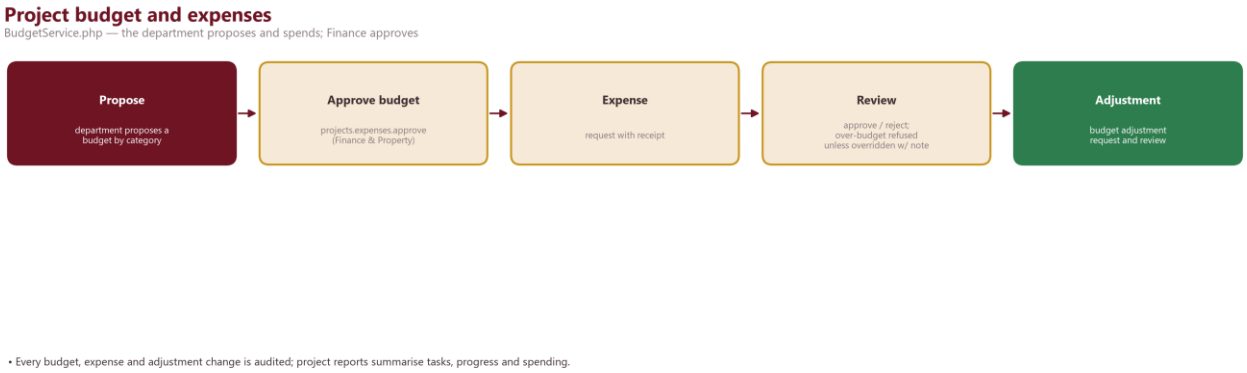
Property requests

Department property request

PropertyRequests.php — stock moves in the same database transaction as the status change



Project budget and expenses



- Every budget, expense and adjustment change is audited; project reports summarise tasks, progress and spending.

25. Security, privacy, authorisation, data validation and file-upload protections

- **Least privilege.** 63 fine-grained permissions; per-user revocations override role grants; the admin UI hides modules without permission and the server enforces every check independently; row-level checks in services.
- **Authentication hardening.** Argon2id hashing, constant-time behaviour for unknown accounts, lockout, IP rate limits, device binding for members, token versioning, per-session revocation, refresh-token reuse detection, uniform responses from password recovery.
- **Encryption at rest.** Sensitive columns (membership form personal fields, chat bodies and meeting links, notification bodies, notes, reasons, review notes, complaint descriptions, IP addresses and user agents) are sealed with an application key on write and opened on read; keys come from the environment (DATA_ENCRYPTION_KEY with rotation support via DATA_ENCRYPTION_OLD_KEYS). Back up the key with the database: without it encrypted data cannot be read. Identifiers, names, phone and e-mail remain plain so that sign-in and search work.
- **Device-side protection.** Secure storage for tokens; AES-256-GCM encryption for downloaded books and queued signatures; FLAG_SECURE on personal screens; the offline directory never contains phone numbers.
- **Validation.** Strict schemas on every route; enumerations for statuses; phone normalisation; length limits; error codes per field.
- **File uploads.** Size limits per kind, declared-type pre-check, byte-level type detection, private storage for personal/internal files, path traversal rejected on public file routes, nosniff and sandbox CSP headers, private no-store caching for sensitive downloads.
- **HTTP headers.** Helmet-equivalent security headers on every response (CSP permitting the Flutter CanvasKit/WASM runtime), `X-Powered-By` removed, CORS restricted to `CORS_ORIGINS`.
- **Privacy in design.** Students see only published marks; special-event programs are not shown on public share pages; links in e-mails to private entities open the app and re-check access; the hidden authority's actions are attributed without revealing identity.
- **Auditability.** Sensitive administrative actions and every export are written to `audit_logs`.
- **Open security concerns (section 31):** plaintext credentials in local project files; production SQL dump inside the project folder; `APP_DEBUG` must be false in production (*requires verification* on the server).

26. Installation, environment configuration, deployment and hosting

26.1 Requirements

Component	Requirement
Server	PHP 8.2+ with pdo_mysql, mbstring, openssl, sodium/argon2, gd or imagick (dompdf), zip, fileinfo; Composer
Database	MariaDB 10.4+ or MySQL 8 (utf8mb4)
Web server	Apache/LiteSpeed (cPanel) or Nginx with document root = `laravel/public`
Client build	Flutter 3.44.x / Dart 3.12, Android SDK, JDK 17

26.2 Environment variables (names only)

Area	Variables
Application	APP_ENV, APP_KEY, APP_DEBUG (false in production), APP_URL, APP_TIMEZONE (Africa/Addis_Ababa), PUBLIC_BASE_URL, CORS_ORIGINS, STORAGE_PATH
Database	DB_CONNECTION=mysql, DB_HOST, DB_PORT, DB_DATABASE, DB_USERNAME, DB_PASSWORD
Tokens	JWT_SECRET, JWT_REFRESH_SECRET, JWT_ACCESS_TTL, JWT_REFRESH_TTL_DAYS (0 = no expiry), JWT_ISSUER, JWT_AUDIENCE
Encryption	DATA_ENCRYPTION, DATA_ENCRYPTION_KEY, DATA_ENCRYPTION_OLD_KEYS
Uploads	MAX_IMAGE_MB, MAX_AUDIO_MB, MAX_DOCUMENT_MB, MAX_BOOK_MB, MAX_APK_MB
Integrations	FIREBASE_, *SMTP_*, TELEGRAM_*, ANDROID_PACKAGE, ANDROID_CERT_SHA256
Jobs	JOBS_ENABLED
Restricted	System-console variables — see Appendix F

26.3 First installation

```
composer install --no-dev --optimize-autoloader
cp .env.production.example .env   # then fill in values (see 26.2)
php artisan key:generate
php artisan mz:encrypt-data --generate-key   # store DATA_ENCRYPTION_KEY in .env AND a safe place
php artisan migrate --force
php artisan db:seed --class=GuidanceSeeder --force
php artisan storage:link
php artisan optimize:clear
```

Note: the baseline migration matches the imported production dump; a brand-new installation from an empty database should be tested before relying on it (requires verification).

26.4 Client builds

```
cd mobile/mobile
flutter pub get && flutter gen-l10n
flutter build apk --release        # build/app/outputs/flutter-apk/app-release.apk
flutter build web --release        # build/web → copied into laravel/public
```

The API base URL defaults to the production host in `core/config/app_config.dart` and can be overridden with `--dart-define`. The web build needs `sqflite_sw.js` and `sqlite3.wasm` next to `index.html`.

26.5 Release routine used by the project

1. Back up the production database (gzip, stored under `~/backups` on the server).
2. Run `deploy.py`: packages backend + web build (+ APK), uploads over SFTP, unzips, runs `migrate --force` and the GuidanceSeeder. It excludes `.env*` and never deletes server files removed locally — delete those by hand.
3. Run `php artisan optimize:clear`; check `/health`, the web root and the API.
4. Register the APK through Admin → App versions (`Releases::publish`: private file, SHA-256, audit); set *mandatory* / *minimum supported build* to force older apps to update.
5. Build the web app only after all source edits are finished (a build fails if sources change mid-compile).

Hosting (from project records): a shared cPanel account with the API and web app on one sub-domain (document root = `public`); the APK download page is `/download` and builds are served from `/download/android/{code}.apk`.

27. Database setup, migrations, backups and recovery

- **Setup.** Create a utf8mb4 database, import the baseline (or the latest backup), then `php artisan migrate --force`. Migrations are additive, idempotent (`hasTable`/`hasColumn`) and reversible; they never delete live rows.
- **Test database.** `mh_app_test` = production dump + migrations; integration tests run inside rolled-back transactions. After pulling unreleased migrations run `DB_DATABASE=mh_app_test php artisan migrate --force` before the integration suite.
- **Backups.** Before every release a compressed dump is taken on the server; the project records list them by release (e.g. before 1.5.3+16). Back up the storage root (private files: signatures, books, lessons, chat, APKs) and the `.env` keys together with the database.
- **Recovery.** Restore the dump and the storage root, restore the same `APP_KEY` / `DATA_ENCRYPTION_KEY` / JWT secrets, run `php artisan migrate --force` (applies only missing migrations) and `optimize:clear`. A migration can be rolled back with `php artisan migrate:rollback --step=1` (each migration implements `down()`).
- **Text repair.** `php artisan mz:repair-text <dump> [--dry-run]` restores Amharic values saved as '?' in items/borrowings from a dump (historical latin1 issue).
- **Re-keying.** `php artisan mz:encrypt-data --dry-run | --rotate | --decrypt` manages encrypted columns; always back up first.
- *No automated off-server backup schedule was found in the code (requires verification with the hosting provider).*

28. Background jobs, scheduled processes, queues and notification delivery

Jobs run **without cron or a queue worker**: after each HTTP response, at most once per 60 s, one request acquires the MySQL named lock `mz:jobs:tick` and runs the due jobs; the last run of each job is kept in `job_runs`. `php artisan mz:jobs [--list] [names] [--force]` runs them by hand. Because jobs depend on traffic, a site with no requests runs no jobs — add an external ping or a cron calling `php artisan mz:jobs` for guaranteed timing.

Job	Schedule	Work
notifications:dispatch	every 60 s	Send pending push / e-mail deliveries
scheduled:dispatch	every 60 s	Release due scheduled notifications (task deadlines, etc.)
programs:reminders	every 5 min	Program offset and interval reminders
borrowings:reminders	every 30 min	Due and overdue borrowing reminders
fines:recalculate	daily 00:30	Recalculate fines of open borrowings
choir:eligibility	daily 02:15	Recompute choir eligibility
cleanup	daily 03:00	Purge old revoked tokens, resets, sync receipts, old read notifications, stale FCM tokens; mark finished programs completed
telegram:webhook / telegram:updates	daily / every 60 s	Register the webhook, or poll updates when no webhook secret is set

Attendance sessions need no job: the scheduled end is applied whenever a session is read. Selection results of open selections are recomputed after relevant writes (deferred to the end of the request).

29. Logging, troubleshooting, known errors and maintenance

Symptom	Likely cause / action
Members 'logged out by themselves'	Fixed in 1.5.3+12–14 (refresh failures no longer sign out; sessions don't expire). If it recurs check `JWT_REFRESH_TTL_DAYS=0` on the server and the device-binding rules.
Amharic shows as ????	Table not utf8mb4 when written; run the utf8mb4 migration and `mz:repair-text`; re-enter values written after the import.
Reminders / pushes not sent	No traffic (jobs are request-driven) or FCM not configured; see Admin → Notification delivery and `mz:jobs --list`.
Removed backend file still active	deploy.py does not delete server files; remove them on the server and `route:clear`.
500 on education endpoints after updating code	New migrations not applied: `php artisan migrate --force`.
Audit-log PDF export fails	Memory exhaustion on very large ranges; narrow the period or use XLSX/CSV.
Web build fails	Sources changed during compile; rebuild after edits finish.
Offline attendance missing	Check the HR conflict queue (`/admin/attendance/conflicts`); captures > 3 days old are refused.

Logs: Laravel log (`storage/logs`), job results in `job_runs.last_result`, audit and activity logs in the database. The app logs overflow errors in debug builds; `/health` returns `{"status":"ok"}` when the database answers.

30. Testing approach and verified test coverage

Both suites were executed on 2026-10-16 on the development machine against the current source tree (1.5.3+19):

Suite	Result	Notes
Flutter (`flutter test`, 24 test files)	**176 passed, 12 skipped, 0 failed**	Skipped = screenshot renderers that run only with `--dart-define=SCREENSHOTS=true` (run separately, they produced the screenshots in this document).
Laravel PHPUnit (`php vendor/bin/phpunit`)	**119 tests: 82 passed, 34 failures, 3 errors** (581 assertions)	All 37 failing tests are Integration tests touching Education/attendance. Their errors trace to the test database missing the two unreleased migrations (table `edu_academic_years` missing; column `config` missing). Unit tests all passed. The last fully green run recorded in the project log is 98 tests / 1,131 assertions at release 1.5.3+16.

Coverage areas (by test file): authentication/sessions/permissions, guidance, module access and security, selections, PDF export, program reminders, attendance sessions and regressions (Mezmur playback, borrowing signatures, contributions, finalisation), education workflow, teacher assignment, learning platform, exports, wedding selection, Amharic items; Flutter: offline resilience and encrypted queue, attendance session card, Ethiopian calendar, time zones, library/registration, notification centre, projects, members, guidance, overflow checks at 360 dp and 130 % text.

To re-establish a green backend run: apply the unreleased migrations to `mh_app_test` (`DB_DATABASE=mh_app_test php artisan migrate --force`) and re-run. This was not done during documentation, to avoid altering the shared test database while development work is in progress.

31. Current limitations, technical debt, incomplete features and risks

#	Item	Impact	Recommendation
1	Plaintext credentials (server, database) in local project files (`.env-ssh`, `deploy.py`, `MEMORY.md`)	High — anyone with the folder can access production	Rotate all credentials; move to environment variables or a secrets manager; remove from files and backups
2	Production SQL dump (`hersolot_mh_prod.sql`) and archives inside the project folder	High — personal data exposure	Move to encrypted storage outside the project; restrict access
3	Unreleased schema changes in the source tree (F0a, F1–F6, builds +17…+19); test DB not migrated (37 failing integration tests)	Medium — risk of deploying untested changes	Migrate the test DB, get a green run, release per plan with backups
4	Request-driven jobs on shared hosting	Medium — reminders late when traffic is low	Add a cron/uptime ping calling `php artisan mz:jobs`
5	deploy.py does not remove deleted files	Medium — stale routes/classes remain live	Use rsync-style sync or a release-folder deploy
6	Offline mode limited to listed operations; admin writes need a connection	Medium	F7 plan
7	Education gaps: outcomes/pass marks, retakes, certificates, engagement scores, assignment deadlines, question banks	Functional gaps	F9–F15 plan; decisions D7, D8 needed
8	Physical library quantity-based (no per-copy IDs)	Low/Medium	F17, decision D6
9	Selection service bonus cannot add points under the current rule	Business-rule ambiguity	Decide the intended rule
10	Unused tables from withdrawn department criteria	Low	Keep for history or drop after archiving
11	Permission keys referenced but not in the catalogue (`education.view`, `programs.view`); `guidance.manage` only in DB	Low	Align Catalog.php
12	Audit-log PDF export memory use	Low	Stream/paginate PDF generation
13	iOS build not verified; OpenAPI file may be outdated	Low	Verify or remove
14	Environment example lists legacy variable names	Low	Update `.env.production.example`

32. Recommended future improvements (proposals, not existing functionality)

- Complete the approved roadmap in order: +17 (F0a, F5) → +18 (F1–F3) → +19 (F4, F6) → +20 platform-wide offline (F7) → +21 results pipeline (F8–F12) → +22 certificates, dashboard, scores (F13–F15) → +23 libraries (F16–F17) → +24 settings, edit pen, guidance, responsive polish (F18–F21).
- Introduce secrets management and credential rotation; add automated off-site database and storage backups with periodic restore tests.
- Add a real scheduler (cron) for jobs; consider a queue for e-mail/push at scale.
- Add CI: run Flutter and PHPUnit suites on every change against a freshly migrated database; static analysis (Pint, flutter analyze) as gates.
- Monitoring: uptime checks on /health, error tracking, and alerting on job failures recorded in job_runs.
- Refresh the OpenAPI specification from the Laravel routes and publish it for integrators.
- Consider an iOS release once Apple developer access and push certificates are available.

33. Glossary

Term	Meaning
መዝሙር (Mezmur)	Spiritual hymn of the Ethiopian Orthodox Tewahedo Church
ወረብ (Wereb)	Liturgical chant/dance service; one of the two holiday service categories
Selection (holiday service selection)	The scoring process choosing members for holiday Mezmur service
Occurrence	One dated instance of a (possibly recurring) program
Attendance taker	The one person assigned by HR to record attendance for an occurrence
Session	The attendance window of one occurrence (scheduled → active → ended)
Excuse (permission request)	A member's request to be excused from an occurrence
Correction	A reviewed change to attendance after the session closed
Member group	Member type (children, youth & adult, married, distance...) used for targeting and rules
Period	Ethiopian month `YYYY-MM` used for contributions
Pagume (ጳጉሜ)	13th Ethiopian month (5–6 days); counted with Nehase for payments
Offline pack	Data downloaded to a phone for offline work
Sync queue	Operations made offline and replayed later through /sync
Seal / encrypted column	A database column stored encrypted at rest
Module gate	Server-side switch that disables the Education module for some members
Release / build	App version `1.5.3+N`; N is the build number used for forced updates

Appendix A. Screen and route map of the admin panel

`/admin` sub-routes: users, users/:id, mezmurs, mezmurs/edit/:id, categories, ads, pages, pages/edit/:id, programs, programs/edit/:id, programs/:id/participants, programs/:id/attendance, attendance, attendance/corrections, attendance/rules (signature policy), attendance/conflicts, permissions (excuses), import, departments, payments, selections, member-ids, member-groups, weddings, property-requests, library-requests, borrowings/direct-policy, program-categories, change-requests, releases, notify, notify/:id, notification-health, items, borrowings, borrowings/new, borrowings/bulk, borrowings/:id, fines, reports, audit, activity, admins, admins/:id, roles, settings, guidance, chat, chat/:id, chat/:id/private, chat/:id/user/:uid.

Appendix B. Settings keys

Key	Purpose
fines.policy	Fine amount per day/week, grace days, maximum, currency
borrowing.reminders	Days before due, overdue repeat days, default loan days
borrowing.direct	Member groups allowed to borrow directly, approval, limits
programs.reminder_defaults	Default reminder settings and offsets for new programs
notifications.retention_days	Retention of read notifications (30–3650, default 365)
payments.policy	Currency, start period, grace days
attendance.permissions	Excuse requests: enabled, deadline hours, monthly/quarterly limits, weight
attendance.signature	Member groups exempt from signatures
participation.policy	Maximum monthly participation score
choir.policy	Choir eligibility weights, minimums, window and sources
member_id.format	Member ID prefixes and display style
modules.policy	Per-group module access (Education switch)

Appendix C. Data dictionary

Columns as read from `information_schema` (PK = primary key, UNI = unique, MUL = indexed). Foreign keys follow each table.

`users`

Every account (members and administrators): identity, sign-in credentials (hashed), status, trusted device, member number and group.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
full_name	varchar(150)	NO	MUL	
phone	varchar(20)	YES	UNI	NULL
email	varchar(190)	YES	UNI	NULL
password_hash	varchar(255)	NO		
profile_image	varchar(255)	YES		NULL
status	enum('active','disabled','pending')	NO		'active'
account_type	enum('user','admin')	NO	MUL	'user'
role_id	int(10) unsigned	YES	MUL	NULL
created_by	int(10) unsigned	YES	MUL	NULL
device_id	varchar(128)	YES	MUL	NULL
device_name	varchar(150)	YES		NULL
device_bound_at	datetime	YES		NULL
token_version	int(10) unsigned	NO		0
failed_login_count	int(10) unsigned	NO		0
locked_until	datetime	YES		NULL
last_login_at	datetime	YES		NULL
password_changed_at	datetime	YES		NULL
must_change_password	tinyint(1)	NO		0
totp_secret_enc	varchar(255)	YES		NULL
created_at	datetime	NO	MUL	current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL
telegram_chat_id	bigint(20)	YES	UNI	NULL
telegram_username	varchar(64)	YES		NULL
telegram_linked_at	datetime	YES		NULL
member_uuid	varchar(36)	YES	UNI	NULL
member_type	enum('student','worker')	YES		NULL
member_group	varchar(40)	NO	MUL	'youth_adult'
can_login	tinyint(1)	NO		1
membership_status	enum('active','deceased','inactive','university','abroad')	NO		'active'
membership_status_date	date	YES		NULL
membership_status_detail	text	YES		NULL
spiritual_level	tinyint(3) unsigned	YES		NULL
member_no	int(10) unsigned	YES	UNI	NULL

Foreign keys: `created_by` → `users.id` (on delete set null); `role_id` → `roles.id` (on delete set null)

Unique: `users_email_unique` (email); `users_member_no_unique` (member_no); `users_member_uuid_unique` (member_uuid); `users_phone_unique` (phone); `users_telegram_chat_id_unique` (telegram_chat_id)

`roles`

Administrative roles with a level; department roles carry department_id. One hidden system role exists (restricted).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
slug	varchar(64)	NO	UNI	
name	varchar(120)	NO		
description	varchar(500)	YES		NULL
level	int(11)	NO	MUL	
is_hidden	tinyint(1)	NO		0
is_system	tinyint(1)	NO		0
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
is_active	tinyint(1)	NO		1
department_id	int(10) unsigned	YES	MUL	NULL

Foreign keys: `department_id` → `departments.id` (on delete set null)

Unique: `roles_slug_unique` (slug)

`permissions`

The permission catalogue (63 keys) mirrored from app/Mz/Auth/Catalog.php.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
key	varchar(80)	NO	UNI	
group_key	varchar(40)	NO		
description	varchar(255)	YES		NULL

Unique: `permissions_key_unique` (key)

`role_permissions`

Permissions granted to each role.

Column	Type	Null	Key	Default / extra
role_id	int(10) unsigned	NO	PRI	
permission_id	int(10) unsigned	NO	PRI	

Foreign keys: `permission_id` → `permissions.id` (on delete cascade); `role_id` → `roles.id` (on delete cascade)

`user_permissions`

Direct permission grants to a single account.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
permission_id	int(10) unsigned	NO	PRI	
granted_by	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()

Foreign keys: `granted_by` → `users.id` (on delete set null); `permission_id` → `permissions.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`user_permission_revocations`

Per-account revocations that override role grants.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
permission_id	int(10) unsigned	NO	PRI	
revoked_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `permission_id` → `permissions.id` (on delete cascade); `revoked_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`refresh_tokens`

Rotating refresh tokens (stored as hashes) grouped by session family; device and client metadata.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
token_hash	varchar(64)	NO	UNI	
family_id	varchar(36)	NO	MUL	
scope	enum('app','system')	NO		'app'
device_id	varchar(128)	YES		NULL
expires_at	datetime	NO	MUL	
revoked_at	datetime	YES		NULL
replaced_by	bigint(20) unsigned	YES		NULL
ip	text	YES		NULL
user_agent	text	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

Unique: `refresh_tokens_token_hash_unique` (token_hash)

`revoked_access_tokens`

Access-token ids (jti) revoked before expiry (sign-out of one session).

Column	Type	Null	Key	Default / extra
jti	char(36)	NO	PRI	
user_id	int(10) unsigned	YES		NULL
expires_at	datetime	NO	MUL	

`password_resets`

Hashed 6-digit reset codes with expiry and attempt counter.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
code_hash	varchar(64)	NO		
attempts	int(10) unsigned	NO		0
expires_at	datetime	NO		
used_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

`device_history`

Device bind / login / rejected-login / reset events per account.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
device_id	varchar(128)	YES		NULL
device_name	varchar(150)	YES		NULL
event	enum('bound','login','login_rejected','reset','force_logout')	NO		
actor_id	int(10) unsigned	YES	MUL	NULL
ip	text	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `actor_id` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`user_preferences`

Theme, language, audio, reminder time, lyrics font scale and notification topics.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
notifications_enabled	tinyint(1)	NO		1
notification_topics	text	YES		NULL
audio_autoplay	tinyint(1)	NO		0
download_audio	tinyint(1)	NO		1
theme	enum('system','light','dark')	NO		'system'
reminder_time	varchar(5)	NO		'18:00'
language	varchar(8)	NO		'am'
lyrics_font_scale	int(10) unsigned	NO		100
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

`user_preferred_categories`

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	

Column	Type	Null	Key	Default / extra
category_id	int(10) unsigned	NO	PRI	

Foreign keys: `category_id` → `categories.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`telegram_link_tokens`

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
token_hash	varchar(64)	NO	UNI	
expires_at	datetime	NO		
used_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

Unique: `telegram_link_tokens_token_hash_unique` (token_hash)

`telegram_state`

Column	Type	Null	Key	Default / extra
key	varchar(40)	NO	PRI	
value	bigint(20)	NO		

`fcm_tokens`

Firebase push tokens per device.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
token	varchar(255)	NO	UNI	
device_id	varchar(128)	YES		NULL
platform	enum('android','ios','web')	NO		'android'
last_seen_at	datetime	NO		current_timestamp()
invalidated_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

Unique: `fcm_tokens_token_unique` (token)

`member_profiles`

The membership form (birth, baptism, address, emergency contact, education, profession, confession father...). Many columns are encrypted at rest.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
gender	enum('male','female')	YES		NULL
mother_name	text	YES		NULL
birth_date	date	YES		NULL

Column	Type	Null	Key	Default / extra
birth_diocese	text	YES		NULL
birth_city	text	YES		NULL
birth_woreda	text	YES		NULL
birth_kebele	text	YES		NULL
birth_place_name	text	YES		NULL
christian_name	text	YES		NULL
baptism_church	text	YES		NULL
baptism_date	date	YES		NULL
address_woreda	text	YES		NULL
house_number	text	YES		NULL
phone_home	text	YES		NULL
emergency_name	text	YES		NULL
emergency_phone_home	text	YES		NULL
emergency_phone_mobile	text	YES		NULL
abinet_education	enum('qes','deacon','none','other')	YES		NULL
abinet_other	text	YES		NULL
theology_education	enum('masters','degree','diploma','certificate')	YES		NULL
secular_education	enum('doctorate','masters','degree','diploma','certificate')	YES		NULL
institution_name	text	YES		NULL
field_of_study	text	YES		NULL
employer_type	enum('private','government')	YES		NULL
profession_id	int(10) unsigned	YES	MUL	NULL
profession_category	varchar(100)	YES	MUL	NULL
profession	text	YES		NULL
profession_other	varchar(255)	YES		NULL
experience_profession	text	YES		NULL
church_service	text	YES		NULL
school_name	text	YES		NULL
confession_father_name	text	YES		NULL
confession_father_phone	text	YES		NULL
marital_status	enum('married','single')	YES		NULL
marriage_type	enum('teklil','kurban','other')	YES		NULL
marriage_other	text	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

`member_groups`

Member types (Children, Youth & Adult, Married, distance...) used for targeting, attendance and payments.

Column	Type	Null	Key	Default / extra
key	varchar(40)	NO	PRI	

Column	Type	Null	Key	Default / extra
name	varchar(100)	NO		
name_en	varchar(100)	YES		NULL
description	varchar(500)	YES		NULL
sort_order	int(11)	NO		0
is_active	tinyint(1)	NO		1
built_in	tinyint(1)	NO		0
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

`id_sequences`

Counters for permanent member numbers.

Column	Type	Null	Key	Default / extra
name	varchar(40)	NO	PRI	
value	bigint(20) unsigned	NO		0

`profile_change_requests`

Member requests to change name, phone or profile fields, awaiting approval.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
old_full_name	varchar(150)	NO		
old_phone	varchar(20)	YES		NULL
full_name	varchar(150)	YES		NULL
phone	varchar(20)	YES		NULL
changes	mediumtext	YES		NULL
note	text	YES		NULL
status	enum('pending','approved','rejected','cancelled')	NO	MUL	'pending'
reason	text	YES		NULL
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `reviewed_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`profile_change_history`

Append-only history of applied profile changes (old → new).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
field	varchar(60)	NO		

Column	Type	Null	Key	Default / extra
old_value	mediumtext	YES		NULL
new_value	mediumtext	YES		NULL
source	enum('request','admin','self')	NO		
request_id	int(10) unsigned	YES	MUL	NULL
changed_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `changed_by` → `users.id` (on delete set null); `request_id` → `profile_change_requests.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`professions`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
category	varchar(100)	NO	MUL	
name	varchar(150)	NO		
is_active	tinyint(1)	NO	MUL	1
sort_order	int(11)	NO		0
created_at	timestamp	YES		NULL
updated_at	timestamp	YES		NULL

Unique: `professions_category_name_unique` (category,name)

`departments`

The 7 church departments (Amharic + English names, code prefix).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(150)	NO		
description	varchar(1000)	YES		NULL
is_active	tinyint(1)	NO		1
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
name_en	varchar(150)	YES		NULL
code_prefix	varchar(8)	YES		NULL
sort_order	tinyint(3) unsigned	NO		99

Foreign keys: `created_by` → `users.id` (on delete set null)

`user_departments`

Department membership and role label of each member.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
department_id	int(10) unsigned	NO	MUL	
assigned_by	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
role	enum('member','leader')	NO		'member'

Foreign keys: `assigned_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `user_departments_user_id_department_id_unique` (user_id,department_id)

department_admins

Department administrators and their department role.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
department_id	int(10) unsigned	NO	MUL	
role_id	int(10) unsigned	NO	MUL	
assigned_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `assigned_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete cascade); `role_id` → `roles.id` (on delete restrict); `user_id` → `users.id` (on delete cascade)

Unique: `department_admins_user_id_department_id_role_id_unique` (user_id,department_id,role_id)

department_participation

Monthly participation score per member per department (recorded by the department leader).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
period	varchar(7)	NO		
note	text	YES		NULL
recorded_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
score	decimal(4,1)	YES		NULL

Foreign keys: `department_id` → `departments.id` (on delete cascade); `recorded_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `department_participation_department_id_user_id_period_unique` (department_id,user_id,period)

department_participation_history

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	MUL	
period	varchar(7)	NO		
old_score	decimal(4,1)	YES		NULL
new_score	decimal(4,1)	NO		
note	text	YES		NULL
changed_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `changed_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`mezmurs`

Hymns: title, lyrics, optional LRC timing, audio/cover, category, status, counters, content version.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
title	varchar(255)	NO	MUL	
lyrics	mediumtext	NO		
lrc	mediumtext	YES		NULL
description	varchar(2000)	YES		NULL
category_id	int(10) unsigned	YES	MUL	NULL
cover_image	varchar(255)	YES		NULL
audio_path	varchar(255)	YES		NULL
audio_mime	varchar(60)	YES		NULL
audio_size	int(10) unsigned	YES		NULL
audio_duration_sec	int(10) unsigned	YES		NULL
status	enum('draft','published','archived')	NO	MUL	'draft'
is_featured	tinyint(1)	NO		0
view_count	int(10) unsigned	NO		0
favorite_count	int(10) unsigned	NO		0
download_count	int(10) unsigned	NO		0
content_version	int(10) unsigned	NO		1
published_at	datetime	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO	MUL	current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL
youtube_url	varchar(255)	YES		NULL
study_on	date	YES	MUL	NULL

Foreign keys: `category_id` → `categories.id` (on delete set null); `created_by` → `users.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

`categories`

Hierarchical Mezmur categories (parent_id, depth), with images and 'today' highlighting (study_on).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
parent_id	int(10) unsigned	YES	MUL	NULL
name	varchar(150)	NO		
description	varchar(1000)	YES		NULL
image	varchar(255)	YES		NULL
icon	varchar(60)	YES		NULL
display_order	int(11)	NO		0
is_active	tinyint(1)	NO	MUL	1
depth	int(10) unsigned	NO		0
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
study_on	date	YES	MUL	NULL

Foreign keys: `created_by` → `users.id` (on delete set null); `parent_id` → `categories.id` (on delete restrict); `updated_by` → `users.id` (on delete set null)

`tags`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(80)	NO	UNI	

Unique: `tags_name_unique` (name)

`mezmur_tags`

Column	Type	Null	Key	Default / extra
mezmur_id	int(10) unsigned	NO	PRI	
tag_id	int(10) unsigned	NO	PRI	

Foreign keys: `mezmur_id` → `mezmurs.id` (on delete cascade); `tag_id` → `tags.id` (on delete cascade)

`favorites`

Member favourite Mezmurs.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
mezmur_id	int(10) unsigned	NO	PRI	
created_at	datetime	NO		current_timestamp()

Foreign keys: `mezmur_id` → `mezmurs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`downloads`

Mezmurs downloaded for offline use (with or without audio).

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	

Column	Type	Null	Key	Default / extra
mezmur_id	int(10) unsigned	NO	PRI	
with_audio	tinyint(1)	NO		0
content_version	int(10) unsigned	NO		1
downloaded_at	datetime	NO		current_timestamp()

Foreign keys: `mezmur_id` → `mezmurs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`advertisements`

Home banners linked to pages or programs, scheduled by date.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
kind	enum('advertisement','event','announcement','campaign','info')	NO		'advertisement'
title	varchar(255)	NO		
short_description	varchar(500)	YES		NULL
image	varchar(255)	YES		NULL
starts_at	datetime	YES		NULL
ends_at	datetime	YES		NULL
display_order	int(11)	NO	MUL	0
is_active	tinyint(1)	NO	MUL	1
page_id	int(10) unsigned	YES	MUL	NULL
program_id	int(10) unsigned	YES	MUL	NULL
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null); `page_id` → `pages.id` (on delete set null); `program_id` → `programs.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

`ad_attachments`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
ad_id	int(10) unsigned	NO	MUL	
kind	enum('file','link')	NO		
title	varchar(255)	YES		NULL
path	varchar(255)	YES		NULL
url	varchar(1000)	YES		NULL
mime	varchar(100)	YES		NULL
size	int(10) unsigned	YES		NULL
display_order	int(11)	NO		0
created_at	datetime	NO		current_timestamp()

Foreign keys: `ad_id` → `advertisements.id` (on delete cascade)

pages

Internal content pages with call-to-action and optional program link.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
title	varchar(255)	NO		
summary	varchar(500)	YES		NULL
cover_image	varchar(255)	YES		NULL
content	mediumtext	NO		
event_date	datetime	YES		NULL
location	varchar(255)	YES		NULL
cta_label	varchar(80)	YES		NULL
cta_action	varchar(30)	NO		'none'
cta_target_id	int(10) unsigned	YES		NULL
cta_url	varchar(500)	YES		NULL
program_id	int(10) unsigned	YES	MUL	NULL
status	enum('draft','published')	NO	MUL	'draft'
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null); `program_id` → `programs.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

page_images

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
page_id	int(10) unsigned	NO	MUL	
path	varchar(255)	NO		
caption	varchar(255)	YES		NULL
display_order	int(11)	NO		0

Foreign keys: `page_id` → `pages.id` (on delete cascade)

programs

Church programs/events: schedule, recurrence, audience, registration, attendance and reminder settings.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
title	varchar(255)	NO		
description	text	YES		NULL
image	varchar(255)	YES		NULL
location	varchar(255)	YES		NULL
starts_at	datetime	NO	MUL	
ends_at	datetime	NO	MUL	
has_end_time	tinyint(1)	NO		1

Column	Type	Null	Key	Default / extra
requires_registration	tinyint(1)	NO		0
requires_attendance	tinyint(1)	NO		1
capacity	int(10) unsigned	YES		NULL
audience	enum('all','participants')	NO		'all'
status	enum('draft','published','cancelled','completed')	NO	MUL	'draft'
reminder_enabled	tinyint(1)	NO		1
is_mandatory_reminder	tinyint(1)	NO		0
reminder_days_before	int(10) unsigned	NO		1
reminder_title	varchar(255)	YES		NULL
reminder_body	varchar(1000)	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
event_type	varchar(40)	YES		NULL
category_id	int(10) unsigned	YES	MUL	NULL
recurrence	enum('once','weekly','monthly','twice_monthly','ethiopian_days')	NO	MUL	'once'
monthly_rule	enum('day_of_month','nth_weekday')	YES		NULL
recurrence_until	date	YES		NULL
member_groups	varchar(1000)	YES		NULL
ethiopian_days	varchar(60)	YES		NULL
is_additional	tinyint(1)	NO		0
counts_for_selection	tinyint(1)	NO		1
on_time_minutes	smallint(5) unsigned	YES		NULL
late_minutes	smallint(5) unsigned	YES		NULL
reminder_offsets	varchar(60)	YES		NULL
reminder_interval_days	smallint(5) unsigned	YES		NULL
reminder_interval_start	date	YES		NULL
reminder_interval_last_at	datetime	YES		NULL

Foreign keys: `category_id` → `program_categories.id` (on delete set null); `created_by` → `users.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

`program_categories`

Program types with colour, default reminder offsets and repeat interval.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(120)	NO		
name_en	varchar(120)	YES		NULL
color	varchar(9)	YES		NULL
reminder_offsets	varchar(60)	YES		NULL
reminder_interval_days	smallint(5) unsigned	YES		NULL
display_order	int(11)	NO		0

Column	Type	Null	Key	Default / extra
is_active	tinyint(1)	NO		1
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

`program_participants`

Invited / registered participants of participant-only programs.

Column	Type	Null	Key	Default / extra
program_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
status	enum('invited','registered','cancelled')	NO		'registered'
source	enum('self','admin')	NO		'self'
added_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `added_by` → `users.id` (on delete set null); `program_id` → `programs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`program_attendance_takers`

The one default attendance taker of each program.

Column	Type	Null	Key	Default / extra
program_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
added_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `added_by` → `users.id` (on delete set null); `program_id` → `programs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `pat_one_taker` (program_id)

`program_occurrence_overrides`

Cancelled or rescheduled occurrences of recurring programs.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
program_id	int(10) unsigned	NO	MUL	
occurrence_date	date	NO		
action	enum('cancelled','rescheduled')	NO		
starts_at	datetime	YES		NULL
ends_at	datetime	YES		NULL
reason	varchar(500)	YES		NULL
changed_by	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `changed_by` → `users.id` (on delete set null); `program_id` → `programs.id` (on delete cascade)

Unique: `program_occurrence_overrides_program_id_occurrence_date_unique` (program_id,occurrence_date)

`program_reminder_choices`

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
program_id	int(10) unsigned	NO	PRI	
offsets	varchar(60)	NO		''
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `program_id` → `programs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

`reminders`

Personal reminders (local-first, synced) for programs, borrowings or free text.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
program_id	int(10) unsigned	YES	MUL	NULL
client_uuid	varchar(36)	NO		
title	varchar(255)	NO		
note	varchar(500)	YES		NULL
remind_at	datetime	NO		
repeat	enum('none','daily','weekly')	NO		'none'
is_enabled	tinyint(1)	NO		1
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL
borrowing_id	int(10) unsigned	YES	MUL	NULL

Foreign keys: `borrowing_id` → `borrowings.id` (on delete cascade); `program_id` → `programs.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `reminders_user_id_client_uuid_unique` (user_id,client_uuid); `reminders_user_id_program_id_remind_at_unique` (user_id,program_id,remind_at)

`attendance`

One mark per member per program occurrence: status, capture time, elapsed seconds, method, signature link, sync metadata.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
program_id	int(10) unsigned	NO	MUL	

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	MUL	
status	enum('present','late','absent','excused')	NO		'present'
attended_at	datetime	YES		NULL
client_captured_at	datetime	YES		NULL
received_at	datetime	YES		NULL
elapsed_seconds	int(11)	YES		NULL
time_source	enum('server','device','correction')	YES		NULL
notes	text	YES		NULL
recorded_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
occurrence_date	date	NO		
session_id	int(10) unsigned	YES	MUL	NULL
via	enum('qr','manual')	YES		NULL
signature_id	int(10) unsigned	YES		NULL
signature_exempt	tinyint(1)	NO		0
source	enum('online','offline')	YES		NULL
client_event_id	char(36)	YES	UNI	NULL

Foreign keys: `program_id` → `programs.id` (on delete cascade); `recorded_by` → `users.id` (on delete set null); `updated_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `attendance_client_event` (client_event_id); `attendance_occurrence_unique` (program_id,occurrence_date,user_id)

`attendance_sessions`

One session per program occurrence: taker, status, start/end times and sources.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
program_id	int(10) unsigned	NO	MUL	
occurrence_date	date	NO		
taker_id	int(10) unsigned	YES	MUL	NULL
assigned_by	int(10) unsigned	YES		NULL
assigned_at	datetime	YES		NULL
status	enum('scheduled','active','ended','cancelled')	NO		'scheduled'
started_at	datetime	YES		NULL
started_by	int(10) unsigned	YES		NULL
start_source	enum('server','offline')	YES		NULL
client_started_at	datetime	YES		NULL
client_uuid	char(36)	YES	UNI	NULL
ended_at	datetime	YES		NULL
ended_by	int(10) unsigned	YES		NULL
end_reason	enum('taker','scheduled_end','hr')	YES		NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `program_id` → `programs.id` (on delete cascade); `taker_id` → `users.id` (on delete set null)

Unique: `attendance_sessions_client` (client_uuid); `attendance_sessions_occurrence` (program_id,occurrence_date)

`attendance_signatures`

Signature images (private files, SHA-256) captured for Present marks.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
attendance_id	bigint(20) unsigned	YES	MUL	NULL
user_id	int(10) unsigned	NO		
path	varchar(255)	NO		
sha256	char(64)	NO		
signed_at	datetime	NO		
captured_by	int(10) unsigned	YES		NULL
client_event_id	char(36)	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

`attendance_sync_conflicts`

HR review queue for offline attendance the server could not apply.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
kind	enum('record','session_start','session_end')	NO		
code	varchar(60)	NO		
program_id	int(10) unsigned	NO	MUL	
occurrence_date	date	NO		
user_id	int(10) unsigned	YES		NULL
submitted_by	int(10) unsigned	NO		
client_event_id	char(36)	YES	UNI	NULL
client_captured_at	datetime	YES		NULL
captured_at	datetime	YES		NULL
payload	longtext	YES		NULL
signature_path	varchar(255)	YES		NULL
signature_sha256	char(64)	YES		NULL
status	enum('open','accepted','rejected')	NO	MUL	'open'
resolved_by	int(10) unsigned	YES		NULL
resolved_at	datetime	YES		NULL
resolution_note	varchar(1000)	YES		NULL
attendance_id	bigint(20) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()

Column	Type	Null	Key	Default / extra
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Unique: `attendance_sync_conflicts_event` (client_event_id)

`attendance_closures`

Early closure of an occurrence's attendance.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
program_id	int(10) unsigned	NO	MUL	
occurrence_date	date	NO		
closed_by	int(10) unsigned	YES	MUL	NULL
closed_at	datetime	NO		current_timestamp()

Foreign keys: `closed_by` → `users.id` (on delete set null); `program_id` → `programs.id` (on delete cascade)

Unique: `attendance_closures_program_id_occurrence_date_unique` (program_id,occurrence_date)

`attendance_corrections`

Correction requests after closure, with review outcome.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
program_id	int(10) unsigned	NO	MUL	
occurrence_date	date	NO		
user_id	int(10) unsigned	NO	MUL	
original_status	varchar(20)	NO		
requested_status	enum('present','late','excused','absent')	YES		NULL
requested_attended_at	datetime	YES		NULL
reason	text	NO		
status	enum('pending','approved','rejected')	NO	MUL	'pending'
requested_by	int(10) unsigned	YES	MUL	NULL
reviewed_by	int(10) unsigned	YES	MUL	NULL
review_note	text	YES		NULL
final_status	varchar(20)	YES		NULL
reviewed_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `program_id` → `programs.id` (on delete cascade); `requested_by` → `users.id` (on delete set null); `reviewed_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`permission_requests`

Members' requests to be excused from a program occurrence (approve, reject, cancel).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment

Column	Type	Null	Key	Default / extra
program_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
reason	text	YES		NULL
status	enum('pending','approved','rejected','cancelled')	NO	MUL	'pending'
period_month	varchar(7)	NO		
period_quarter	varchar(7)	NO		
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
review_note	text	YES		NULL
cancelled_by	int(10) unsigned	YES		NULL
cancelled_at	datetime	YES		NULL
cancel_reason	text	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
occurrence_date	date	NO		

Foreign keys: `program_id` → `programs.id` (on delete cascade); `reviewed_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `permreq_occurrence_unique` (program_id,occurrence_date,user_id)

`edu_assessments`

Assessments of a course (max score, weight, date).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
course_id	int(10) unsigned	NO	MUL	
title	varchar(150)	NO		
max_score	decimal(8,2)	NO		100.00
weight	decimal(5,2)	NO		0.00
held_on	date	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `course_id` → `edu_courses.id` (on delete cascade)

`edu_assignments`

Assignments (individual or group) with attachment and due date.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
course_id	int(10) unsigned	NO	MUL	
title	varchar(200)	NO		
instructions	text	YES		NULL
due_at	datetime	YES		NULL
max_score	decimal(8,2)	NO		100.00

Column	Type	Null	Key	Default / extra
is_group	tinyint(1)	NO		0
attachment	varchar(255)	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `course_id` → `edu_courses.id` (on delete cascade); `created_by` → `users.id` (on delete set null)

`edu_attendance`

Legacy class attendance (to be frozen read-only when class attendance moves to programs).

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
class_id	int(10) unsigned	NO	MUL	
session_date	date	NO		
user_id	int(10) unsigned	NO	MUL	
status	enum('present','late','absent','excused')	NO		'present'
via	enum('manual','qr')	NO		'manual'
recorded_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `class_id` → `edu_classes.id` (on delete cascade); `recorded_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `edu_attendance_class_id_session_date_user_id_unique` (class_id,session_date,user_id)

`edu_categories`

Learning categories (Holy Bible, Church History, Liturgy & Order, Hagiography).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(150)	NO		
name_en	varchar(150)	YES		NULL
description	varchar(2000)	YES		NULL
image	varchar(255)	YES		NULL
icon_key	varchar(40)	YES		NULL
sort_order	int(11)	NO		0
status	enum('active','archived')	NO	MUL	'active'
created_by	int(10) unsigned	YES		NULL
updated_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

edu_class_staff

Column	Type	Null	Key	Default / extra
class_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
role	enum('attendance_taker')	NO	PRI	'attendance_taker'

Foreign keys: `class_id` → `edu_classes.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

edu_classes

Classes (term, academic year, semester, department, streams, registration window, capacity).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(150)	NO		
description	varchar(2000)	YES		NULL
term	varchar(80)	YES		NULL
academic_year	varchar(20)	YES		NULL
semester	varchar(40)	YES		NULL
department_id	int(10) unsigned	YES		NULL
streams	varchar(500)	YES		NULL
registration_opens_at	datetime	YES		NULL
registration_closes_at	datetime	YES		NULL
allow_regular	tinyint(1)	NO		1
allow_distance	tinyint(1)	NO		1
capacity	int(10) unsigned	YES		NULL
status	enum('active','archived')	NO		'active'
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

edu_complaint_messages

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
complaint_id	int(10) unsigned	NO	MUL	
author_id	int(10) unsigned	NO		
body	text	NO		
is_internal	tinyint(1)	NO		0
status_change	varchar(30)	YES		NULL
created_at	datetime	NO		current_timestamp()

edu_complaints

Academic complaints with reference EC-YYYY-NNNNNN, private attachment and status.

Column	Type	Null	Key	Default / extra

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
ref	varchar(24)	YES	UNI	NULL
user_id	int(10) unsigned	NO	MUL	
category	enum('teacher','marks','course_admin','enrollment','placement','study_mode','other')	NO		
class_id	int(10) unsigned	YES		NULL
course_id	int(10) unsigned	YES		NULL
assessment_id	int(10) unsigned	YES		NULL
teacher_user_id	int(10) unsigned	YES		NULL
subject	varchar(200)	NO		
description	text	NO		
attachment_path	varchar(255)	YES		NULL
attachment_name	varchar(255)	YES		NULL
status	enum('submitted','under_review','awaiting_information','resolved','rejected','escalated')	NO	MUL	'submitted'
assigned_to	int(10) unsigned	YES		NULL
resolved_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Unique: `edu_complaints_ref` (ref)

`edu_course_enrollments`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
course_id	int(10) unsigned	NO	MUL	
status	enum('added','dropped')	NO		
changed_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Unique: `edu_course_enrollments_unique` (user_id,course_id)

`edu_course_teachers`

Teacher assignments with status history (only active teachers enter marks).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
course_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
status	enum('active','suspended','removed')	NO		'active'
assigned_by	int(10) unsigned	YES		NULL
assigned_at	datetime	NO		current_timestamp()
changed_by	int(10) unsigned	YES		NULL
changed_at	datetime	YES		NULL

Column	Type	Null	Key	Default / extra
note	varchar(500)	YES		NULL

edu_courses

Subjects taught in a class (optionally linked to a curriculum subject).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
class_id	int(10) unsigned	NO	MUL	
subject_id	int(10) unsigned	YES	MUL	NULL
name	varchar(150)	NO		
code	varchar(30)	YES		NULL
description	varchar(2000)	YES		NULL
teacher_user_id	int(10) unsigned	YES	MUL	NULL
progress	tinyint(3) unsigned	NO		0
completed_at	date	YES		NULL
status	enum('active','inactive','archived')	NO		'active'
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `class_id` → `edu_classes.id` (on delete cascade); `teacher_user_id` → `users.id` (on delete set null)

edu_enrollments

Class enrolment with program (regular / distance) and stream.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
class_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
program	enum('regular','distance')	NO		'regular'
stream	varchar(80)	YES		NULL
status	enum('active','withdrawn')	NO		'active'
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `class_id` → `edu_classes.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `edu_enrollments_class_id_user_id_unique` (class_id,user_id)

edu_featured

Featured banner entries on the Education home.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
target_type	enum('subject','lesson','class')	NO		
target_id	int(10) unsigned	NO		
title	varchar(200)	YES		NULL

Column	Type	Null	Key	Default / extra
subtitle	varchar(300)	YES		NULL
image	varchar(255)	YES		NULL
starts_at	datetime	YES		NULL
ends_at	datetime	YES		NULL
sort_order	int(11)	NO		0
is_active	tinyint(1)	NO	MUL	1
created_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

edu_group_members

Column	Type	Null	Key	Default / extra
group_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	

Foreign keys: `group_id` → `edu_groups.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

edu_groups

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
course_id	int(10) unsigned	NO	MUL	
name	varchar(100)	NO		
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `course_id` → `edu_courses.id` (on delete cascade)

edu_journey

Append-only student journey events.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
event	varchar(40)	NO		
class_id	int(10) unsigned	YES		NULL
course_id	int(10) unsigned	YES		NULL
assessment_id	int(10) unsigned	YES		NULL
ref_type	varchar(30)	YES		NULL
ref_id	int(10) unsigned	YES		NULL
old_value	varchar(255)	YES		NULL
new_value	varchar(255)	YES		NULL
actor_id	int(10) unsigned	YES		NULL
reason	text	YES		NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()

edu_lesson_materials

Lesson files (private) or external links (video).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
lesson_id	int(10) unsigned	NO	MUL	
kind	enum('file','link','video_link')	NO		
title	varchar(200)	NO		
path	varchar(255)	YES		NULL
original_name	varchar(255)	YES		NULL
mime	varchar(120)	YES		NULL
size	int(10) unsigned	YES		NULL
sha256	char(64)	YES		NULL
url	varchar(1000)	YES		NULL
sort_order	int(11)	NO		0
uploaded_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `lesson_id` → `edu_lessons.id` (on delete cascade)

edu_lesson_progress

Per-member lesson start/completion and bookmarks.

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
lesson_id	int(10) unsigned	NO	PRI	
started_at	datetime	NO		current_timestamp()
last_opened_at	datetime	NO		current_timestamp()
completed_at	datetime	YES		NULL
bookmarked	tinyint(1)	NO		0

Foreign keys: `lesson_id` → `edu_lessons.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

edu_lessons

Lessons: body, scripture references, objectives, teacher notes, publication status.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
subject_id	int(10) unsigned	NO	MUL	
unit_id	int(10) unsigned	YES	MUL	NULL
title	varchar(200)	NO		
summary	varchar(1000)	YES		NULL
body	longtext	YES		NULL

Column	Type	Null	Key	Default / extra
scripture_refs	varchar(1000)	YES		NULL
objectives	text	YES		NULL
teacher_notes	text	YES		NULL
est_minutes	smallint(5) unsigned	YES		NULL
sort_order	int(11)	NO		0
status	enum('draft','published','archived')	NO		'draft'
published_at	datetime	YES		NULL
author_id	int(10) unsigned	YES		NULL
updated_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `subject_id` → `edu_subjects.id` (on delete cascade); `unit_id` → `edu_units.id` (on delete set null)

edu_quiz_answers

Column	Type	Null	Key	Default / extra
attempt_id	int(10) unsigned	NO	PRI	
question_id	int(10) unsigned	NO	PRI	
response	text	YES		NULL
is_correct	tinyint(1)	YES		NULL
points_awarded	decimal(6,2)	YES		NULL
graded_by	int(10) unsigned	YES		NULL
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `attempt_id` → `edu_quiz_attempts.id` (on delete cascade); `question_id` → `edu_quiz_questions.id` (on delete cascade)

edu_quiz_attempts

Attempts with auto/manual score and transfer to the score sheet.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
quiz_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
attempt_no	tinyint(3) unsigned	NO		
started_at	datetime	NO		current_timestamp()
expires_at	datetime	YES		NULL
submitted_at	datetime	YES		NULL
status	enum('in_progress','submitted','graded')	NO		'in_progress'
auto_score	decimal(8,2)	YES		NULL
manual_score	decimal(8,2)	YES		NULL
max_score	decimal(8,2)	YES		NULL
needs_grading	tinyint(1)	NO		0
transferred_at	datetime	YES		NULL

Foreign keys: `quiz_id` → `edu_quizzes.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `edu_quiz_attempts_unique` (quiz_id,user_id,attempt_no)

edu_quiz_questions

Quiz questions with type, options, answer, points.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
quiz_id	int(10) unsigned	NO	MUL	
type	enum('single','multiple','true_false','short_text')	NO		
prompt	text	NO		
options	text	YES		NULL
answer	text	YES		NULL
explanation	text	YES		NULL
points	decimal(6,2)	NO		1.00
sort_order	int(11)	NO		0

Foreign keys: `quiz_id` → `edu_quizzes.id` (on delete cascade)

edu_quizzes

Quizzes (practice or linked to an assessment), timing, attempts, review policy.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
course_id	int(10) unsigned	NO	MUL	
lesson_id	int(10) unsigned	YES	MUL	NULL
assessment_id	int(10) unsigned	YES	MUL	NULL
title	varchar(200)	NO		
instructions	text	YES		NULL
opens_at	datetime	YES		NULL
closes_at	datetime	YES		NULL
time_limit_minutes	smallint(5) unsigned	YES		NULL
max_attempts	tinyint(3) unsigned	NO		1
scoring	enum('best','last')	NO		'best'
shuffle	tinyint(1)	NO		0
review_policy	enum('never','after_submit','after_close','practice')	NO		'after_close'
status	enum('draft','published','closed')	NO		'draft'
created_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `assessment_id` → `edu_assessments.id` (on delete set null); `course_id` → `edu_courses.id` (on delete cascade); `lesson_id` → `edu_lessons.id` (on delete set null)

edu_requests

Student requests (class/stream/study-mode change, add/drop) applied on approval.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
ref	varchar(24)	YES	UNI	NULL
user_id	int(10) unsigned	NO	MUL	
type	enum('class_change','stream_change','study_mode_change','add_class','drop_class','add_subject','drop_subject','other')	NO		
class_id	int(10) unsigned	YES		NULL
target_class_id	int(10) unsigned	YES		NULL
course_id	int(10) unsigned	YES		NULL
current_value	varchar(120)	YES		NULL
requested_value	varchar(120)	YES		NULL
reason	text	YES		NULL
status	enum('pending','approved','rejected','cancelled','implemented')	NO	MUL	'pending'
reviewed_by	int(10) unsigned	YES		NULL
reviewed_at	datetime	YES		NULL
review_note	text	YES		NULL
implemented_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Unique: `edu_requests_ref` (ref)

`edu_schedule`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
class_id	int(10) unsigned	NO	MUL	
course_id	int(10) unsigned	YES	MUL	NULL
weekday	tinyint(3) unsigned	NO	MUL	
frequency	enum('weekly','monthly')	NO		'weekly'
week_of_month	tinyint(3) unsigned	YES		NULL
program	enum('all','regular','distance')	NO		'all'
mode	enum('in_person','online')	NO		'in_person'
starts	varchar(5)	NO		
ends	varchar(5)	NO		
location	varchar(150)	YES		NULL

Foreign keys: `class_id` → `edu_classes.id` (on delete cascade); `course_id` → `edu_courses.id` (on delete cascade)

`edu_score_history`

Every mark change (old/new score and status, actor, reason).

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
assessment_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO		
action	varchar(20)	NO		
old_score	decimal(8,2)	YES		NULL
new_score	decimal(8,2)	YES		NULL
old_status	varchar(20)	YES		NULL
new_status	varchar(20)	YES		NULL
actor_id	int(10) unsigned	YES		NULL
reason	text	YES		NULL
created_at	datetime	NO		current_timestamp()

edu_scores

Marks with draft → submitted → approved/rejected → published workflow.

Column	Type	Null	Key	Default / extra
assessment_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
score	decimal(8,2)	YES		NULL
remark	text	YES		NULL
status	enum('draft','submitted','approved','rejected','published')	NO	MUL	'draft'
submitted_by	int(10) unsigned	YES		NULL
submitted_at	datetime	YES		NULL
reviewed_by	int(10) unsigned	YES		NULL
reviewed_at	datetime	YES		NULL
review_note	text	YES		NULL
published_by	int(10) unsigned	YES		NULL
published_at	datetime	YES		NULL
recorded_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `assessment_id` → `edu_assessments.id` (on delete cascade); `recorded_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

edu_subjects

Curriculum subjects with visibility (enrolled-only / all) and sequential option.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
category_id	int(10) unsigned	YES	MUL	NULL
name	varchar(150)	NO		
name_en	varchar(150)	YES		NULL
code	varchar(30)	YES		NULL

Column	Type	Null	Key	Default / extra
description	varchar(4000)	YES		NULL
objectives	text	YES		NULL
cover_image	varchar(255)	YES		NULL
visibility	enum('enrolled','members')	NO		'enrolled'
sequential	tinyint(1)	NO		0
status	enum('draft','published','archived')	NO		'draft'
sort_order	int(11)	NO		0
created_by	int(10) unsigned	YES		NULL
updated_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `category_id` → `edu_categories.id` (on delete set null)

`edu_submissions`

Student submissions with grading and feedback.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
assignment_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
group_id	int(10) unsigned	YES	MUL	NULL
text	mediumtext	YES		NULL
file	varchar(255)	YES		NULL
file_name	varchar(255)	YES		NULL
score	decimal(8,2)	YES		NULL
feedback	text	YES		NULL
graded_by	int(10) unsigned	YES	MUL	NULL
graded_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `assignment_id` → `edu_assignments.id` (on delete cascade); `graded_by` → `users.id` (on delete set null); `group_id` → `edu_groups.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `edu_submissions_assignment_id_user_id_unique` (assignment_id,user_id)

`edu_units`

Units inside a subject.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
subject_id	int(10) unsigned	NO	MUL	
title	varchar(200)	NO		
description	varchar(2000)	YES		NULL
sort_order	int(11)	NO		0

Column	Type	Null	Key	Default / extra
status	enum('draft','published','archived')	NO		'published'
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `subject_id` → `edu_subjects.id` (on delete cascade)

`holiday_selections`

Holiday Mezmur service selections: period, member groups, JSON configuration, status, finalisation and publication.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(150)	NO		
from_date	date	NO		
to_date	date	NO		
member_groups	varchar(1000)	YES		NULL
config	longtext	NO		
status	enum('draft','open','review','finalized','cancelled')	NO	MUL	'draft'
is_published	tinyint(1)	NO		0
published_at	datetime	YES		NULL
published_by	int(10) unsigned	YES		NULL
required_departments	text	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL
finalized_by	int(10) unsigned	YES	MUL	NULL
finalized_at	datetime	YES		NULL
computed_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null); `finalized_by` → `users.id` (on delete set null)

`selection_results`

Per-applicant computed result: components, total, qualified, overrides, service category and rank.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
components	longtext	NO		
base_score	decimal(6,2)	NO		
bonus	decimal(5,2)	NO		0.00
total	decimal(6,2)	NO		
qualified	tinyint(1)	NO		
outcome	varchar(30)	YES		NULL
rank	int(10) unsigned	YES		NULL
score_exact	decimal(10,4)	YES		NULL

Column	Type	Null	Key	Default / extra
config_version_id	int(10) unsigned	YES		NULL
failed	longtext	YES		NULL
missing	longtext	YES		NULL
override	enum('qualified','not_qualified')	YES		NULL
override_reason	text	YES		NULL
service_category	varchar(100)	YES		NULL
service_rank	int(10) unsigned	YES		NULL
category_assigned_by	int(10) unsigned	YES		NULL
category_assigned_at	datetime	YES		NULL
overridden_by	int(10) unsigned	YES	MUL	NULL
overridden_at	datetime	YES		NULL
computed_at	datetime	NO		current_timestamp()

Foreign keys: `overridden_by` → `users.id` (on delete set null); `selection_id` → `holiday_selections.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `selection_results_selection_id_user_id_unique` (selection_id,user_id)

`selection_reviews`

Each required department's review decision.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
department_id	int(10) unsigned	YES		NULL
department_name	varchar(150)	YES		NULL
reviewer_id	int(10) unsigned	YES		NULL
decision	varchar(30)	NO		
comments	text	YES		NULL
config_version	varchar(64)	YES		NULL
created_at	timestamp	YES		NULL
updated_at	timestamp	YES		NULL

`selection_feedback`

Members' feedback on a published result.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO		
comments	text	NO		
status	varchar(30)	NO		'submitted'
reviewed_by	int(10) unsigned	YES		NULL
response	text	YES		NULL
created_at	timestamp	YES		NULL
updated_at	timestamp	YES		NULL

selection_config_versions

Created by the withdrawn department-criteria feature (unused); reused by in-development config versioning.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
version	int(10) unsigned	NO		
status	enum('draft','submitted','approved','rejected','superseded')	NO		'draft'
allocation	longtext	NO		
passing	longtext	NO		
departments	longtext	NO		
hash	varchar(64)	YES		NULL
created_by	int(10) unsigned	YES		NULL
submitted_at	datetime	YES		NULL
decided_by	int(10) unsigned	YES		NULL
decided_at	datetime	YES		NULL
comment	text	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Unique: selection_config_versions_unique (selection_id,version)

selection_scores

Withdrawn department-criteria feature (unused, kept for history).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
criterion_key	varchar(60)	NO		
user_id	int(10) unsigned	NO		
raw_score	decimal(8,2)	YES		NULL
participated	tinyint(1)	NO		1
status	enum('draft','submitted','approved','rejected')	NO		'draft'
entered_by	int(10) unsigned	YES		NULL
entered_at	datetime	YES		NULL
reviewed_by	int(10) unsigned	YES		NULL
reviewed_at	datetime	YES		NULL
note	text	YES		NULL

Unique: selection_scores_unique (selection_id,criterion_key,user_id)

selection_evaluators

Withdrawn department-criteria feature (unused, kept for history).

Column	Type	Null	Key	Default / extra

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
selection_id	int(10) unsigned	NO	MUL	
criterion_key	varchar(60)	NO		
user_id	int(10) unsigned	NO		
assigned_by	int(10) unsigned	YES		NULL
created_at	datetime	NO		current_timestamp()

Unique: `selection_evaluators_unique` (selection_id,criterion_key,user_id)

`choir_eligibility`

Daily-computed choir eligibility score per member (choir.policy).

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
score	decimal(6,2)	NO		
eligible	tinyint(1)	NO		0
breakdown	longtext	YES		NULL
eligible_since	datetime	YES		NULL
computed_at	datetime	NO		

Foreign keys: `user_id` → `users.id` (on delete cascade)

`choir_selections`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
performance_type	enum('wereb_mezmur','mezmur_only')	NO		
season	varchar(120)	YES		NULL
notes	text	YES		NULL
status	enum('active','removed')	NO	MUL	'active'
selected_by	int(10) unsigned	YES	MUL	NULL
selected_at	datetime	NO		current_timestamp()
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `selected_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`wedding_services`

Wedding Mezmur services; legacy eligibility columns kept but ignored.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
title	varchar(200)	NO		
service_at	datetime	NO	MUL	
location	varchar(200)	YES		NULL
notes	text	YES		NULL

Column	Type	Null	Key	Default / extra
requires_attendance	tinyint(1)	NO		0
min_attendance	decimal(5,2)	YES		NULL
requires_exam	tinyint(1)	NO		0
min_exam	decimal(5,2)	YES		NULL
lookback_months	int(10) unsigned	NO		3
category_ids	varchar(200)	YES		NULL
status	enum('planned','completed','cancelled')	NO		'planned'
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

`wedding_service_members`

Members selected for a wedding service.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
service_id	int(10) unsigned	NO	MUL	
user_id	int(10) unsigned	NO	MUL	
status	enum('selected','participated','absent','withdrawn')	NO		'selected'
snapshot	longtext	YES		NULL
note	text	YES		NULL
selected_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `selected_by` → `users.id` (on delete set null); `service_id` → `wedding_services.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

Unique: `wedding_service_members_service_id_user_id_unique` (service_id,user_id)

`payments`

Monthly contributions by Ethiopian period; one payment may cover 1–12 months; status and verification.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
period	varchar(7)	NO	MUL	
amount	decimal(12,2)	NO		
currency	varchar(8)	NO		'ETB'
method	enum('cash','bank','telebirr','other')	NO		'cash'
reference	text	YES		NULL
paid_on	date	NO		
receipt	varchar(255)	YES		NULL

Column	Type	Null	Key	Default / extra
note	text	YES		NULL
recorded_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
months	int(10) unsigned	NO		1
coverage_end	varchar(7)	NO		
member_type	enum('student','worker')	YES		NULL
monthly_rate	decimal(12,2)	YES		NULL
status	enum('recorded','verified')	NO		'recorded'
verified_by	int(10) unsigned	YES	MUL	NULL

Foreign keys: `recorded_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade); `verified_by` → `users.id` (on delete set null)

`payment_rates`

Monthly rate per member type (student / worker) with effective date.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
member_type	enum('student','worker')	NO	MUL	
monthly_amount	decimal(12,2)	NO		
currency	varchar(8)	NO		'ETB'
effective_from	date	NO		
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

`items`

Property and library items with stock, condition, domain (education/mezmur), department and borrowing rules.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(200)	NO	MUL	
author	varchar(200)	YES		NULL
code	varchar(60)	NO	UNI	
category	varchar(100)	YES	MUL	NULL
description	varchar(1000)	YES		NULL
quantity	int(10) unsigned	NO		1
available_quantity	int(10) unsigned	NO		1
condition	varchar(100)	YES		NULL
status	enum('active','inactive','retired')	NO	MUL	'active'
image	varchar(255)	YES		NULL
location	varchar(200)	YES		NULL
notes	varchar(1000)	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL
domain	enum('mezmur','timhirt')	NO	MUL	'mezmur'
kind	enum('physical','digital')	NO	MUL	'physical'
department_id	int(10) unsigned	YES	MUL	NULL
direct_borrowable	tinyint(1)	NO		0
max_quantity	int(10) unsigned	YES		NULL
max_days	int(10) unsigned	YES		NULL
renewable	tinyint(1)	NO		0
file_path	varchar(255)	YES		NULL
file_mime	varchar(100)	YES		NULL
file_size	int(10) unsigned	YES		NULL
downloads	int(10) unsigned	NO		0

Foreign keys: `created_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

Unique: `items_code_unique` (code)

`item_categories`

Column	Type	Null	Key	Default / extra
key	varchar(40)	NO	PRI	
name	varchar(100)	NO		
name_en	varchar(100)	YES		NULL
code	varchar(6)	NO		
sort_order	int(11)	NO		0
is_active	tinyint(1)	NO		1
built_in	tinyint(1)	NO		0
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

`borrowings`

Individual loans: quantities, due date, return condition, fine and fine overrides.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
item_id	int(10) unsigned	NO	MUL	
batch_id	int(10) unsigned	YES	MUL	NULL
quantity	int(10) unsigned	NO		1
borrowed_on	date	NO		
due_on	date	NO	MUL	
status	enum('pending_approval','active','returned','partially_returned','lost','rejected')	NO	MUL	'active'

Column	Type	Null	Key	Default / extra
notes	text	YES		NULL
issued_by	int(10) unsigned	YES	MUL	NULL
returned_on	date	YES		NULL
returned_quantity	int(10) unsigned	NO		0
return_condition	varchar(100)	YES		NULL
return_notes	text	YES		NULL
returned_to	int(10) unsigned	YES	MUL	NULL
fine_amount	decimal(10,2)	NO		0.00
fine_overridden	tinyint(1)	NO		0
fine_override_reason	text	YES		NULL
fine_overridden_by	int(10) unsigned	YES	MUL	NULL
fine_status	enum('none','pending','paid','waived')	NO		'none'
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
approved_by	int(10) unsigned	YES	MUL	NULL
approved_at	datetime	YES		NULL
rejection_reason	text	YES		NULL
renewals	int(10) unsigned	NO		0
issue_condition	varchar(100)	YES		NULL
property_request_id	int(10) unsigned	YES		NULL

Foreign keys: `approved_by` → `users.id` (on delete set null); `batch_id` → `borrowing_batches.id` (on delete set null); `fine_overridden_by` → `users.id` (on delete set null); `issued_by` → `users.id` (on delete set null); `item_id` → `items.id` (on delete restrict); `returned_to` → `users.id` (on delete set null); `user_id` → `users.id` (on delete restrict)

borrowing_batches

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
created_by	int(10) unsigned	YES	MUL	NULL
record_count	int(10) unsigned	NO		0
notes	text	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

borrowing_signatures

Signatures captured when an item is handed over.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
borrowing_id	int(10) unsigned	NO	UNI	
user_id	int(10) unsigned	NO	MUL	
signature_path	varchar(255)	NO		
sha256	varchar(64)	NO		

Column	Type	Null	Key	Default / extra
signed_at	datetime	NO		
captured_by	int(10) unsigned	YES	MUL	NULL

Foreign keys: `borrowing_id` → `borrowings.id` (on delete cascade); `captured_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `borrowing_signatures_borrowing_id_unique` (borrowing_id)

`return_conditions`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
label	varchar(100)	NO	UNI	
marks_lost	tinyint(1)	NO		0
restock	tinyint(1)	NO		1
display_order	int(11)	NO		0
is_active	tinyint(1)	NO		1

Unique: `return_conditions_label_unique` (label)

`library_requests`

Members' requests for physical books, linked to the resulting borrowing.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
item_id	int(10) unsigned	NO	MUL	
quantity	int(10) unsigned	NO		1
note	text	YES		NULL
needed_on	date	YES		NULL
status	enum('requested','approved','rejected','issued','cancelled')	NO	MUL	'requested'
review_note	text	YES		NULL
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
borrowing_id	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `borrowing_id` → `borrowings.id` (on delete set null); `item_id` → `items.id` (on delete restrict); `reviewed_by` → `users.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

`property_requests`

Department property requests through review, issue and return.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	

Column	Type	Null	Key	Default / extra
item_id	int(10) unsigned	YES	MUL	NULL
item_name	varchar(200)	YES		NULL
quantity	int(10) unsigned	YES		NULL
purpose	varchar(1000)	NO		
needed_on	date	YES		NULL
return_on	date	YES		NULL
status	enum('draft','submitted','under_review','approved','rejected','issued','partially_issued','returned','cancelled')	NO	MUL	'submitted'
issued_quantity	int(10) unsigned	NO		0
returned_quantity	int(10) unsigned	NO		0
requested_by	int(10) unsigned	YES	MUL	NULL
reviewed_by	int(10) unsigned	YES	MUL	NULL
review_note	text	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `department_id` → `departments.id` (on delete restrict); `item_id` → `items.id` (on delete set null); `requested_by` → `users.id` (on delete set null); `reviewed_by` → `users.id` (on delete set null)

property_request_events

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
request_id	int(10) unsigned	NO	MUL	
action	varchar(30)	NO		
from_status	varchar(20)	YES		NULL
to_status	varchar(20)	YES		NULL
quantity	int(10) unsigned	YES		NULL
note	text	YES		NULL
actor_id	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `actor_id` → `users.id` (on delete set null); `request_id` → `property_requests.id` (on delete cascade)

project_budget_adjustments

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
current_budget	decimal(14,2)	YES		NULL
amount	decimal(14,2)	NO		
reason	text	NO		

Column	Type	Null	Key	Default / extra
requested_by	int(10) unsigned	YES	MUL	NULL
status	enum('pending','approved','rejected','cancelled')	NO		'pending'
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
review_note	text	YES		NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `project_id` → `projects.id` (on delete cascade); `requested_by` → `users.id` (on delete set null); `reviewed_by` → `users.id` (on delete set null)

project_budget_categories

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
name	varchar(120)	NO		
name_en	varchar(120)	YES		NULL
sort_order	smallint(5) unsigned	NO		0
is_active	tinyint(1)	NO		1
created_at	datetime	NO		current_timestamp()

project_budget_lines

Column	Type	Null	Key	Default / extra
project_id	int(10) unsigned	NO	PRI	
category_id	int(10) unsigned	NO	PRI	
amount	decimal(14,2)	NO		0.00

Foreign keys: `category_id` → `project_budget_categories.id` (on delete restrict); `project_id` → `projects.id` (on delete cascade)

project_comments

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
entity_type	enum('issue','suggestion','task')	NO	MUL	
entity_id	int(10) unsigned	NO		
user_id	int(10) unsigned	YES	MUL	NULL
body	mediumtext	NO		
created_at	datetime	NO		current_timestamp()

Foreign keys: `project_id` → `projects.id` (on delete cascade); `user_id` → `users.id` (on delete set null)

project_expenses

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
category_id	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
amount	decimal(14,2)	NO		
currency	varchar(8)	NO		'ETB'
description	text	NO		
expense_date	date	NO		
requested_by	int(10) unsigned	YES	MUL	NULL
paid_by	text	YES		NULL
receipt_path	varchar(255)	YES		NULL
receipt_mime	varchar(100)	YES		NULL
receipt_name	varchar(150)	YES		NULL
status	enum('pending','approved','rejected','recorded','cancelled')	NO		'pending'
over_budget	tinyint(1)	NO		0
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
review_note	text	YES		NULL
notes	text	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `category_id` → `project_budget_categories.id` (on delete set null); `project_id` → `projects.id` (on delete cascade); `requested_by` → `users.id` (on delete set null); `reviewed_by` → `users.id` (on delete set null)

`project_files`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
task_id	int(10) unsigned	YES	MUL	NULL
issue_id	int(10) unsigned	YES	MUL	NULL
path	varchar(255)	NO		
mime	varchar(100)	NO		
size	int(10) unsigned	NO		
name	varchar(150)	YES		NULL
uploaded_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `issue_id` → `project_issues.id` (on delete cascade); `project_id` → `projects.id` (on delete cascade); `task_id` → `project_tasks.id` (on delete cascade); `uploaded_by` → `users.id` (on delete set null)

`project_issues`

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
task_id	int(10) unsigned	YES	MUL	NULL
department_id	int(10) unsigned	NO	MUL	
reporter_id	int(10) unsigned	YES	MUL	NULL

Column	Type	Null	Key	Default / extra
title	varchar(200)	NO		
description	mediumtext	YES		NULL
priority	enum('low','medium','high','urgent')	NO		'medium'
status	enum('open','under_review','assigned','in_progress','resolved','closed')	NO		'open'
assignee_id	int(10) unsigned	YES	MUL	NULL
due_date	date	YES		NULL
resolution	mediumtext	YES		NULL
resolved_at	datetime	YES		NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `assignee_id` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete restrict); `project_id` → `projects.id` (on delete cascade); `reporter_id` → `users.id` (on delete set null); `task_id` → `project_tasks.id` (on delete set null)

project_members

Column	Type	Null	Key	Default / extra
project_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
role	enum('leader','member')	NO		'member'
added_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `added_by` → `users.id` (on delete set null); `project_id` → `projects.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

project_suggestions

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	
project_id	int(10) unsigned	YES	MUL	NULL
author_id	int(10) unsigned	YES	MUL	NULL
title	varchar(200)	NO		
body	mediumtext	YES		NULL
status	enum('submitted','under_review','accepted','rejected','converted')	NO		'submitted'
review_note	text	YES		NULL
reviewed_by	int(10) unsigned	YES	MUL	NULL
reviewed_at	datetime	YES		NULL
converted_task_id	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `author_id` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete restrict); `project_id` → `projects.id` (on delete cascade); `reviewed_by` → `users.id` (on delete set null); `converted_task_id` → `project_tasks.id` (on delete set null)

project_task_assignees

Column	Type	Null	Key	Default / extra
task_id	int(10) unsigned	NO	PRI	
user_id	int(10) unsigned	NO	PRI	
assigned_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `assigned_by` → `users.id` (on delete set null); `task_id` → `project_tasks.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

project_tasks

Tasks with assignees, deadlines, status reasons, progress and versioning (offline conflicts).

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
project_id	int(10) unsigned	NO	MUL	
department_id	int(10) unsigned	NO	MUL	
title	varchar(200)	NO		
description	mediumtext	YES		NULL
start_date	date	YES		NULL
deadline	datetime	YES	MUL	NULL
priority	enum('low','medium','high','urgent')	NO		'medium'
status	enum('not_started','pending','in_progress','blocked','completed','cancelled')	NO		'not_started'
status_reason	text	YES		NULL
pending_reason	enum('approval','materials','finance','another_task','another_department','paused','other')	YES		NULL
progress	tinyint(3) unsigned	NO		0
weight	smallint(5) unsigned	NO		1
notes	mediumtext	YES		NULL
issue_id	int(10) unsigned	YES	MUL	NULL
program_id	int(10) unsigned	YES	MUL	NULL
suggestion_id	int(10) unsigned	YES		NULL
completed_at	datetime	YES		NULL
completed_by	int(10) unsigned	YES	MUL	NULL
completion_note	text	YES		NULL
version	int(10) unsigned	NO		1
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL

Foreign keys: `completed_by` → `users.id` (on delete set null); `created_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete restrict); `issue_id` → `project_issues.id` (on delete set null); `program_id` → `programs.id` (on delete set null); `project_id` → `projects.id` (on delete cascade)

projects

Department projects: leader, dates, status, priority, progress, budgets.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	
title	varchar(200)	NO		
description	mediumtext	YES		NULL
leader_id	int(10) unsigned	YES	MUL	NULL
start_date	date	YES		NULL
deadline	datetime	YES	MUL	NULL
status	enum('planning','not_started','in_progress','on_hold','completed','cancelled')	NO		'planning'
priority	enum('low','medium','high','urgent')	NO		'medium'
progress	tinyint(3) unsigned	NO		0
progress_mode	enum('tasks','weighted','manual')	NO		'tasks'
notes	mediumtext	YES		NULL
allow_late_tasks	tinyint(1)	NO		0
currency	varchar(8)	NO		'ETB'
proposed_budget	decimal(14,2)	YES		NULL
approved_budget	decimal(14,2)	YES		NULL
budget_approved_by	int(10) unsigned	YES	MUL	NULL
budget_approved_at	datetime	YES		NULL
completed_at	datetime	YES		NULL
created_by	int(10) unsigned	YES	MUL	NULL
updated_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()
deleted_at	datetime	YES		NULL

Foreign keys: `budget_approved_by` → `users.id` (on delete set null); `created_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete restrict); `leader_id` → `users.id` (on delete set null); `updated_by` → `users.id` (on delete set null)

chat_messages

Department group and private messages, attachments, meeting links and Task-Bot posts.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
department_id	int(10) unsigned	NO	MUL	
thread_key	varchar(40)	NO		
sender_id	int(10) unsigned	NO	MUL	
recipient_id	int(10) unsigned	YES	MUL	NULL
kind	enum('text','image','audio','video','file','meeting','task_bot')	NO		'text'
body	mediumtext	YES		NULL
attachment_path	varchar(255)	YES		NULL
attachment_mime	varchar(100)	YES		NULL

Column	Type	Null	Key	Default / extra
attachment_size	int(10) unsigned	YES		NULL
attachment_name	text	YES		NULL
meeting_url	text	YES		NULL
meeting_at	datetime	YES		NULL
deleted_at	datetime	YES		NULL
deleted_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()
task_id	int(10) unsigned	YES	MUL	NULL
bot_data	longtext	YES		NULL

Foreign keys: `deleted_by` → `users.id` (on delete set null); `department_id` → `departments.id` (on delete cascade); `recipient_id` → `users.id` (on delete cascade); `sender_id` → `users.id` (on delete cascade); `task_id` → `project_tasks.id` (on delete set null)

chat_reads

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
department_id	int(10) unsigned	NO	PRI	
thread_key	varchar(40)	NO	PRI	
last_read_id	bigint(20) unsigned	NO		0
created_at	datetime	NO		current_timestamp()
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `department_id` → `departments.id` (on delete cascade); `user_id` → `users.id` (on delete cascade)

notifications

In-app notifications per user with push / e-mail delivery status and dedupe key.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
type	varchar(40)	NO		
topic	varchar(20)	YES		NULL
title	varchar(255)	NO		
body	text	NO		
entity_type	varchar(40)	YES		NULL
entity_id	int(10) unsigned	YES		NULL
data	longtext	YES		NULL
broadcast_id	int(10) unsigned	YES	MUL	NULL
resent_from_id	bigint(20) unsigned	YES		NULL
sent_by	int(10) unsigned	YES		NULL
dedupe_key	varchar(150)	YES	UNI	NULL
delivery_status	enum('pending','sent','failed','skipped')	NO	MUL	'pending'
sent_at	datetime	YES		NULL
email_status	enum('none','pending','sent','failed','skipped')	NO	MUL	'none'

Column	Type	Null	Key	Default / extra
emailed_at	datetime	YES		NULL
read_at	datetime	YES		NULL
created_at	datetime	NO	MUL	current_timestamp()

Foreign keys: `broadcast_id` → `notification_broadcasts.id` (on delete set null); `user_id` → `users.id` (on delete cascade)

Unique: `notifications_dedupe_key_unique` (dedupe_key)

`notification_broadcasts`

Admin messages to one, many, a filtered group or everyone.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
sender_id	int(10) unsigned	YES	MUL	NULL
kind	enum('announcement','reminder','program_info','admin_message')	NO		
title	varchar(255)	NO		
body	varchar(2000)	NO		
target_type	enum('user','selected','filter','all')	NO		
target_filter	longtext	YES		NULL
entity_type	varchar(40)	YES		NULL
entity_id	int(10) unsigned	YES		NULL
recipient_count	int(10) unsigned	NO		0
created_at	datetime	NO		current_timestamp()
resent_from_id	int(10) unsigned	YES		NULL
channels	varchar(40)	NO		'push'

Foreign keys: `sender_id` → `users.id` (on delete set null)

`scheduled_notifications`

Time-based notifications stored ahead and released by the dispatcher job.

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
due_at	datetime	NO		
type	varchar(40)	NO		
title	varchar(255)	NO		
body	text	NO		
entity_type	varchar(40)	YES		NULL
entity_id	int(10) unsigned	YES		NULL
data	longtext	YES		NULL
source_type	varchar(40)	NO	MUL	
source_id	int(10) unsigned	NO		
dedupe_key	varchar(150)	NO	UNI	
status	enum('pending','sent','cancelled')	NO	MUL	'pending'
sent_at	datetime	YES		NULL

Column	Type	Null	Key	Default / extra
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

Unique: `scheduled_notifications_dedupe_unique` (dedupe_key)

guidance_articles

Help-centre articles (bilingual), contextual help per screen and tours.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
slug	varchar(100)	NO	UNI	
category	varchar(60)	NO	MUL	
title	varchar(255)	NO		
summary	text	YES		NULL
content	longtext	NO		
target_role	varchar(40)	NO	MUL	'all'
target_screen	varchar(100)	YES	MUL	NULL
required_permission	varchar(80)	YES	MUL	NULL
steps	longtext	YES		NULL
tips	longtext	YES		NULL
faq_items	longtext	YES		NULL
tour_id	varchar(60)	YES	MUL	NULL
icon	varchar(60)	YES		NULL
sort_order	int(11)	NO		0
status	varchar(20)	NO	MUL	'published'
created_by	int(10) unsigned	YES		NULL
updated_by	int(10) unsigned	YES		NULL
created_at	timestamp	YES		NULL
updated_at	timestamp	YES		NULL

Unique: `guidance_articles_slug_unique` (slug)

user_guidance_progress

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	NO	MUL	
tour_id	varchar(60)	NO		
completed	tinyint(1)	NO		0
dismissed	tinyint(1)	NO		0
completed_at	datetime	YES		NULL
created_at	timestamp	YES		NULL
updated_at	timestamp	YES		NULL

Unique: `user_guidance_progress_user_id_tour_id_unique` (user_id,tour_id)

settings

Validated JSON settings (fines, payments, attendance, borrowing, reminders, member ID format...).

Column	Type	Null	Key	Default / extra
key	varchar(80)	NO	PRI	
value	longtext	NO		
updated_by	int(10) unsigned	YES	MUL	NULL
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

Foreign keys: `updated_by` → `users.id` (on delete set null)

audit_logs

Administrative audit trail (actor, action, entity, metadata, IP, user agent).

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
actor_id	int(10) unsigned	YES	MUL	NULL
is_hidden	tinyint(1)	NO	MUL	0
action	varchar(80)	NO	MUL	
entity_type	varchar(40)	YES	MUL	NULL
entity_id	int(10) unsigned	YES		NULL
metadata	longtext	YES		NULL
ip	text	YES		NULL
user_agent	text	YES		NULL
created_at	datetime	NO	MUL	current_timestamp()

Foreign keys: `actor_id` → `users.id` (on delete set null)

activity_logs

Member activity events (logins, plays, downloads...).

Column	Type	Null	Key	Default / extra
id	bigint(20) unsigned	NO	PRI	auto_increment
user_id	int(10) unsigned	YES	MUL	NULL
action	varchar(60)	NO	MUL	
entity_type	varchar(40)	YES	MUL	NULL
entity_id	int(10) unsigned	YES		NULL
metadata	longtext	YES		NULL
client_event_id	varchar(36)	YES		NULL
occurred_at	datetime	NO		current_timestamp()
created_at	datetime	NO		current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete set null)

Unique: `activity_logs_user_id_client_event_id_unique` (user_id,client_event_id)

job_runs

Last run of each background job.

Column	Type	Null	Key	Default / extra
name	varchar(64)	NO	PRI	

Column	Type	Null	Key	Default / extra
last_run_at	datetime	YES		NULL
last_day	date	YES		NULL
last_result	varchar(2000)	YES		NULL
updated_at	datetime	NO		current_timestamp() on update current_timestamp()

sync_receipts

Idempotency receipts of offline operations (op_id per user).

Column	Type	Null	Key	Default / extra
user_id	int(10) unsigned	NO	PRI	
op_id	varchar(36)	NO	PRI	
status	varchar(20)	NO		
result	longtext	YES		NULL
created_at	datetime	NO	MUL	current_timestamp()

Foreign keys: `user_id` → `users.id` (on delete cascade)

app_releases

Published APKs: version, size, SHA-256, notes, mandatory flag, minimum supported build.

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
platform	enum('android')	NO	MUL	'android'
version_name	varchar(32)	NO		
version_code	int(10) unsigned	NO		
file_path	varchar(255)	NO		
file_size	bigint(20) unsigned	NO		
sha256	varchar(64)	NO		
release_notes	text	YES		NULL
min_supported_code	int(10) unsigned	NO		0
is_mandatory	tinyint(1)	NO		0
is_published	tinyint(1)	NO		1
download_count	int(10) unsigned	NO		0
created_by	int(10) unsigned	YES	MUL	NULL
created_at	datetime	NO		current_timestamp()

Foreign keys: `created_by` → `users.id` (on delete set null)

Unique: `app_releases_platform_version_code_unique` (platform,version_code)

migrations

Column	Type	Null	Key	Default / extra
id	int(10) unsigned	NO	PRI	auto_increment
migration	varchar(255)	NO		
batch	int(11)	NO		

Appendix D. Unreleased schema changes in the source tree

Migration	Changes
2026_10_20_000001_academic_structure	New tables edu_academic_years, edu_terms, edu_levels (Grades 1–12 seeded), program_edu_courses; edu_classes + academic_year_id, term_id, level_id, allow_multiple; edu_enrollments + academic_year_id, exclusive, exclusive_key; edu_courses + term_id, member_groups, mode, pass_mark
2026_10_20_000002_program_edu_attendance	New tables program_edu_classes, attendance_edu_classes; edu.attendance_cutover setting; attendance_sync_conflicts supports class conflicts; selection_config_versions + config

Appendix E. Source documents consulted

- MEMORY.md — implementation history (contains credentials: handle as restricted).
- GUIDANCE_IMPLEMENTATION_AUDIT_AND_DEBUG_LOG.md — move-by-move audit log up to the 1.5.3+16 release.
- SELECTION.md — holiday selection guide (restored single-configuration logic).
- prompt.md — Education & Attendance implementation plan F0–F21 with decisions D1–D9, reviewed 2026-10-16.

Appendix F. Restricted information

The platform contains a hidden top-level authority (separate sign-in, separate token scope and additional factors) used for recovery and governance. Its routes, commands and configuration are intentionally excluded from this document and from the presentation. They are described in the separately distributed file `RESTRICTED_Technical_Appendix.md`, which must be shared only with the system owner and the responsible technical administrator.